Supplementary Information

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Part I. Methodology

S1. Data Collection and Sample Construction

S1.1. Overview of the data curation pipeline

Our goal was to assemble a corpus of news articles in which a known politician is pictured, so that the stance of the headline and of the image toward that politician could be compared. To our knowledge, no existing large scale dataset links news articles shared on Facebook to the identity of the people shown in their images, so we built one. We collected every article that a broad set of partisan U.S. news outlets shared on Facebook over five years, downloaded the accompanying images, identified the politicians pictured with a face-recognition pipeline, and rated the stance of each headline and image with a large language model. Each step narrows the corpus. Some records are excluded by design because they fall outside the study's scope, such as images without a face or headlines that are not about the pictured politician. Others are lost unintentionally because data could not be retrieved or a step missed something it should have kept, such as broken image links or politician faces the recognizer failed to identify. We distinguish the two throughout, because only unintended losses can make the sample unrepresentative of the population the study targets: articles by these outlets that picture one of the target politicians.

Fig. S1, which reproduces Fig. 1 of the main text, summarizes the pipeline. The main sequence of steps runs from left to right across the middle of the figure, with the number of records retained at each step in blue. The panels above it show the outlet pool (*Data scope*), the reference gallery used for identity matching (*Reference faces*), and the last reductions to the analytic sample. The panels below it show how the face detector and embedding model were selected (panels A and B), the stance prompts, and the human validation of the stance ratings.

Curated datasets. Besides the NewsWhip records, the pipeline relies on three ground-truth datasets that we curated ourselves:

1. **Human-coded news images:** 11,100 news images of 270 public figures other than the target politicians, coded by trained coders for whether each contains a clear, identifiable face (S1.4; Fig. S1, panel A; Table S2).
2. **Photograph pairs:** 1,000 same-person and 1,000 different-person pairs, sampled from the manually verified images of the same 270 figures (S1.4; Fig. S1, panel B).
3. **Reference gallery:** 11,170 reference photographs of 1,311 public figures, including the target politicians. The list of people was compiled by hand to span industries, gender, and race/ethnicity; their photographs were collected and screened automatically (S1.6; Fig. S1, *Reference faces*).

Three human validations. Each automated step that decides which articles enter the sample, or how they are coded, was checked against human judgment:

1. **Face detection.** The detector's output was compared with the coders' judgments of the 11,100 news images. YOLOv8 found 94.5% of the faces that coders judged clear and identifiable (S1.4; Table S3A).
2. **Identity matching.** For each target politician, 95–97 matched images were inspected by hand. Precision ranged from .823 to 1.000 for the eight retained politicians, and two politicians with low precision

were excluded (S1.6; Table S4).

3. **Stance classification.** Crowd raters (n = 393) rated the headlines and images of 288 articles separately, about 8.2 raters per item. The model's ratings tracked the average human rating (Spearman ρ = .69 for headlines and .72 for images; S2; Table S5).

The pipeline proceeds in seven steps.

1. **Source selection (S1.2).** Defines the population of outlets: an initial pool of 63 U.S. news outlets spanning the partisan spectrum.
2. **Article retrieval (S1.3).** For every article we need the headline, the image, and the engagement it received. NewsWhip supplied the headline, image link, and Facebook engagement for every article the outlets shared between 2017 and 2021: 9,359,596 articles from 51 of the 63 outlets. *Lost unintentionally:* the 12 outlets that NewsWhip does not cover, and four outlet-year files that were not retrieved.
3. **Face-model selection (S1.4).** Retrieval returns millions of image links, and most of the images show no person or people outside the study, so coding them by hand is not feasible. Two automatic tools are needed: a face detector that finds faces in an image, and an embedding model that converts each face into a numeric vector such that photographs of the same person lie close together. We narrowed the candidates using published benchmarks and then evaluated them on news images of 270 public figures other than the target politicians, hand-coded by trained coders (Fig. S1, panel A). We compared each candidate detector's output with the coders' judgment of whether an image contains a clear, identifiable face. We then ran a discriminant analysis on same-person and different-person photograph pairs, testing how well each detector and embedding model combination separates the two. No records are excluded at this step.
4. **Face detection and image download (S1.5).** Because the study concerns how people are depicted, only images showing a human face are relevant. We therefore ran the face detector during the download, so that only images with a face were stored, and embedded each detected face in the same pass, so that the stored output was ready for identity matching (4,224,177 images; 11,663,296 faces). *Excluded by design:* images without a detectable face. *Lost unintentionally:* images whose links no longer worked, including nearly all images from three outlets, which therefore dropped out of the study.
5. **Identity matching (S1.6).** To study people of known identity, each face must be linked to a person. We curated a gallery of reference photographs of 1,311 public figures, including the target politicians (Fig. S1, *Reference faces*), and matched each stored face to the person in the gallery it most resembles. A face was kept only if that person was a target politician and the match was confident (227,929 matched rows). Because the gallery is broad, a face of someone else, such as a politician's family member, is matched to that person instead of being forced onto the most similar politician. Because the discriminant analysis measures how well the models tell two faces apart, not how well they identify a given politician in news images, we checked the precision of the matches separately by manually inspecting a sample of matched faces for each politician; missed faces were not checked. *Excluded by design:* faces of everyone else, low-confidence matches, and two politicians (Jeff Sessions, Lindsey Graham) whose faces were identified poorly in the manual check. *Lost unintentionally:* target politicians' faces that the pipeline missed, for example in profile or at small sizes.
6. **Stance detection (S2).** A large language model rated how favorably the headline and the image portray the pictured politician, and coded whether the headline refers to that politician at all and whether

the politician can be identified in the image. The ratings were validated against crowd ratings of the same headlines and images (S2).

7. **Analytic sample (S1.7).** Repeated rows for images showing more than one target politician were collapsed to one record per article (214,383 articles). Using the stance coding from step 6, articles whose headline does not refer to the pictured politician were then removed (*excluded by design:* 30,853), as were articles whose image the model judged not to show the politician identifiably (*excluded by design:* 14). The analytic sample has 183,516 articles from 48 outlets.

Two of these steps, article retrieval and image download, also determine which outlets remain in the sample. S1.8 describes these source-level exclusions and assesses whether they, and image-download failures within retained outlets, bias the estimates.

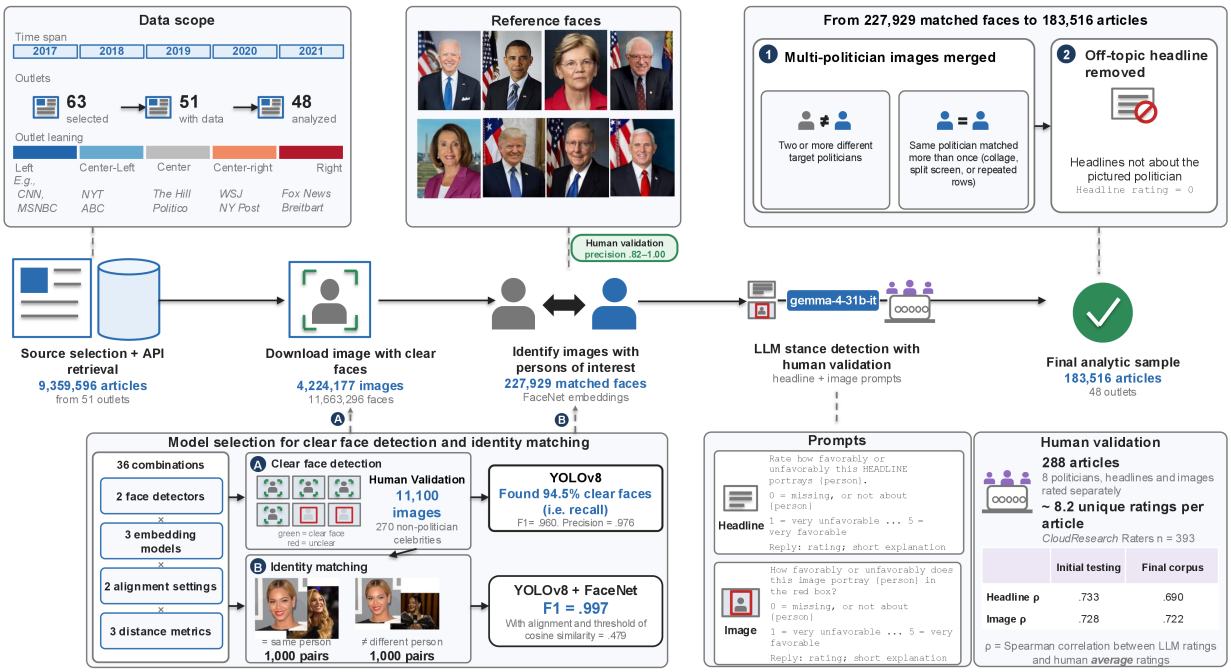


Fig. S1. Data collection, image processing, and stance classification pipeline. This figure reproduces Fig. 1 of the main text. We include it here because it gives a schematic overview of the pipeline that S1–S2 document step by step. The pipeline has four main steps: (1) retrieve news articles, (2) keep images with clear faces, (3) identify images showing the target politicians, and (4) rate headline and image stance. It relies on three curated ground-truth datasets: human-coded news images of public figures other than the target politicians, same-person and different-person photograph pairs composed from those images, and a reference gallery that includes the target politicians. Human judgment was used to check face detection, identity matching, and stance classification. Blue numbers give the records retained at each step. **(1) Source selection and retrieval (S1.2–S1.3).** We selected 63 U.S. news outlets spanning the partisan spectrum (leaning coded by Media Bias/Fact Check; example outlets shown for each leaning) and retrieved their 2017–2021 articles, with headlines, image links, and Facebook engagement counts, through the NewsWhip API, which covered 51 of them. **(2) Clear-face detection (S1.4–S1.5).** Images were downloaded only if the face detector found a face. YOLOv8 found 94.5% of the faces that trained coders judged clear and identifiable in news images of 270 public figures other than the target politicians (panel A). **(3) Identity matching**

(S1.4, S1.6). Among 36 configurations of detectors, embedding models, alignment settings, and distance metrics, YOLOv8 with FaceNet embeddings best distinguished same-person from different-person photograph pairs (panel B). The F1 shown (.997) is for the best configuration, which used face alignment; the deployed pipeline used the same detector and embedding without alignment (F1 = .994). Each detected face was matched against the reference gallery, and manual inspection gave a precision of .82 to 1.00 for the eight retained politicians (Table S4). **(4) Stance classification and human validation (S2).** gemma-4-31b-it rated the favorability of each headline and each image toward the pictured politician, using separate prompts; headlines not about that politician, and images in which the politician could not be identified, were coded 0 and removed. Crowd raters judged headlines and images separately, and the automated ratings tracked the average human ratings (Spearman ρ = .69–.73). **Final sample (S1.7).** Images showing more than one target politician were merged into one record per article, and articles with off-topic headlines were removed, leaving 183,516 articles from 48 outlets.

S1.2. Source selection

Initial outlet pool. We initially compiled a total of 63 U.S. news outlets based on Pew Research Center's surveys of the sources Americans use for political news, namely its 2014 study of political polarization and media habits (Mitchell et al., 2014) and its 2020 Election News Pathways project (Jurkowitz et al., 2020), and on prior research on partisan news audiences. The pool spans national newspapers (e.g., *New York Times*, *Wall Street Journal*), broadcast and cable networks (e.g., NBC, CNN, Fox News), public broadcasters (PBS, NPR), and digital-native outlets with liberal and conservative audiences (e.g., HuffPost, Daily Caller, Daily Wire). Each outlet was coded for partisan leaning (Media Bias/Fact Check, MBFC; S3.1) and media type.

Of the 63 outlets, 48 reached the analytic sample; they are listed in Table S1. The other 15 were excluded at the retrieval stages described below (S1.3 and S1.5). The coverage bias analysis is documented in S1.8.

Table S1. News outlets in the analytic sample

Outlet	MBFC leaning	Type	Articles (NewsWhip)	Images with faces (% of articles)	Analytic articles
AlterNet	Left	Digital Native	51,112	14,927 (29%)	1,626
CNN	Left	Cable News	295,377	210,093 (71%)	13,958
Daily Beast	Left	Digital Native	97,394	54,835 (56%)	2,779
MSNBC	Left	Cable News	94,661	74,168 (78%)	6,196
Newsweek	Left	Magazine	188,198	126,699 (67%)	7,189
Salon	Left	Digital Native	52,585	39,216 (75%)	4,328
Slate	Left	Digital Native	51,675	29,218 (57%)	1,248
Vox	Left	Digital Native	36,028	30,157 (84%)	1,990
ABC	Center-left	Network TV	473,434	212,143 (45%)	6,399
Business Insider	Center-left	Digital Native	386,697	142,469 (37%)	5,703
BuzzFeed	Center-left	Digital Native	299,548	164,060 (55%)	663
CBS	Center-left	Network TV	238,999	164,404 (69%)	8,819
HuffPost	Center-left	Digital Native	186,329	124,962 (67%)	6,881
Mother Jones	Center-left	Magazine	24,806	14,150 (57%)	1,149
NBC	Center-left	Network TV	167,747	111,952 (67%)	5,909
NPR	Center-left	Digital Native	139,969	52,913 (38%)	1,222
NYT	Center-left	Newspaper	493,921	208,283 (42%)	3,762
PBS	Center-left	Network TV	68,054	37,819 (56%)	1,986
The Atlantic	Center-left	Magazine	31,060	19,341 (62%)	731
The Guardian	Center-left	Newspaper	524,648	346,459 (66%)	3,660
USA Today	Center-left	Newspaper	570,865	347,530 (61%)	7,154
Washington Post	Center-left	Newspaper	459,838	32,861 (7%)	953
Politico	Center	Digital Native	97,419	56,529 (58%)	5,462
The Hill	Center	Digital Native	252,956	13,402 (5%)	1,469
NY Post	Center-right	Newspaper	286,516	73,113 (26%)	2,871
RealClearPolitics	Center-right	Digital Native	108,667	43,689 (40%)	4,046
WSJ	Center-right	Newspaper	220,862	90,854 (41%)	3,069
Washington Times	Center-right	Newspaper	644,695	210,988 (33%)	7,164
Daily Caller	Right	Digital Native	171,050	24,417 (14%)	1,557
Daily Wire	Right	Digital Native	66,955	24,800 (37%)	2,477
Epoch Times	Right	Digital Native	171,199	99,961 (58%)	4,530
Fox News	Right	Cable News	288,210	131,752 (46%)	8,020
Free Beacon	Right	Digital Native	32,248	24,932 (77%)	2,137
National Review	Right	Magazine	58,961	34,800 (59%)	3,423
NewsBusters	Right	Digital Native	37,095	29,543 (80%)	1,116
Newsmax	Right	Cable News	212,011	12,868 (6%)	749
OANN	Right	Cable News	81,891	419 (1%)	29
PJ Media	Right	Digital Native	174,789	9,512 (5%)	1,478
RedState	Right	Digital Native	52,253	16,636 (32%)	2,213
Right Scoop	Right	Digital Native	18,915	16,332 (86%)	2,517
The Blaze	Right	Digital Native	51,309	17,208 (34%)	1,302
The Federalist	Right	Magazine	24,844	5,215 (21%)	408
Townhall	Right	Digital Native	262,581	99,080 (38%)	6,492
Western Journal	Right	Digital Native	82,406	57,216 (69%)	6,038
Breitbart	Extreme right	Digital Native	768,265	513,195 (67%)	16,292
Gateway Pundit	Extreme right	Digital Native	65,722	29,603 (45%)	2,325
Infowars	Extreme right	Digital Native	58,830	12,790 (22%)	932
Political Insider	Extreme right	Digital Native	20,823	4,917 (24%)	1,095
Total (48 outlets)			9,244,417	4,212,430 (46%)	183,516

Notes. Leaning categories follow Media Bias/Fact Check (MBFC); 'least biased' is coded Center. Articles = article records retrieved from NewsWhip, 2017–2021. Images with faces = images downloaded with at least one face detected by YOLOv8, as a share of articles; failures were mostly HTTP 403, 404, or 410 errors on the original image URLs. Analytic articles = deduplicated articles with valid headline and image stance (N = 183,516). The 15 outlets in the initial pool that are not in the analytic sample are listed in S1.8.

S1.3. NewsWhip retrieval

The study requires, for each article, the headline, the image, and a measure of audience engagement, for all outlets in the pool over the same period. Through a paid academic subscription (initiated March 2022), we accessed NewsWhip's historical dataset of articles shared on Facebook. NewsWhip records engagement for each article rather than for each Facebook post, so an article that an outlet shared more than once appears only once in the data; the article is therefore the unit of observation and analysis throughout. For every article published by the accessible outlets between 2017 and 2021, NewsWhip supplied:

- the headline and blurb;
- the image URL;
- the publication time;
- an aggregate Facebook engagement count combining reactions, comments, and shares, which cannot be separated by interaction type.

NewsWhip queries return at most 5,000 articles per request. To check completeness, we compared yearly, monthly, and weekly query windows and found that weekly windows captured more than 99% of available content for nearly all outlets, so articles were retrieved in weekly windows. Engagement is cumulative up to the retrieval date.

NewsWhip returned articles for 51 of the 63 outlets in the pool, 9,359,596 article records in total. The other 12 outlets are not covered by NewsWhip: the BBC, CNS News, Daily Kos, The Economist, the *Los Angeles Times*, *The New Yorker*, NowThis, ThinkProgress, Univision, Vice, *The Weekly Standard*, and Zero Hedge (S1.8). Of the 51 retrieved outlets, 48 reached the analytic sample, contributing 9,244,417 articles (Table S1).

Four outlet-year files are missing from the retrieval: Fox News 2020, *Washington Times* 2017, BuzzFeed 2021, and NewsBusters 2020. We kept these four outlets because each is missing only one of the five years and still contributes articles from the other four.

S1.4. Face-model selection

NewsWhip supplies links to images, not the images themselves, and most linked images either show no person or show people outside the study. Two tasks therefore had to be automated: detecting faces, so that only images with a person are kept (S1.5), and identifying whose face it is (S1.6). Both rely on the same two components, a face detector and an embedding model that represents each face as a vector so that two photographs of the same person are close and photographs of different people are far apart. We selected these components first, before any news image was processed.

Published benchmarks. Candidate components were taken from the DeepFace library, whose authors benchmark combinations of detectors, embedding models, distance metrics, and alignment on the Labeled Faces in the Wild (LFW) dataset (Huang et al., 2007; Serengil & Özpınar, 2024). In that benchmark, FaceNet-family embeddings and VGG-Face approach or exceed the 97.5% verification accuracy of human judges. Detectors are usually compared on the WIDER FACE dataset (Yang et al., 2016); YuNet, for example, reaches 81.1% average precision on its hard subset (Wu et al., 2023). We used these results to narrow the candidates to three detectors (YOLOv8, Jocher et al., 2023; YuNet; FastMTCNN, Zhang et al., 2016) and three embedding models (FaceNet and FaceNet512, Schroff et al., 2015; VGG-Face, Parkhi et al., 2015). The published benchmarks are not sufficient for our purpose, however, for two reasons. First, LFW consists of mostly frontal, well-lit, centered photographs, whereas news images also contain small, turned, or partly covered faces, crowds, and graphics. Second, the pipeline does not need every face, only faces clear enough to be identified. We therefore evaluated the candidates on a more stringent benchmark built from the study's own corpus.

Benchmark public figures. The people in the benchmark come from a list of public figures compiled for a broader, ongoing research program on how computer-vision systems detect, identify, and frame human faces. The list began with 6,558 people collected from public sources spanning politics, entertainment, science, business, and other domains (for example, rosters of members of Congress, award winners, and lists of prominent figures), with each person's domain recorded from the source. Because many of them were not prominent enough to yield reliable images, we kept the 1,307 whose names appeared at least 50 times in the news headlines, which became the basis of the reference gallery (S1.6). Each person's gender and race/ethnicity were obtained by querying GPT-4o and recorded manually. From these 1,307, we drew a stratified sample of 270 people, spread as evenly as possible across combinations of gender and race/ethnicity, so that the face models could be evaluated across demographic groups rather than on the groups that dominate news coverage.

Human-coded news images. We drew 11,250 news images from 2017–2021 showing the 270 public figures, with up to 50 images per person. They span film and television, politics, music, sports, journalism, and business, as well as gender and race/ethnicity (Table S2). None of them is one of the ten target politicians, so the benchmark is independent of the identity-matching evaluation (S1.6). Six trained coders each coded a share of the images on three questions:

1. whether the image contains a clear, identifiable human face. Faces that were heavily covered (for example, an athlete in a ski mask and goggles), too small or blurred to recognize, or drawn rather than photographed were coded as no face;
2. if so, whether it shows the named person;
3. if it does, whether the face is suitable as a reference: both eyes, nose, and mouth visible, facing the

camera, and not covered.

Because the images were collected for the named figures, nearly all of them show a person. The benchmark therefore tests whether a detector finds the faces that a human could identify, not whether it can tell images with people from images without. Excluding 30 rows flagged as coding errors and 120 without a face code, 11,100 images were coded; 9,995 contain a clear face. Of these, 7,892 show the named person, and 6,309 of them (262 people) are also of good quality. That roughly one face image in five (2,084) shows someone other than the named person illustrates why the identity of the pictured person cannot be inferred from news text and had to be established from the face.

Face detection. We ran YOLOv8 and YuNet on every coded image and compared their output with the coders' judgment (Table S3A). The main measure is recall, the share of clear faces that the detector found. YOLOv8 found 94.5% of them (95% CI [94.0, 94.9]); YuNet found 51.5% [50.5, 52.4]. This ranking reverses the one on LFW, where YuNet paired with FaceNet slightly outperforms YOLOv8 (96.1% vs. 91.9% accuracy, Euclidean distance with alignment; Serengil & Özpınar, 2024). The reversal shows that performance on clean benchmark photographs does not carry over to news images, and it is the reason for evaluating the detectors on the corpus itself.

Precision is less informative here. About 90% of the benchmark images contain a clear face, and the remaining images are hard cases: photographs of a person whose face coders judged not identifiable. There are no images without people, which make up most of the corpus. Precision and the false-positive rate depend on this composition, so they do not describe how the detector performs on the corpus; we report them in Table S3A for completeness. YOLOv8 flagged a face in 230 of the 1,105 images coded as having none (21%). Most of these are probably faces that coders judged too unclear rather than detections in empty scenes; coders' notes on some of them mention drawings, artwork, or faces printed on objects. For the pipeline, recall is the quantity that matters: a detected face that does not belong to a target politician is discarded at identity matching (S1.6), whereas a missed face is lost.

Curated pairs. From the verified images, we assembled 1,000 pairs of photographs showing the same person and 1,000 pairs showing different people, sampled from the manually verified images of the 270 public figures (Fig. S1, panel B). Images in which no candidate detector found a face, and four images that turned out to be drawings, were replaced.

Discriminant analysis. For each candidate configuration, we detected the face in each photograph, embedded it, computed the distance between the two faces in each pair, and asked how well that distance separates same-person from different-person pairs. The three detectors were crossed with the three embedding models, face alignment on or off, and three distance metrics (cosine similarity, Euclidean, and L2-normalized Euclidean distance), all run through DeepFace. FastMTCNN was dropped because it timed out or detected non-faces, leaving 36 configurations (Table S3B). For each configuration, we scanned thresholds over the range where the same-person and different-person distributions overlap and kept the threshold with the highest F1; a finer scan over the full range gave nearly identical results.

Selection. YOLOv8 found far more of the human-coded faces than YuNet (Table S3A) and outperformed it in the pair analysis for every embedding; FaceNet was the best embedding (Table S3B). The best configuration, FaceNet with YOLOv8 and alignment, reached F1 = .997 (precision .998, recall .996; cosine threshold 0.479). The deployed pipeline used the same detector and embedding without alignment, to preserve head orientation and expression; that configuration reached F1 = .994 (cosine threshold 0.439). Because thresh-

olds were tuned on the same pairs used to compute F1, these are development-set estimates.

These F1 scores describe how well a configuration tells whether two photographs show the same person. They are not estimates of how accurately the pipeline identifies the target politicians in news images, which involves different photographs, a reference set, and a matching rule; the precision of the matches was checked separately (S1.6).

Limitation: single coding. Each benchmark image was coded by one coder, so inter-coder reliability cannot be estimated, and the face codes are a single human judgment rather than a consensus. Judging whether a face is "clear and identifiable" leaves room for discretion, and coders appear to have differed somewhat in where they drew the line: YOLOv8's recall computed separately against each coder ranged from .87 to .98. Because YOLOv8's advantage over YuNet holds for every coder, the choice of detector does not depend on these differences, but the overall recall should be read as approximate.

Table S2. Public figures in the face-detection benchmark

Characteristic	Category	People	%	Images
Primary public domain	Film and television	96	35.6	4,383
	Politics and government	58	21.5	2,325
	Music	44	16.3	1,936
	Sports	33	12.2	1,035
	Journalism and commentary	17	6.3	734
	Business and technology	5	1.9	99
	Other	17	6.3	738
Gender	Woman	120	44.4	5,220
	Man	148	54.8	5,946
	Nonbinary	2	0.7	84
Race/ethnicity	White	48	17.8	1,857
	Black	59	21.9	2,465
	Hispanic/Latino	62	23.0	2,619
	Asian	48	17.8	2,045
	Middle Eastern/North African	4	1.5	147
	Multiracial	49	18.1	2,117

Notes. The 270 public figures whose news images form the human-coded benchmark (Section S1.4; 11,250 images, up to 50 per person) and from whose verified images the photograph pairs were drawn. All are part of the hand-curated list of 1,311 public figures used for the reference gallery (Section S1.6), which was selected to span industries, gender, and race/ethnicity. None is one of the ten target politicians; other politicians and officials are included. Domain is each person's primary public role in 2017–2021; Other includes activists, authors, lawyers, chefs, fashion designers, models, members of royal families, and people in the news. Gender and race/ethnicity follow how each person is publicly identified; people widely described as having parents of different racial or ethnic backgrounds are coded Multiracial, and Latin American and Spanish heritage is coded Hispanic/Latino. Percentages may not sum to 100 because of rounding.

Table S3. Face detection and embedding model selection

A. Detection of clear, identifiable faces in human-coded news images.

Detector	Clear faces (human)	Found	Missed	Recall [95% CI]	Flagged in no-clear-face images	Precision	F1
YOLOv8 (deployed)	9,995	9,446	549	0.945 [0.940, 0.949]	230 of 1,105	0.976	0.960
YuNet	9,995	5,143	4,852	0.515 [0.505, 0.524]	70 of 1,105	0.987	0.676

B. Same- versus different-person discrimination (36 configurations).

Embedding	Detector	Alignment	Distance	Threshold	Precision	Recall	F1
FaceNet (best)	YOLOv8	Yes	Cosine similarity	0.479	0.998	0.996	0.997
FaceNet	YOLOv8	Yes	L2-normalized Euclidean	1.017	0.998	0.996	0.997
FaceNet	YOLOv8	Yes	Euclidean	11.828	0.998	0.992	0.995
FaceNet (deployed)	YOLOv8	No	Cosine similarity	0.439	0.992	0.996	0.994
FaceNet	YOLOv8	No	L2-normalized Euclidean	1.060	0.992	0.996	0.994
FaceNet512	YOLOv8	No	Cosine similarity	0.455	0.992	0.995	0.994
FaceNet512	YOLOv8	No	L2-normalized Euclidean	1.041	0.992	0.995	0.994
FaceNet512	YOLOv8	Yes	Cosine similarity	0.425	0.988	0.998	0.993
FaceNet512	YOLOv8	Yes	L2-normalized Euclidean	1.069	0.988	0.998	0.993
FaceNet	YOLOv8	No	Euclidean	12.282	0.989	0.994	0.992
FaceNet512	YOLOv8	No	Euclidean	23.278	0.987	0.990	0.989
FaceNet512	YOLOv8	Yes	Euclidean	23.089	0.988	0.987	0.987
VGG-Face	YOLOv8	Yes	Euclidean	1.205	0.982	0.989	0.986
VGG-Face	YOLOv8	Yes	L2-normalized Euclidean	1.205	0.982	0.989	0.986
VGG-Face	YOLOv8	Yes	Cosine similarity	0.259	0.978	0.992	0.985
VGG-Face	YOLOv8	No	Cosine similarity	0.278	0.981	0.986	0.984
VGG-Face	YOLOv8	No	Euclidean	1.203	0.981	0.986	0.984
VGG-Face	YOLOv8	No	L2-normalized Euclidean	1.203	0.981	0.986	0.984
FaceNet	YuNet	No	Euclidean	12.191	0.967	0.949	0.958
FaceNet512	YuNet	No	Cosine similarity	0.435	0.987	0.920	0.952
FaceNet512	YuNet	No	L2-normalized Euclidean	1.058	0.988	0.919	0.952
FaceNet	YuNet	No	Cosine similarity	0.455	0.996	0.912	0.952
FaceNet	YuNet	No	L2-normalized Euclidean	1.046	0.995	0.912	0.951
FaceNet512	YuNet	No	Euclidean	24.438	0.937	0.965	0.951
VGG-Face	YuNet	No	Euclidean	1.203	0.980	0.910	0.943
VGG-Face	YuNet	No	L2-normalized Euclidean	1.203	0.980	0.910	0.943
VGG-Face	YuNet	No	Cosine similarity	0.265	0.977	0.911	0.943
FaceNet	YuNet	Yes	Euclidean	12.180	0.887	0.879	0.883
FaceNet512	YuNet	Yes	Euclidean	24.615	0.856	0.907	0.881
FaceNet512	YuNet	Yes	Cosine similarity	0.356	0.951	0.794	0.865
FaceNet512	YuNet	Yes	L2-normalized Euclidean	1.111	0.961	0.787	0.865
FaceNet	YuNet	Yes	Cosine similarity	0.456	0.972	0.771	0.860
FaceNet	YuNet	Yes	L2-normalized Euclidean	1.033	0.972	0.771	0.860
VGG-Face	YuNet	Yes	Cosine similarity	0.263	0.951	0.771	0.851
VGG-Face	YuNet	Yes	Euclidean	1.207	0.955	0.768	0.851
VGG-Face	YuNet	Yes	L2-normalized Euclidean	1.207	0.955	0.768	0.851

Notes. Panel A: 11,250 images from the study's news corpus (2017–2021) showing 270 public figures, none of them target politicians, were each coded by one of 6 trained coders for whether the image contains a clear, identifiable human face; heavily covered, unrecognizably small or blurred, and drawn faces were coded as no face. 30 rows flagged as coding errors and 120 rows without a face code are excluded (n = 11,100; 9,995 with a clear face). Recall = share of clear faces the detector found (95% Wilson CI). Nearly all images show a person, and the images without a clear face are hard cases (a person whose face is not identifiable) rather than images without people, so precision depends on this composition and does not describe performance on the corpus; it is shown for completeness, as is F1, the harmonic mean of precision and recall, which inherits the same dependence. Each image was coded once, so inter-coder reliability could not be estimated; YOLOv8 recall computed separately for each coder ranged from 0.87 to 0.98. Candidate detectors and embeddings were preselected from published benchmarks (Serengil & Özpınar, 2024; Wu et al., 2023). Panel B: discriminant analysis run November 2024 on manually verified same-person and

different-person pairs of celebrity figures (1,000 same-person and 1,000 different-person pairs from 270 individuals). Three detectors (YOLOv8, YuNet, FastMTCNN), three embedding models, alignment on or off, and three distance metrics were crossed; FastMTCNN was dropped because it timed out or detected non-faces, leaving 36 configurations. For each configuration, thresholds were scanned in 100 equal steps over the region where the same-person and different-person distance distributions overlap, and the threshold with the highest F1 was kept; a 500-step scan over the full range gave very similar thresholds and F1. For cosine similarity, values at or above the threshold indicate the same person; for the Euclidean distances, values at or below it. Rows are sorted by F1. Thresholds were chosen on the same pairs used to compute F1, so these are development-set estimates. The deployed pipeline used FaceNet with YOLOv8 without alignment, to preserve head orientation and expression (F1 = .994 with cosine similarity). F1 refers to classifying pairs as same or different person; it is not the accuracy of identifying the target politicians in news images, which was checked manually (Table S4).

S1.5. Face detection and image download

This step turns image links into the faces used for identification. Only images showing a person are relevant, so detection and embedding were built into the download itself rather than run afterward on a full image archive: each image was downloaded, checked for faces, and kept only if at least one face was found, and every face was embedded at the same time.

Images were downloaded from the NewsWhip URLs with a 10-second timeout. When a URL failed, the pipeline queried the Internet Archive's Wayback Machine and, if an archived copy existed, downloaded it instead. YOLOv8 ran during download, and only images with at least one detected face were stored. Each detected face was cropped with 5% padding into a separate clip and embedded with FaceNet (`DeepFace.represent, model_name = "Facenet", detector_backend = "yolov8", align = False`). Alignment was disabled to preserve head orientation and expression.

In total, 4,224,177 images with at least one face and 11,663,296 face clips were stored; 4,212,430 of the images came from the 48 analytic outlets (Table S1). Download success, the share of articles for which an image with a face was obtained, was 46% across the 48 analytic outlets but varied widely between them, from under 10% at OANN, PJ Media, *The Hill*, Newsmax, and the *Washington Post* to more than 80% at Vox and Right Scoop (Table S1). This measure combines an intended exclusion with an unintended loss that the download log does not separate: images that were downloaded but contained no detectable face, and images that could not be downloaded at all. The log's remarks attribute most low-success outlet-years to HTTP 403, 404, or 410 errors on the original image URLs, that is, to failed downloads. About 4% of the stored images were recovered from the Wayback Machine. Section S1.8 examines whether these failures bias the estimates.

Detection accuracy. The human-coded benchmark (S1.4; Table S3A) indicates how much the face filter itself loses. On news images from the same corpus, YOLOv8 missed 5.5% of the clear, identifiable faces that coders found. Detections that are not a clear face of a target politician are harmless for the analysis, because they are discarded at identity matching (S1.6). Missed images are lost for good. They reduce the sample, but they would bias the estimates only if missed faces differed systematically in how the politician is depicted, for example if detection failed more often for faces that are turned away or partly covered. This cannot be ruled out; it is the same concern raised for unmatched faces in S1.6. The benchmark and the download used the same detector settings, except that faces in the downloaded images were not aligned (S1.4).

S1.6. Identity matching for the target politicians

Most stored faces belong to people outside the study, and the study concerns politicians whose identity, and therefore party, is known. This step links faces to the target politicians. It uses the reference gallery (Fig. S1, *Reference faces*), a set of reference photographs of known people, illustrated in the figure with the target politicians. The models selected in S1.4 (Fig. S1, panels A and B) supply the embeddings. The step proceeds in four parts, described in turn below: choosing the target politicians, building the reference gallery, matching faces against it, and checking the matches by hand.

1. Choosing the targets. We targeted the ten U.S. political figures mentioned most often in the corpus headlines: Barack Obama, Mike Pence, Donald Trump, Joe Biden, Elizabeth Warren, Nancy Pelosi, Bernie Sanders, Mitch McConnell, Jeff Sessions, and Lindsey Graham.

2. Building the reference gallery. The gallery covers 1,311 individuals. It is built from the list of public figures described in S1.4: the 1,307 people mentioned at least 50 times in the news headlines, with the target politicians added where the frequency filter missed them (for example, Bernie Sanders, listed in the source rosters under his legal name, Bernard Sanders). Besides the target politicians, the gallery therefore includes other politicians and officials, entertainers, musicians, athletes, journalists, and business figures, many of whom appear alongside the target politicians in the news. Its breadth matters for matching: a face of someone other than a target politician can be matched to that person instead of being forced onto the most similar politician (Table S2 describes the 270 gallery members used in the face benchmark, S1.4). For each individual, we automatically collected and screened reference photographs using the following procedure:

a. Retrieved the first 20 Google Images results for the individual's name.
b. Retained only photographs containing exactly one frontal face, with the eyes, nose, mouth, and chin visible.
c. Selected an anchor photograph by comparing the first two eligible photographs with up to ten other candidates. The photograph verified as depicting the same individual most frequently was selected as the anchor.
d. Retained the anchor and up to nine additional photographs verified as depicting the same individual.
e. Generated FaceNet and YOLOv8 embeddings for the retained photographs using the settings described in S1.5, and averaged them into one reference embedding per person.

Same-person verification in steps (c) and (d) was performed using FaceNet through DeepFace (cosine distance, default verification threshold, without alignment). The resulting gallery contained 11,170 reference photographs, with a median of ten per person.

By including a broader set of known individuals, the procedure allowed detected faces to match people other than the target politicians. For example, a photograph of Melania Trump could be matched to her own reference photographs rather than incorrectly assigned to Donald Trump. Because Melania Trump was not a target politician, the match would then be discarded. This was particularly useful for distinguishing target politicians from family members, people with similar appearances, and other public figures who frequently appeared in the same news coverage.

The broader gallery reduced the risk of false identification but did not eliminate it. Individuals not represented in the gallery could still be incorrectly assigned to a target politician if their facial similarity exceeded the matching threshold. We therefore applied additional similarity and detection-confidence criteria, described

below.

3. Matching faces to the reference set. For each face detected in a news image:

a. its FaceNet embedding (S1.5) was compared with the averaged reference embedding of every person in the gallery (cosine similarity);
b. the face was assigned to the single person it resembled most, using one averaged embedding per person that combined the embeddings of all of that person's reference photographs;
c. it was kept as a politician match only if all three of the following held: that person was a target politician; the similarity score exceeded 0.60; and the face-detection confidence exceeded 0.74. Lower detection confidence typically indicates a small, blurred, or partly occluded face.

The 0.60 cut-off is stricter than the F1-optimal threshold from the pair analysis (0.439), trading some recall for precision. The two are not directly comparable, because matching compares a news-image face with a reference set rather than with a single photograph.

4. Checking the matches manually. The detector and embedding model were selected (S1.4) before any corpus image was processed. Because the discriminant analysis does not measure how accurately the target politicians are identified, we also checked the matches by hand once matching was completed. For each of the ten politicians, we manually inspected 95–97 images sampled from the faces the pipeline had matched to that politician and counted how many actually showed them. The share correct is the precision of the match (Table S4). Precision was high for eight politicians: Barack Obama 1.000, Elizabeth Warren .989, Nancy Pelosi .989, Donald Trump .979, Mike Pence .958, Joe Biden .948, Mitch McConnell .938, and Bernie Sanders .823. It was far lower for Jeff Sessions (.385) and Lindsey Graham (.271), who were therefore excluded; every retained politician had a precision of .823 or higher. This is a spot check rather than a systematic evaluation. It also measures only false positives, that is, faces wrongly matched to a politician. We did not check for false negatives, that is, faces of a target politician that the pipeline failed to match, so recall is unknown. Missed faces reduce the sample but would bias the estimates only if the pipeline's ability to recognize a politician depended on how that politician is depicted, for example if unflattering or unusual photographs were matched less often.

The image stance coding (S2.3) added a second, automated screen on every match. The model was told which politician the image was supposed to show and coded 0 when it could not confidently identify that politician. Of the 227,929 matched faces, 60 (in 60 articles) were coded 0, most of them for Bernie Sanders (28) and Elizabeth Warren (17); in 42 of these articles the headline was also not about the matched politician. Fourteen of the articles with a relevant headline were removed for this reason (S1.7). The low number is consistent with the high precision of the manual check, although the model, like the manual check, could only flag false positives.

Table S4. Manual precision check of identity matching and retained observations by politician

Politician	Party	Images checked	Correct	Precision	Decision	Matched rows	Unique articles	Analytic articles
Donald Trump	R	96	94	.979	Retained	87,497	84,512	76,984
Joe Biden	D	97	92	.948	Retained	64,755	62,432	56,998
Nancy Pelosi	D	95	94	.989	Retained	16,718	16,037	12,145
Mike Pence	R	96	92	.958	Retained	15,475	13,913	7,275
Bernie Sanders	D	96	79	.823	Retained	12,509	9,359	8,080

Politician	Party	Images checked	Correct	Precision	Decision	Matched rows	Unique articles	Analytic articles
Mitch McConnell	R	96	90	.938	Retained	11,627	10,534	7,623
Barack Obama	D	95	95	1.000	Retained	10,090	9,371	7,883
Elizabeth Warren	D	95	94	.989	Retained	9,258	8,225	6,528
Jeff Sessions	R	96	37	.385	Excluded	—	—	—
Lindsey Graham	R	96	26	.271	Excluded	—	—	—

Notes. Precision = share of the images the pipeline matched to each politician that actually showed that politician, from a manual check of 95–97 matched images per politician. False negatives (faces of a politician that the pipeline did not match) were not checked, so recall is unknown. Matched rows = rows in the stance-coded file (227,929 total); unique articles after removing repeated rows (214,383); analytic articles = valid headline and image stance (183,516). Unique-article counts reflect the assignment of each multi-politician article to one politician (Section S1.7). A face was assigned to a politician when YOLOv8 detection confidence exceeded 0.74 and the top-1 identity cosine similarity exceeded 0.60 (Section S1.6).

S1.7. From matched faces to the analytic sample

This subsection follows the matched faces through to the analytic sample. One of the steps, removing articles whose headline is not about the pictured politician, relies on the stance coding, which is described in S2. We report its effect on the sample here so that all exclusions are documented in one place; the coding itself is explained in S2.

Matched rows. Identity matching produced 227,929 face clips matched to the eight retained politicians (Table S4).

One record per article. Matching operates on face clips, so an image showing two target politicians yields two rows with the same article identifier. Removing repeated rows leaves 214,383 unique articles. Of the articles with more than one row, 11,259 had two or more different politicians and 727 repeated the same politician. Deduplication kept the first row of each article, so for multi-politician articles the politician analyzed is determined by file order. S5.10 shows that excluding all multi-politician articles leaves the results unchanged.

Headline relevance and valid ratings. The stance coding (S2) rated each headline and image for favorability toward the pictured politician. It coded a headline 0 if it was missing or not about that politician, and an image 0 if the politician could not be identified in it. Of the 214,383 unique articles:

- 30,853 headlines were coded 0 and removed by design, because a headline about someone else cannot be compared with the image as a portrayal of the same person;
- 183,530 articles remained with a headline referring to the politician pictured;
- 14 of these had an image coded 0 and were removed by design, because the image could not be rated as a portrayal of that politician;
- **183,516 articles** from 48 outlets have valid ratings on both modalities and form the analytic sample.

The articles that failed the relevance filter include stories about another person that are illustrated with a photograph of one of the eight politicians. Across all matched faces, 60 images were coded 0 (S1.6). Analyses that use a subset of the analytic sample are described where they are introduced.

S1.8. Source-level exclusion and coverage bias

Two kinds of missing data could make the sample unrepresentative: whole outlets lost during retrieval, and images that could not be downloaded for the outlets that remain.

Source-level exclusions. The pipeline's download log, which records articles, downloaded images, and faces for every outlet-year, shows why 15 of the 63 outlets in the initial pool (S1.2) are not in the analytic sample:

1. **No NewsWhip coverage (12 outlets).** The BBC, CNS News, Daily Kos, The Economist, the *Los Angeles Times*, *The New Yorker*, NowThis, ThinkProgress, Univision, Vice, *The Weekly Standard*, and Zero Hedge are absent from the log: NewsWhip returned no articles for them (S1.3).
2. **Image retrieval failure (3 outlets).** NewsWhip returned articles for these outlets, but their images could not be downloaded for all or most of the study period (S1.5). For The Intercept (9,664 articles) and the *Washington Examiner* (32,033 articles, all from 2021), almost no images could be downloaded (9 and 0 images with a face), mostly because of HTTP 404 errors or invalid links. For *Time* (73,482 articles), the 2017–2019 image links returned HTTP 410 errors (resource no longer available), so images with a face were obtained for only 16% of articles (11,738), nearly all from 2020–2021.

Composition of the retained outlets. Exclusions fell somewhat more on left-leaning outlets. Of the 31 outlets coded left of center (Extreme left, Left, Center-left), 22 (71%) were retained, compared with 24 of 28 (86%) right-of-center outlets (Center-right, Right, Extreme right); this difference is not statistically significant (Fisher's exact test, P = .22). The loss is concentrated at the far left: the only outlet coded Extreme left (NowThis) and three of the 11 coded Left (Daily Kos, ThinkProgress, *The New Yorker*) are excluded, so no Extreme-left outlet is in the analytic sample, whereas four of the six Extreme-right outlets are. Every other leaning category is represented. By media type, magazines were retained least often (5 of 8) and cable networks most often (5 of 6).

Most excluded outlets (12 of 15) were lost because NewsWhip did not cover them, which reflects the vendor's coverage rather than anything about the outlets' content. The main consequence is that findings generalize to outlets from the center-left to the far right more securely than to the far left, where the sample rests on Left and Center-left outlets only.

Image coverage within retained outlets. Download success, the share of articles for which an image with a face was obtained, differs systematically by outlet leaning. The median outlet share is 69% for Left, 56% for Center-left, 32% for Center, 36% for Center-right, 37% for Right, and 34% for Extreme-right outlets; across the 48 outlets, download success declines with leaning from left to right (Spearman $\rho = -.35$, P = .015). Coverage also varies over time within outlets, because image links break at different rates across outlets and years.

Two features of this measure limit how much of the variation is bias. First, download success counts images without a detectable face as losses too (S1.5), so part of the variation may reflect how often outlets use photographs of people rather than retrieval failure. Second, the failures themselves are mostly broken or expired links (HTTP 403, 404, 410), which depend on each outlet's content-management and archiving practices rather than on the content of individual articles. Bias would arise only if, within an outlet, articles whose images survived differ in stance or engagement from articles whose images were lost. S5.10 tests this by re-weighting articles by their outlet-year download success and by restricting the sample to well-covered

outlet-years and outlets.

S2. Headline and Image Stance: Measurement and Validation

S2.1–S2.3 describe how headline and image stance were defined and rated; S2.4–S2.8 report the human validation of these ratings.

S2.1. Construct, coding scale, and valence–extremity variables

Construct. Stance is the evaluative favorability that a headline or image expresses toward a specific political figure. Sentiment, by contrast, is the general emotional tone of the content regardless of target (S3.3). Headlines and images were coded on the same scale:

- 1 = very unfavorable;
- 2 = somewhat unfavorable;
- 3 = neutral;
- 4 = somewhat favorable;
- 5 = very favorable;
- 0 = could not be rated for the target politician: for headlines, missing or not about the politician; for images, the politician not identifiable.

These are the response options of the human-validation survey (S2.4), which labeled the midpoint "neither unfavorable nor favorable"; the zero-shot headline prompt below uses the same midpoint label, and the few-shot headline and image prompts label it "neutral". The prompts used to rate the corpus labeled the endpoints "very unfavorable" (1) and "very favorable" (5) for both headlines and images.

Valence and extremity. Ratings were rescaled to $H = (h - 3)/2$ and $I = (i - 3)/2$, where h and i are the five-point headline and image ratings. Both run from –1 (very unfavorable) through 0 (neutral) to +1 (very favorable). The sign gives *valence*; $|H|$ and $|I|$ (0, 0.5, or 1) give *extremity*. A rating is *extreme* when h or i equals 1 or 5, that is, when $|H| = 1$ or $|I| = 1$. Distributions are reported in S4.

S2.2. Headline stance classification

Model and input. Headline stance was rated by Gemma 4 31B instruction-tuned (`gemma-4-31b-it`, Google Generative AI API; Gemma Team, 2026) from the headline alone. The same model rated image stance (S2.3). Why we chose this model over larger or commercial alternatives is explained in S2.6.

Prompt development. We compared a plain zero-shot prompt with a few-shot prompt on the human-validation stimuli (S2.5). The few-shot prompt adds two elements to the zero-shot prompt:

- **Framing rules.** Short instructions for headlines whose stance toward the politician differs from their surface wording: sarcasm, scare quotes, and mockery count as unfavorable, and a headline that attacks someone else while mentioning the politician is not favorable on that basis alone. The rules also tell the model not to use prior opinions about the politician.
- **Five worked examples.** One headline from the human-validation stimuli for each scale point, with its score and a one-line reason:

Politician	Score	Reason given
McConnell	1	Hostile or sarcastic framing
Sanders	2	Critical claim
Sanders	3	Neutral primary coverage
Warren	4	Advancing an agenda
Warren	5	Amplifying a positive message

The few-shot prompt improved agreement only marginally (development-set ρ = .733 vs. .707 for the zero-shot prompt; S2.5). The corpus headlines were rated with the few-shot prompt. Both prompts are reproduced verbatim below; `{person}` and `{headline}` are filled in for each article.

Zero-shot prompt (development only):

```
You are a media-bias analyst. Below is the HEADLINE of a news article that concerns the U.S.
↪ political figure {person}.

Task: Rate how favorably or unfavorably this HEADLINE portrays {person}.

Headline: {headline}

Favorability scale:
0 = the headline is missing/empty, OR does not mention or concern {person} at all
1 = very unfavorable
2 = somewhat unfavorable
3 = neither unfavorable nor favorable
4 = somewhat favorable
5 = very favorable

IMPORTANT: If the headline is empty or does not concern {person}, immediately answer 0.
↪ Otherwise give at most a few short sentences of reasoning, then commit. Never loop or keep
↪ re-reading the headline.
```

```
END your reply with a single line of raw JSON (no markdown, no code fences) in EXACTLY this
↪ form:
{"score": <integer 0-5>, "explanation": "<one concise sentence, no line breaks>"}
```

Few-shot prompt (used for the corpus):

```
You are a media-bias analyst. Below is the HEADLINE of a news article that concerns the U.S.
↪ political figure {person}.

Task: Rate how favorably or unfavorably this HEADLINE portrays {person}.

Framing rules (apply before scoring):
- Judge how the headline frames {person}, not surface sentiment words alone.
- Sarcasm, scare quotes, mockery, and attack headlines about {person} are unfavorable to
↪ {person}.
- Headlines that "throw" {person} into a fight, "smear" {person}/allies, or celebrate damage to
↪ {person} are unfavorable.
- Do NOT score favorable just because the headline attacks someone else while mentioning
↪ {person}.
- Do NOT use your prior opinions about {person}.

Examples (headline-only; scores already decided):

Person: Mitch McConnell
Headline: Mitch McConnell throws Joe Biden right into Democrat’s faces!!
{"score": 1, "explanation": "sarcastic/hostile framing of McConnell, not praise"}

Person: Bernard Sanders
Headline: Bernie Sanders will scare away voters with truth about 'Medicare-for-all' tax burden,
↪ Andy Puzder says
{"score": 2, "explanation": "critical claim that Sanders will scare voters"}

Person: Bernard Sanders
Headline: 5 takeaways from the New Hampshire Democratic primary
{"score": 3, "explanation": "factual/neutral primary coverage"}

Person: Elizabeth Warren
Headline: Warren rolls out plan that would rein in private-equity firms
{"score": 4, "explanation": "presents Warren advancing a substantive agenda"}

Person: Elizabeth Warren
Headline: Sen. Warren: Every family deserves high-quality child care - Source
{"score": 5, "explanation": "clearly sympathetic / amplifying Warren's positive message"}

Now rate this new headline.

Headline: {headline}
```

```
Favorability scale:
0 = the headline is missing/empty, OR does not mention or concern {person} at all
1 = very unfavorable
2 = somewhat unfavorable
3 = neutral
4 = somewhat favorable
5 = very favorable

IMPORTANT: If the headline is empty or does not concern {person}, immediately answer 0.
 ↪  Otherwise give at most a few short sentences of reasoning, then commit. Never loop or keep
 ↪  re-reading the headline.

END your reply with a single line of raw JSON (no markdown, no code fences) in EXACTLY this
 ↪  form:
{"score": <integer 0-5>, "explanation": "<one concise sentence, no line breaks>"}
```

Inference settings. Ratings used temperature 0 and a 4,096-token output limit, and the score was taken from the last JSON object in the response. If no score could be parsed, a shorter prompt requesting an immediate JSON answer was sent at temperature 0.4 and then 0.8, each with a 512-token limit. API errors were retried up to five times with exponential back-off. Ratings that remained unparseable were treated as missing. Because API-served models can change, the rating file and model explanations are archived for replication.

S2.3. Image stance classification

Model and input. Image stance was rated by the same model as headline stance, Gemma 4 31B instruction-tuned (`gemma-4-31b-it`, Google Generative AI API). Using one model for both modalities means that differences between headline and image ratings do not reflect different raters. For each article, the full news image was retrieved from the stored copy and sent as a JPEG together with the text prompt. The image was not marked; instead, the prompt named the politician the image was supposed to show and asked the model to judge only that person when several people appear. Because the model had to locate the named politician itself, the prompt also let it answer 0 when it could not confidently identify that politician, which adds an automated check on identity matching (S1.6).

Prompt. The prompt, reproduced verbatim below, restricts the judgment to visual cues; `{person}` is filled in for each article.

```
You are a visual media analyst. The image is a news photograph that is supposed to show the
 ↪ U.S. political figure {person}.

Task: Rate how favorably this photograph VISUALLY portrays {person}. Base your judgment ONLY on
 ↪ visible cues: facial expression, gaze, body language, gesture, camera angle, framing,
 ↪ lighting, image quality, and the overall impression the shot creates. Do NOT consider any
 ↪ text or caption, and do NOT use your prior opinions about this person. If more than one
 ↪ person appears, judge only {person}.

Favorability scale:
0 = {person} is NOT clearly present, or you cannot confidently identify {person} in the image
 ↪ (e.g. a collage/crowd where you are unsure which face is them, or they simply are not
 ↪ there)
1 = very unfavorable  - clearly unflattering, negative, weak, angry, or unsympathetic depiction
2 = somewhat unfavorable
3 = neutral           - ordinary, plain, or mixed; neither flattering nor unflattering
4 = somewhat favorable
5 = very favorable    - clearly flattering, positive, strong, confident, or sympathetic
 ↪ depiction

IMPORTANT: Do NOT agonize over identification. If you cannot confidently locate {person} within
 ↪ a sentence or two, immediately answer 0 and move on. Never loop or keep re-examining the
 ↪ image. Give at most a few short sentences of reasoning, then commit to an answer.

END your reply with a single line of raw JSON (no markdown, no code fences) in EXACTLY this
 ↪ form:
{"score": <integer 0-5>, "explanation": "<one concise sentence, no line breaks>"}
```

Inference settings. The settings matched those for headlines (S2.2): temperature 0, a 4,096-token output limit, and the score taken from the last JSON object in the response. If no score could be parsed, usually because the model kept re-examining an image in which it could not find the politician, a shorter prompt requesting an immediate JSON answer on the same 0–5 scale was sent at temperature 0.4 and then 0.8, each with a 512-token limit. API errors were retried up to five times with exponential back-off. Images that

could not be retrieved or returned no valid response were logged and left unrated.

Each rating is accompanied by a one-sentence explanation. These refer to facial expression, body language, camera angle, lighting, and setting, which indicates that the model evaluated the whole image rather than the face alone. Code 0 comes with explanations such as “[politician] is not present in the image”; 60 of the 227,929 matched images received it, 14 of them in articles that would otherwise have entered the analytic sample (S1.6–S1.7).

S2.4. Human validation: procedure, stimuli, and participants

Procedure. The validation ratings were collected within a larger survey experiment fielded on Qualtrics in March 2026. Only the favorability ratings of headlines and images are reported here; the survey's other conditions, stimuli, and measures pertain to other ongoing projects and are omitted. Headlines and images were rated separately. A headline was shown on its own, as a screenshot of the headline text, and an image on its own, with the focal politician's face marked by a red box so that raters knew whom to judge (Fig. S2); the model was instead given the politician's name (S2.3). Each rating therefore reflects one modality only. Participants were asked to *focus on evaluating how the content portrays the political figure, NOT your own personal opinions or feelings toward the politician.* Each stimulus was followed by the question *How favorable or unfavorable does this news content portray [politician]?* with the note *(If [politician] is not mentioned by name, please rate based on how the content implies or reflects on them.)* The five response options were *very unfavorable*, *somewhat unfavorable*, *neither unfavorable nor favorable*, *somewhat favorable*, and *very favorable* (coded 1–5), the same scale as the automated stance ratings (S2.1). Headline ratings are compared with the automated headline stance and image ratings with the automated image stance.

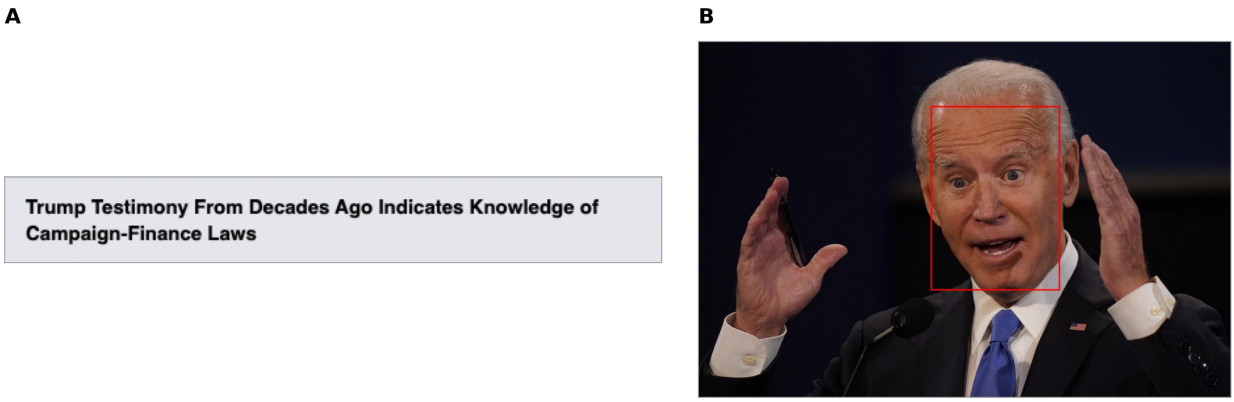


Fig. S2. Example human-validation stimuli. (A) A headline-only stimulus, shown as a screenshot of the headline text. (B) An image-only stimulus, with the focal politician's face (here Joe Biden) marked by a red box. Each stimulus was followed by the favorability question.

Stimuli. The validation set contained 288 articles from the corpus, 36 for each of the eight politicians. Within each politician, the articles were balanced on preliminary headline and image valence: four articles in each of the nine combinations of unfavorable, neutral, and favorable headline and image, or 32 articles per combination across politicians. The preliminary codes came from an earlier three-point automated coding (unfavorable, neutral, favorable) with Gemini 2.0 Flash, which rated the headline together with the article's first paragraph, and the image; they were used only to balance the sample and are not the stance measure used in the analyses (S2.1–S2.3). Each article supplied one headline stimulus and one image stimulus, and each image was used once. The stimuli come from 40 outlets and all five years of the study period (2017: 38 articles; 2018: 49; 2019: 60; 2020: 85; 2021: 56), and 261 of the 288 headlines name the politician. Because five of the eight politicians are Democrats, 180 stimuli feature Democrats and 108 feature Republicans.

Participants. Participants were U.S. adults recruited through CloudResearch; each was paid $5. The study was approved by the authors' institutional review board (protocol details redacted for peer review). Respondents were excluded if they failed an attention check, did not finish, or completed the survey in under 240

seconds (about the 3rd percentile of completion time); a survey preview was also removed. The validation ratings come from 393 participants who passed these checks. Recruitment for the larger survey was benchmarked to U.S. adult demographics, using targets drawn from the framework of Mercer, Lau, and Kennedy (2018), which combines nine population datasets, and adjusted to the 2020 electorate as described by Pew Research Center (Gramlich, 2020). Table S5 compares the validation raters with the full survey sample and with these benchmarks. The validation raters closely resemble the full survey sample. Compared with the benchmarks, both are close in age but include more college graduates and fewer people without a high school diploma, and somewhat more non-Hispanic White and fewer Hispanic respondents. Because the validation compares the model with the average human judgment of each stimulus, these differences matter mainly if favorability judgments differ by education or ethnicity; partisanship, the characteristic most likely to color such judgments, is examined in S2.7.

Table S5. Validation raters, full survey sample, and population benchmarks

Characteristic	Validation raters (n = 393)	Full survey sample (n = 1,554)	Benchmark
Gender			
Male	—	—	49%
Female	—	—	50%
Non-binary, transgender, or prefer not to answer	—	—	1%
Age			
Mean (SD), years	46.3 (15.8)	46.5 (15.5)	—
18–24	8.7%	7.9%	13%
25–34	18.3%	18.0%	18%
35–44	23.4%	22.4%	16%
45–54	15.0%	17.4%	18%
55–64	18.8%	19.4%	16%
65 and older	15.8%	14.9%	19%
Race and ethnicity			
White, non-Hispanic	68.2%	68.7%	61%
Black, non-Hispanic	14.5%	13.6%	12%
Hispanic or Latino/a (any race)	12.2%	11.8%	17%
Asian, non-Hispanic	3.3%	3.4%	5%
Other, non-Hispanic	1.8%	2.5%	4%
Education			
Less than high school	0.3%	0.6%	13%
High school graduate	22.6%	20.0%	28%
Some college or associate degree	30.5%	32.6%	31%
Bachelor's degree	32.8%	32.6%	18%
Postgraduate degree	13.7%	14.0%	10%
Party identification			
Democrat (including leaners)	50.1%	50.6%	—
Independent / no preference	12.5%	12.4%	—
Republican (including leaners)	37.4%	36.9%	—
Ideology			
Liberal	48.9%	48.5%	—
Moderate	13.2%	12.9%	—
Conservative	36.9%	37.8%	—

Notes. Validation raters are the 393 participants who rated headlines and images for favorability (S2.4); the full survey sample is all 1,554 participants who passed the quality checks in the larger survey, whose recruitment was benchmarked to U.S. adult demographics. Benchmark = population targets for the larger survey, drawn from the framework of Mercer, Lau, and Kennedy (2018), which combines nine population datasets, and adjusted to the 2020 electorate as described by Pew Research Center (Gramlich, 2020). Gender was not recorded in the survey data, so only the benchmark is shown. Age = 2026 minus birth year. Respondents who reported Hispanic, Latino/a, or Spanish origin are counted as Hispanic or Latino/a regardless of race; the other race and ethnicity categories are non-Hispanic, and Other includes American Indian or Alaska Native, Native Hawaiian or Other Pacific Islander, and some other race. Education shares

include a few participants who preferred not to answer, and ideology shares a few who were not sure, so these groups may not sum to 100%. Party identification and ideology have no benchmark.

Stimulus assignment and ratings per stimulus. Stimuli were assigned at random within strata so that every participant saw a balanced set. The 288 headlines were divided into six pools by the politician group (Biden, Obama, Sanders, and Warren, or McConnell, Pelosi, Pence, and Trump) and the preliminary headline valence (unfavorable, neutral, or favorable), with 40 to 56 headlines per pool; the images were divided the same way by preliminary image valence (44 to 52 images per pool). Each participant received one randomly drawn headline from each headline pool and one randomly drawn image from each image pool, with no article used twice for the same participant, and the twelve stimuli were shown in random order. The 393 raters thus provided 2,358 headline ratings and 2,358 image ratings (six of each per rater), and no participant judged both the headline and the image of the same article. Because the draws were random rather than quota-controlled per stimulus, the number of raters varied across stimuli: each headline was rated by 8.2 people on average (*SD* 3.2; median 8; range 2–19) and each image by 8.2 (*SD* 2.7; median 8; range 1–15). Stimulus-level analyses require at least three ratings per stimulus, so that the stimulus mean and mode are not driven by one or two people. This rule set aside 4 headlines and 3 images (8 headline and 4 image ratings), leaving 2,350 headline and 2,354 image ratings for the human–human benchmark (S2.8). The human–AI comparison (S2.5) further omits the 15 headlines that received AI code 0 (judged not to be about the politician; S2.1), leaving 269 headlines and 285 images, each rated by 8.3 people on average.

S2.5. Human–AI agreement

Human benchmark. The AI ratings are compared with the average human judgment of each stimulus. For each headline and each image, we averaged the 1–5 ratings of all raters who judged it, which gives a stimulus mean based on about eight ratings (8.3 on average among compared stimuli; S2.4). This number of raters follows crowdsourcing practice. Averaging a few non-expert ratings per item matches the agreement of a single expert annotator: four on average across annotation tasks, and five for emotional valence (Snow et al., 2008). In news-headline ratings, the average of politically balanced crowds of 8 to 12 laypeople correlated with professional fact-checkers' ratings as well as the fact-checkers correlated with one another (Allen et al., 2021), and crowd ratings of Facebook news headlines recruited through CloudResearch, as ours were, used about nine raters per headline (Allen, Watts, & Rand, 2024). Averaging reduces the idiosyncratic disagreement among individual raters, so the stimulus mean is a more reliable target than any single rating (its reliability is .81 for headlines and .78 for images; S2.8). Agreement is measured as the Spearman correlation between the AI rating and the stimulus mean.

Development-set test. While developing the headline prompt, we ran the zero-shot and the few-shot prompts (S2.2) on the 288 validation headlines only and correlated each run's ratings with the human stimulus means. Two kinds of stimuli were left out of each correlation: the 4 headlines with fewer than three human ratings (S2.4), leaving 284, and the headlines that the run coded 0 (not about the politician). For the few-shot prompt, the five headlines used as worked examples in the prompt were also left out, because the model had seen their scores. The zero-shot prompt reached $\rho = .707$ ($n = 266$; 18 headlines coded 0) and the few-shot prompt $\rho = .733$ ($n = 264$; 15 coded 0, 5 examples). Because the same stimuli were used to choose the prompt and to score it, these are development-set estimates, not independent test performance. The image prompt reached $\rho = .728$ in development.

Post-rating test. After the full corpus had been rated with the final prompts, we took the corpus ratings of the validation stimuli and correlated them with the human stimulus means. These ratings come from a separate run of the model over the whole corpus, not from the development runs. The same exclusions apply, except that the five example headlines are kept, since in the corpus they are ordinary articles:

- headlines: $\rho = .690$ ($n = 269$: 284 with at least three ratings, minus 15 coded 0);
- images: $\rho = .722$ ($n = 285$: 288 minus 3 with fewer than three ratings).

The post-rating headline correlation is slightly below the development estimate because it includes the five example headlines and comes from a separate run. The post-rating estimates describe the ratings actually analyzed.

Comparison with prior benchmarks. Agreement of .69 to .72 with the human stimulus means is in the range that comparable studies treat as strong. For automated text analysis, large language models' ratings of the sentiment and discrete emotions of news headlines correlated with human ratings at $r = .56$ to .77, which Rathje et al. (2024) describe as high and well above dictionary methods ($r = .20$ to .30 across their datasets). For crowd judgments of news, the average layperson rating correlated with professional fact-checkers at $r = .57$ to .66, while the fact-checkers themselves agreed at $r = .62$ (Allen et al., 2021), and crowd accuracy ratings of vaccine-related headlines agreed with expert judgments at $r = .72$ (Allen, Watts, & Rand, 2024). Our correlations are Spearman rather than Pearson coefficients and refer to stance rather than sentiment or accuracy, so the comparison is approximate; but they are of the same size, and they are measured against a human target whose own reliability limits the attainable correlation to about .88 to .90

(S2.8).

S2.6. Choice of rating model

Question. Would a larger or commercial model have measured stance more accurately than `gemma-4-31b-it`? We compared Gemma's corpus ratings with ratings from four alternative models on the same validation stimuli: GPT-5.4 and Gemini 3 Flash (preview), each run 10 times, and GPT-4.1 mini and Gemini 2.5 Flash-Lite, each run once.

Procedure. For each model and modality, we computed the Spearman correlation between the model's rating and the human stimulus mean, using the stimuli that every model rated and that had at least three human ratings (269 headlines, 281 images). For the models run 10 times, the main comparison uses the first run, matching Gemma's single corpus run; the range across runs is also reported. Each model's difference from Gemma was tested with a paired bootstrap that resamples stimuli (5,000 draws), with Holm correction across the eight comparisons.

Model	Headlines ρ [95% CI]	Δ vs. Gemma [95% CI]	Images ρ [95% CI]	Δ vs. Gemma [95% CI]
Gemma 4 31B (used)	**.69 [.61, .76]**	—	**.72 [.66, .78]**	—
GPT-5.4	.73 [.66, .78]	+.04 [−.03, .11]	.73 [.66, .78]	+.00 [−.06, .07]
Gemini 3 Flash (preview)	.73 [.66, .79]	+.04 [−.02, .11]	.76 [.71, .81]	+.04 [−.02, .11]
GPT-4.1 mini	.76 [.70, .82]	+.07 [.00, .15][a]	.71 [.65, .77]	−.01 [−.08, .06]
Gemini 2.5 Flash-Lite	.74 [.67, .79]	+.05 [−.03, .13]	.53 [.45, .60]	−.20 [−.28, −.11][b]

Note. ρ = Spearman correlation with the human stimulus mean. Δ = model ρ minus Gemma ρ on the same stimuli. Across the 10 runs, ρ ranged from .70 to .73 (headlines) and .72 to .74 (images) for GPT-5.4, and from .73 to .74 and .76 to .77 for Gemini 3 Flash. [a] Unadjusted p = .038; Holm-adjusted p = .26. [b] Holm-adjusted p = .002.

Result. No alternative model agreed with human raters significantly better than Gemma after correction for multiple comparisons. The largest gain, GPT-4.1 mini on headlines (+.07), was not significant after Holm correction, and the same model did slightly worse on images. The most recent models, GPT-5.4 and Gemini 3 Flash, gained at most .04 in either modality, within sampling error, and averaging their 10 runs raised headline agreement only to .74 for both. Gemini 2.5 Flash-Lite was significantly worse on images. Gemma's headline correlation here (.69) also comes from a separate corpus run; its development estimate with the same prompt was .73 (S2.5), level with the strongest alternatives.

Why Gemma. Because the alternatives offered no reliable gain in validity, we chose Gemma for three practical reasons:

1. **Cost.** The corpus required about 430,000 ratings (one headline and one image rating for each of about 214,000 articles). Gemma was available free of charge through the free tier of the Gemini API at the time of rating, whereas the commercial models are billed per token.
2. **Reproducibility.** Gemma's weights are openly released, so the ratings can be reproduced with a fixed model version long after commercial API versions are retired; one of the alternatives (Gemini 3 Flash) was itself a preview release.
3. **One model for both modalities.** Gemma 4 accepts both text and images, so headline and image stance were rated by the same model with parallel prompts (S2.2–S2.3).

The Gemini 2.0 Flash ratings in the stimulus file are not included in this comparison because they were the preliminary codes used to stratify the validation sample (S2.4), so they are not independent of stimulus

selection. For completeness, their correlations with the human means are .69 for headlines and .73 for images.

S2.7. Small partisan bias in stance judgments

Concern. If judgments of favorability depend on the judge's own partisanship, stance could reflect who is looking rather than what the content shows. We tested this in the validation survey with the 344 raters who identified as Democrats (197) or Republicans (147), including leaners; the 49 independents were excluded. Each rater judged politicians of both parties.

Model. For each modality, we regressed the 1–5 favorability rating on an *in-party* indicator (the rater's party matches the pictured politician's party), the rater's party, age, education, and race/ethnicity, with stimulus fixed effects and standard errors clustered by rater. The politician's party is the party of the named politician (Biden, Obama, Pelosi, Sanders, and Warren are Democrats; McConnell, Pence, and Trump are Republicans). The stimulus fixed effects absorb everything about the content itself, including the politician's party, so the in-party coefficient measures how much more favorably partisans rate the same article when it features a politician from their own party.

	Headlines	Images
In-party effect, *b* [95% CI]	0.24 [0.14, 0.35]	0.25 [0.14, 0.36]
p	< .001	< .001
Effect relative to rating *SD* (*d*)	0.20	0.23
Variance explained by in-party (ΔR^2)	.008	.010
Variance explained by the stimulus (R^2)	.45	.39
Ratings / raters	2,064 / 344	2,064 / 344

Result. Partisan bias is present and statistically significant: raters judged the same headline or image about a quarter of a point more favorably on the five-point scale when it featured a politician of their own party. The effect is small, however. It amounts to 0.20 to 0.23 standard deviations of the ratings and explains about 1% of their variance, whereas which article was being judged explains 39% to 45%. Shared reading of the content therefore outweighs partisan perception by a factor of roughly 40 to 60 in variance explained.

Predicted ratings by rater and politician party. To show the pattern behind the in-party effect, we also fit a multilevel model for each modality, regressing the 1–5 rating on the rater's party, the politician's party, their interaction, and the same rater demographics, with crossed random intercepts for stimulus and rater (statsmodels `MixedLM`, REML). Fig. S3 shows the predicted ratings for the four combinations, averaged over the raters' demographics. Democratic raters rated Democratic politicians' headlines slightly higher than Republican politicians' headlines (2.92 vs. 2.64), and Republican raters did the reverse (2.77 vs. 2.98); for images the predicted ratings were 3.05 vs. 2.92 for Democratic raters and 2.84 vs. 3.18 for Republican raters. The interaction is significant in both modalities (headlines: $b = 0.49$, $SE = 0.09$; images: $b = 0.47$, $SE = 0.09$; both $p < .001$). It is about twice the in-party coefficient above, because it compares the gaps of both rater groups. On the full rating scale, however, all four predictions lie within about a third of a scale point of one another. The variance components tell the same story: the stimulus intercepts vary far more (variance 0.53 for headlines and 0.37 for images) than the rater intercepts (0.08 and 0.05).

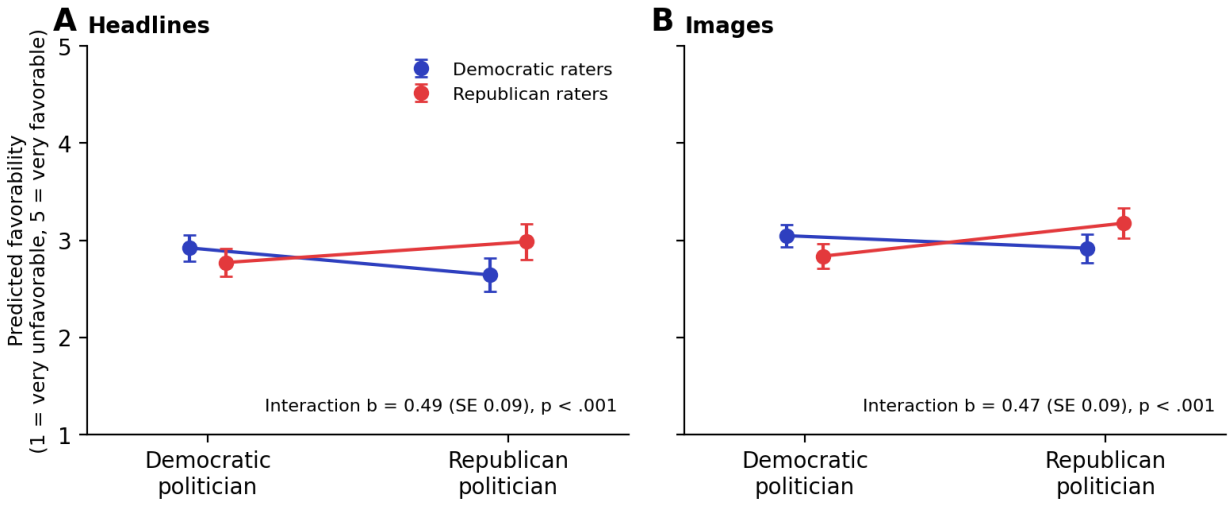


Fig. S3. Predicted favorability ratings by rater party and politician party. Predictions from multilevel models of the 1–5 favorability rating for (A) headlines and (B) images, with rater party × politician party, rater age, education, and race/ethnicity as fixed effects and crossed random intercepts for stimulus and rater. Predictions are averaged over the raters' demographics; bars are 95% confidence intervals for the fixed-effect prediction. 2,064 ratings from 344 Democratic and Republican raters (leaners included) per modality. The *y*-axis spans the full response scale.

Cross-party agreement. Democratic and Republican raters ranked the stimuli similarly. The correlation between the two parties' stimulus means is ρ = .60 for headlines (215 stimuli rated by at least two raters from each party) and ρ = .63 for images (218 stimuli), both $p < .001$, even though each party mean rests on only about four raters.

Does the model side with one party? Gemma's ratings correlate with both parties' stimulus means (Fig. S4). When each party mean is based on the same number of raters per stimulus, so that neither is less noisy, the model agrees slightly more with Democratic raters in both modalities (headlines: .61 vs. .57; images: .64 vs. .57), but neither difference is statistically distinguishable from zero (headlines: .04 [–.06, .15]; images: .07 [–.04, .18]). Without equalizing rater counts, the gap is larger (.65 vs. .58 for headlines and .67 vs. .60 for images), because Democratic raters outnumber Republican raters and their means are therefore more reliable.

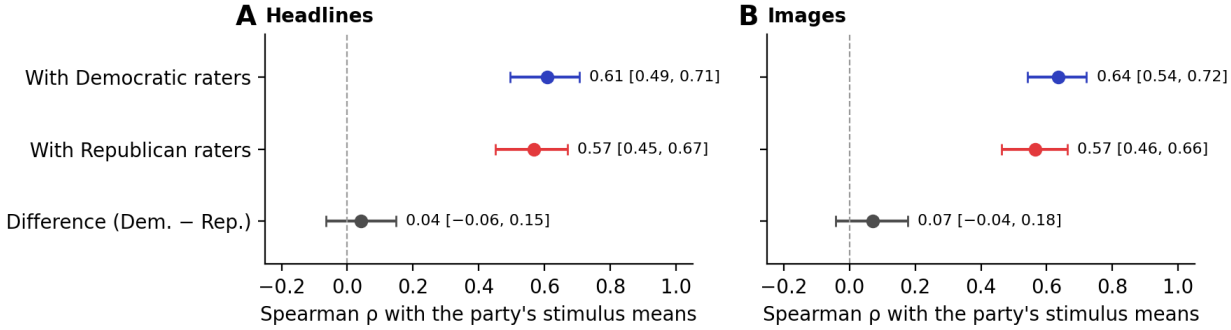


Fig. S4. Agreement of the model's stance ratings with Democratic and Republican raters. Spearman correlations between Gemma's ratings and the stimulus means of Democratic and Republican raters, and

their difference, for (A) headlines (206 stimuli) and (B) images (218 stimuli). Stimuli were rated by at least two raters from each party; headlines that the model coded 0 are excluded. Each party mean uses the same number of raters per stimulus. Points are means over 2,000 draws that subsample equal numbers of raters from each party per stimulus and resample stimuli; bars are 95% intervals across draws, reflecting both sources of variation.

S2.8. Human–human agreement as the ceiling for human–AI agreement

Logic. Because individual raters disagree with one another, no automated measure can agree perfectly with a human average. Two benchmarks show how close Gemma comes to the limit set by human judgment itself.

1. **One human versus the crowd.** For each stimulus, we drew one rater at random and correlated that rater's score with the mean of the remaining raters. We then correlated Gemma's rating with the same leave-one-out mean, so both are scored against an identical target (2,000 draws).
2. **Reliability ceiling.** We split each stimulus's raters at random into two halves, correlated the half means, and applied the Spearman–Brown correction to obtain the reliability of the full stimulus mean. Its square root is the highest correlation any measure can be expected to reach with the observed human mean, given the noise in that mean.

	Headlines	Images
One human vs. remaining raters, ρ [95% range]	.54 [.46, .62]	.48 [.40, .56]
Gemma vs. the same remaining raters, ρ [95% range]	.68 [.66, .70]	.71 [.69, .73]
Draws in which Gemma exceeds the single human	100%	100%
Reliability of the stimulus mean (Spearman–Brown)	.81	.78
Ceiling for correlation with the human mean (√reliability)	.90	.88
Gemma vs. the human mean, ρ	.69	.72
Gemma as a share of the ceiling	77%	82%

Note. Stimuli with at least three human ratings (269 headlines, 285 images; about 8.3 raters per stimulus). Ranges are the 2.5th and 97.5th percentiles across draws.

Result. Gemma agrees with the human consensus more closely than a typical individual human does, in every draw and in both modalities. Its agreement reaches about four-fifths of the theoretical ceiling set by the reliability of the human averages. Most of the remaining gap reflects disagreement among humans rather than model error, so further gains from a different rating model are necessarily limited (see also S2.6).

S3. Covariates

This section describes the outlet-level covariates (S3.1–S3.2) and the article-level covariates (S3.3–S3.5). Definitions, reference dates, coverage, and outlet-level values for all outlet measures are in Table S6. The article-level measures enter the main RQ1 model, model (v), to test whether stance extremity can be explained by general emotional tone, facial expression, or partisan alignment (S5); S3.3 and S3.4 compare stance with headline sentiment and facial expression, showing that they measure different things.

S3.1. Partisan leaning, media type, and extreme-stance prevalence

Partisan leaning. Outlet leaning follows Media Bias/Fact Check (MBFC; Media Bias/Fact Check, n.d.), with "least biased" coded as Center. The 48 analytic outlets fall into six categories: Left, Center-left, Center, Center-right, Right, and Extreme right (Table S1). For partisan congruency (S3.5), leaning is collapsed into Left (Left, Center-left) and Right (Center-right, Right, Extreme right); the two Center outlets have no partisan side. How leaning enters each analysis is described in the model setup of each results section (S5.1, S6.1).

Media type. Media type has five categories: digital native, newspaper, cable news, network TV, and magazine.

Extreme-stance prevalence. An outlet's *extreme-headline share* is the percentage of its analytic articles whose headline stance is 1 or 5. The *extreme-image share* is defined the same way for images. Both are pooled over the eight politicians. Across the 48 analytic outlets, the unweighted mean share is 20.2% for headlines (range 0%–59.6%) and 6.7% for images (range 2.7%–13.8%) (Table S6B). The extremes of both ranges include OANN, whose shares rest on only 29 articles.

Table S6. Outlet characteristics: definitions, dates, coverage, and values

A. Measures.

Measure	Definition	Reference date	Outlets covered
Extreme-headline share	% of an outlet's analytic articles with headline stance 1 or 5	2017–2021 (this study)	48
Extreme-image share	% of articles with image stance 1 or 5	2017–2021 (this study)	48
MBFC leaning	6 categories (Table S1)		48
Media type	Digital native, newspaper, cable, network TV, magazine		48
Facebook followers	Page followers, mean of yearly counts (2018–2021 when 2017 is missing), log scale	2017–2021 (study period); Infowars: NBC News, Aug. 2018	48
NewsGuard trust score	0–100, nine weighted criteria; ≥60 = meets standards	Snapshot 2025-03-18 (post-period)	48
Audience ideological alignment	Mean self-reported ideology of Facebook sharers, −1 to +1 (Bakshy et al. 2015)	2014–2015 data (pre-period)	45 (Daily Wire, Epoch Times, OANN missing)

B. Values for the 48 outlets in the analytic sample.

Outlet	Extreme-headline share (%)	Facebook followers (M)	NewsGuard	Bakshy alignment
ABC	2.8	14.84	87.5	−0.161
AlterNet	40.6	0.86	82.0	−0.880
Breitbart	10.8	4.32	49.5	0.914
Business Insider	13.6	9.11	100.0	−0.059

Outlet	Extreme-headline share (%)	Facebook followers (M)	NewsGuard	Bakshy alignment
BuzzFeed	13.0	11.70	100.0	−0.524
CBS	2.1	5.53	90.0	−0.127
CNN	9.9	31.76	80.0	−0.271
Daily Beast	23.4	2.06	87.5	−0.456
Daily Caller	16.4	5.24	82.0	0.870
Daily Wire	33.1	2.26	57.0	—
Epoch Times	5.8	5.78	17.5	—
Fox News	15.8	17.61	69.5	0.775
Free Beacon	18.2	0.71	72.5	0.878
Gateway Pundit	53.7	0.62	30.0	0.939
HuffPost	24.2	9.38	87.5	−0.618
Infowars	31.5	2.50	7.5	0.600
MSNBC	14.6	2.23	49.5	−0.810
Mother Jones	28.8	1.49	69.5	−0.866
NBC	5.2	10.10	100.0	−0.274
NPR	3.7	6.49	100.0	−0.610
NY Post	16.6	4.29	69.5	0.250
NYT	8.4	16.05	87.5	−0.547
National Review	21.6	1.01	92.5	0.901
NewsBusters	39.3	2.59	92.5	0.917
Newsmax	6.7	1.73	20.0	0.770
Newsweek	15.0	1.33	100.0	−0.273
OANN	0.0	0.64	22.5	—
PBS	2.1	3.62	100.0	−0.539
PJ Media	48.9	0.36	27.5	0.912
Political Insider	35.5	2.76	62.0	0.900
Politico	5.7	1.84	100.0	−0.133
RealClearPolitics	14.4	0.13	62.0	0.662
RedState	59.6	0.49	27.5	0.962
Right Scoop	31.8	0.09	22.0	0.927
Salon	36.4	0.90	100.0	−0.875
Slate	27.8	1.43	87.5	−0.681
The Atlantic	19.8	2.36	100.0	−0.542
The Blaze	25.0	2.07	35.0	0.889
The Federalist	38.2	0.09	12.5	0.856
The Guardian	17.6	8.07	100.0	−0.572
The Hill	8.1	1.44	100.0	0.166
Townhall	30.4	1.20	69.5	0.926
USA Today	8.2	8.46	100.0	−0.064
Vox	18.7	2.61	87.5	−0.659
WSJ	7.5	6.26	100.0	0.275
Washington Post	17.5	6.29	100.0	−0.257
Washington Times	10.4	0.63	62.0	0.698
Western Journal	31.7	4.50	82.0	0.899

Notes. Followers and NewsGuard scores post-date the study period and are treated as approximate indicators of relatively stable outlet characteristics. Bakshy alignment is matched by domain to the top-500 list.

S3.2. Audience size, credibility, and audience ideology

Measure	Definition	Reference date	Coverage
Audience size	Log of the mean yearly follower count of the outlet's main Facebook page	Yearly snapshots, 2017–2021 (study period); Infowars from a 2018 news report	All 48 outlets
Credibility	NewsGuard trust score, 0–100	18 March 2025 snapshot (after the study period)	All 48 outlets
Audience ideology	Mean self-reported ideology of Facebook users who shared the outlet's domain, −1 (liberal) to +1 (conservative)	2014–2015 data (before the study period)	45 outlets

Audience size. Follower counts were recorded once a year, from 2017 to 2021, and averaged for each outlet. Eight outlets (AlterNet, The Guardian, Mother Jones, Newsmax, OANN, PBS, RealClearPolitics, and The Blaze) lack the 2017 count, so their average uses the four years 2018–2021. Infowars has no yearly counts because Facebook unpublished its pages on 6 August 2018. We therefore entered 2.5 million followers, as reported by NBC News at the time of removal (Rosenblatt, 2018). This figure is the combined total across the four Infowars and Alex Jones pages removed ("more than 2.5 million followers"), so it likely overstates the main Infowars page alone.

Credibility. The NewsGuard score (NewsGuard, n.d.) combines nine weighted journalistic criteria. A score of 60 or above means the outlet generally adheres to basic standards of credibility and transparency. Scores were matched to outlets by domain.

Audience ideology. Alignment comes from Bakshy, Messing, and Adamic (2015), matched by domain to their 500 most-shared domains. Daily Wire, Epoch Times, and OANN are not in the top 500. $|\text{alignment}|$, the absolute distance from 0, measures audience one-sidedness. Audience composition may have shifted since 2014–2015.

These three characteristics are strongly correlated with one another and with extreme-headline share (Table S19; S7.1).

S3.3. Headline sentiment

Measure. Headline sentiment was scored with the zero-shot natural language inference (NLI) model `MoritzLaurer/deberta-v3-large-zeroshot-v2.0` (Laurer et al., 2024). Each headline (the premise) was paired with five hypotheses of the form "The tone of this news headline is {very negative, somewhat negative, neutral, somewhat positive, very positive}." We applied a softmax to the five entailment logits and took the probability-weighted mean of –1, –.5, 0, .5, and 1. This gives a continuous score $S \in [-1, 1]$: its sign is the valence of the tone and its absolute value $|S|$ the intensity. Unlike stance, S has no target: it scores the tone of the headline, not its evaluation of the politician.

Model choice. Because sentiment enters the RQ1 models as a control for stance (S5), we chose the sentiment measure by two criteria (Fig. S5). First, it should measure the tone of news headlines accurately: we compared eight pretrained sentiment models by their Pearson correlation with gold valence on the 1,250 SemEval-2007 Task 14 news headlines (Strapparava & Mihalcea, 2007). Second, it should be the most conservative covariate for stance: we compared the models by their Spearman correlation with headline stance on a stance-stratified sample of 3,000 corpus headlines. The NLI model performed best on both:

Model	SemEval-2007 r	Correlation with stance (ρ)
NLI (DeBERTa-v3-large zero-shot)	.82	.60
Cardiff RoBERTa three-class	.80	.50
Cardiff topic-sentiment	.74	.56
Best original SemEval-2007 system	.48	—

Note. All eight models are shown in Fig. S5.

The NLI model is the most accurate measure of headline tone, and because it overlaps most with stance, controlling for it removes the most stance-related variation that could be attributed to tone. Stance effects estimated net of NLI sentiment are therefore the most demanding test among the candidate measures. The Cardiff three-class model is `cardiffnlp/twitter-roberta-base-sentiment-latest` (Loureiro et al., 2022).

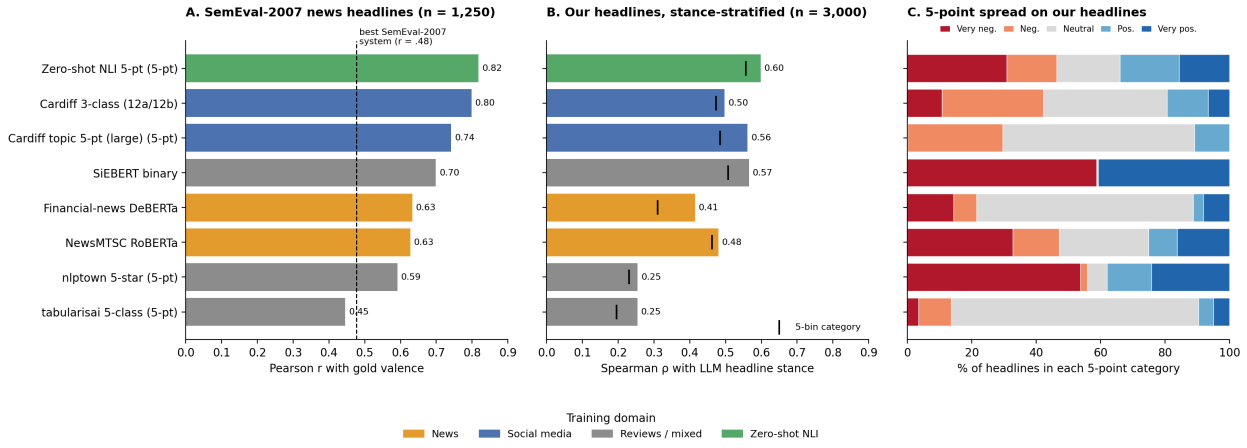


Fig. S5. Headline sentiment: model benchmark. (A) Pearson correlation of eight pretrained sentiment models with gold valence for the 1,250 SemEval-2007 Task 14 news headlines; the dashed line marks the best system in the original competition (r = .48). (B) Spearman correlation of each model's score with

headline stance in a stance-stratified sample of 3,000 corpus headlines; tick marks show the five-category version. (C) Distribution of each model's five-category output on these headlines. Color = model training domain.

Headline stance versus sentiment. Headline stance and sentiment are related but distinct constructs. Sentiment (S) is the emotional tone of the headline as a whole; stance (H) is how the headline portrays the pictured politician. The two correlate moderately in the 183,487 articles with both measures (Spearman ρ = .52), and their magnitudes correlate at ρ = .46 (|H| with |S|). A correlation of .52 means that tone explains roughly a quarter of the rank variance in stance.

Headline stance	Very negative	Somewhat negative	Neutral	Somewhat positive	Very positive	*n*
Very unfavorable	73.1	15.8	4.6	4.7	1.9	25,454
Somewhat unfavorable	33.2	33.1	20.0	11.7	1.9	54,250
Neutral	16.1	16.1	50.5	15.7	1.6	62,697
Somewhat favorable	17.0	10.6	15.4	40.1	17.0	36,499
Very favorable	19.1	3.9	3.0	19.3	54.7	4,587

Note. Row percentages of headline sentiment (most probable of the five NLI categories) within each headline stance level.

Tone and stance point in opposite directions for 11.0% of articles (20,182). Among headlines that are negative in tone, 12.4% are favorable in stance; among headlines that are positive in tone, 20.5% are unfavorable in stance. Conditioning on stance instead, 27.0% of favorable headlines are negative in tone, 11.4% of unfavorable headlines are positive in tone, and 32.2% of neutral-stance headlines are negative in tone. Reading the discordant headlines shows three recurring patterns. In each example below, the bracket gives the pictured politician, the outlet, the year, and the stance rating on the 1–5 scale; all examples are classified as *very negative* or *very positive* in tone.

1. **Negative tone, favorable portrayal.** The politician defends, grieves, condemns a wrongdoer, or is presented as the victim of an attack:
   - "Liberal media exploits hurricane Harvey to attack President Trump" [Trump, Fox News, 2017; stance 5]
   - "Donald Trump's baseless Joe Biden conspiracy fantasy should make Americans feel sick" [Biden, *USA Today*, 2019; stance 5]
   - "'I'll miss you dearly': Sen. Warren announces her oldest brother died from coronavirus" [Warren, *USA Today*, 2020; stance 5]
   - "Joe Biden Condemns Harvey Weinstein As 'Disgusting' And 'Immoral'" [Biden, BuzzFeed, 2017; stance 4]
2. **Positive tone, unfavorable portrayal.** Sarcasm, mockery, or celebration of the politician's setback:
   - "Trump: 'This Is A Great Day' For George Floyd" [Trump, HuffPost, 2020; stance 1]
   - "Biden has figured out how to solve all his border woes!" [Biden, Right Scoop, 2021; stance 1]
   - "Winning: Pelosi, Schumer Are 'Normalizing' Donald Trump" [Pelosi, Breitbart, 2017; stance 1]
   - "Gag: 'View' Pays Loving Tribute to 'Adorable' Biden on Presidents Day" [Biden, NewsBusters, 2021; stance 1]
   - "Trump Gets His Payback Against McConnell and Ryan" [McConnell, NBC, 2017; stance 1]

3. **Negative tone, neutral portrayal.** The politician reports or reacts to negative events, or is quoted criticizing someone else, without the headline taking a side:
    - "Criminal group that hacked law firm threatens to release Trump documents" [Trump, NBC, 2020; stance 3]
    - "Nancy Pelosi criticizes Trump's early coronavirus response, says president thought 'it's going to magically disappear'" [Pelosi, *Newsweek*, 2020; stance 3]

These patterns are what a whole-text tone measure cannot capture: the target of the negativity, and whether surface wording is meant literally. The last McConnell example also shows that one headline can be good news for one actor and bad news for another. Stance is rated toward the pictured politician, so the same tone can imply opposite stances depending on who is pictured.

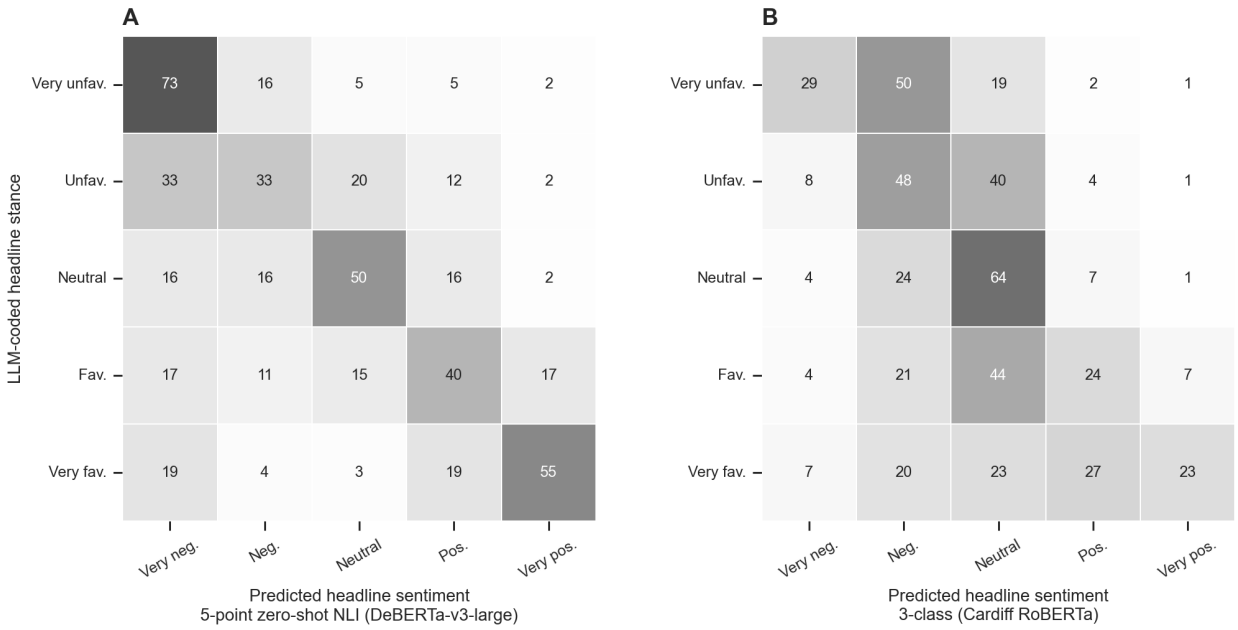


Fig. S6. Headline stance versus headline sentiment. (A) Row-normalized cross-tabulation (%) of headline stance against five-category NLI sentiment. (B) The same against Cardiff RoBERTa three-class sentiment.

S3.4. Facial expression

Measure. Facial expression is coded, like headline sentiment, for both valence and intensity. For the focal politician's face clip, DeepFace's facial-expression classifier (Serengil & Özpınar, 2024), trained on FER-2013 (Goodfellow et al., 2013), returned probabilities for seven emotions: anger, disgust, fear, happiness, sadness, surprise, and neutral. Facial-expression valence is

$$F = p_{happy} - (p_{angry} + p_{disgust} + p_{fear} + p_{sad}),$$

and its absolute value $|F|$ measures the intensity of the expression. Surprise and neutral are excluded from the index.

Coverage. Emotion probabilities exist for every analytic article. For 2,021 articles, however, the stored face count is zero. 181,466 analytic articles have a detected face.

Alternatives. Two alternative codings avoid the valence weights: the dominant emotion, and the six non-neutral emotion probabilities. Replacing F and $|F|$ with the six emotion probabilities leaves the model (v) estimates unchanged (S5.6).

Image stance versus facial expression. Image stance and facial valence correlate only weakly (ρ = .25; Pearson r = .35), and their magnitudes are almost uncorrelated (III with |F|: ρ = .06).

- Most faces in every image-stance category have negative valence. The share in the most negative valence bin is 68% for very unfavorable images and 23% for very favorable images.
- Very favorable images are the only category with many positive-valence faces (55%).
- Image stance varies with the dominant emotion but is not determined by it. Of faces whose dominant emotion is sadness, 49% appear in images rated somewhat unfavorable and 26% in images rated somewhat favorable.

Image stance therefore captures more than facial expression, including posture, camera angle, and setting (S2.3).

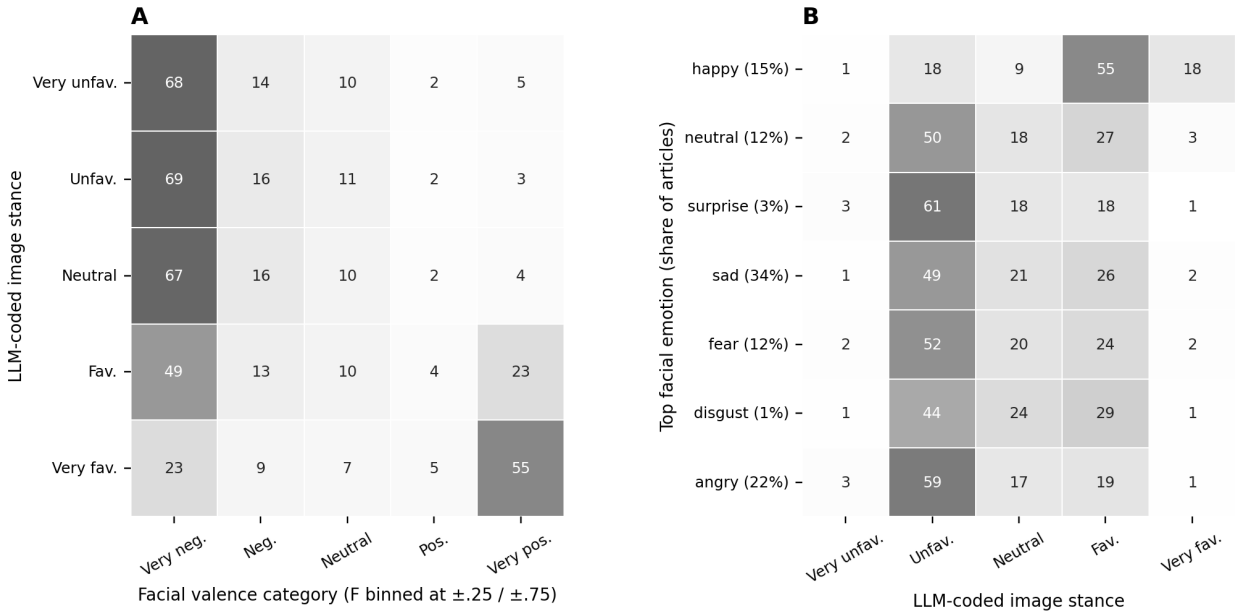


Fig. S7. Image stance versus facial expression. (A) Row-normalized cross-tabulation (%) of image stance against facial valence F, binned at ±.25 and ±.75. (B) Image stance by the face's dominant emotion; row labels give each emotion's share of articles.

S3.5. Partisan congruency

Partisan congruency combines outlet side and politician party. Let $G = +1$ when they match (Left or Center-left outlets with Democratic politicians; Center-right, Right, or Extreme-right outlets with Republican politicians) and $G = -1$ otherwise. Center outlets (Politico, The Hill) are excluded because congruency is undefined for them.

Congruent stance is $H_{cong} = G \cdot H$ and $I_{cong} = G \cdot I$. Positive H_{cong} means a headline favorable toward a co-partisan or unfavorable toward an opponent, the portrayal expected to appeal to the outlet's audience.

Congruency is defined for 176,585 analytic articles from the 46 outlets with a partisan side.

Part II. Results

S4. Descriptive Patterns of Multimodal Political Stance

This section is descriptive: it reports tabulations and rank correlations of the stance measures defined in S2.1, without regression models. All percentages use the analytic sample (N = 183,516 articles) unless noted.

S4.1. Stance distributions, cross-modal alignment, and joint extremity

Distributions. Headline stance is centered on neutral (34.2%), while image stance is concentrated on the somewhat unfavorable category (47.7%) (Fig. S8A–B). Extreme ratings (1 or 5) are much more common for headlines (16.4%) than for images (6.1%).

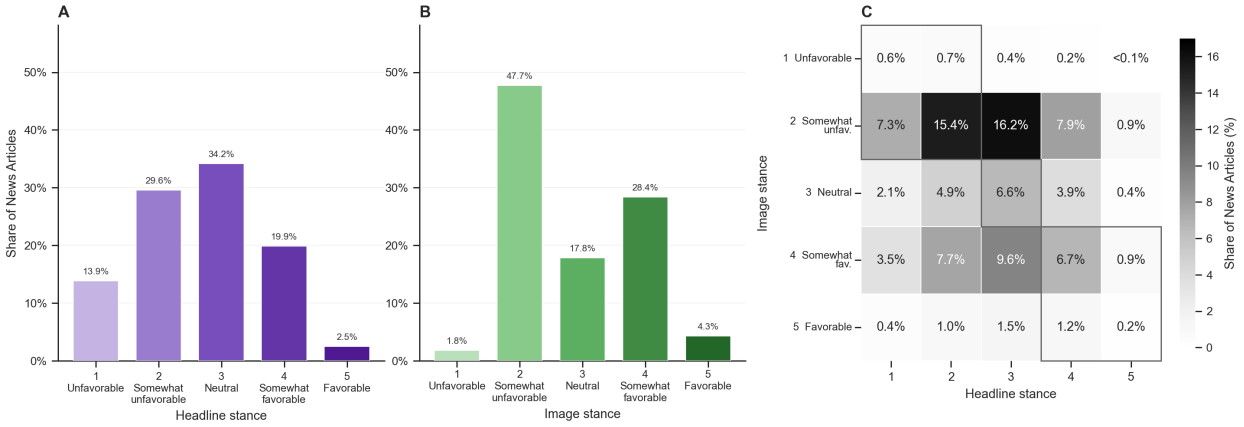


Fig. S8. Distribution of headline and image stance. (A) Headline stance and (B) image stance on the five-point favorability scale. (C) Joint distribution of headline (columns) and image (rows) stance; outlined blocks mark the unfavorable–unfavorable and favorable–favorable regions. All panels show percentages of the analytic sample (N = 183,516 articles).

Cross-modal alignment. Headline and image stance are only weakly associated (Fig. S8C): Spearman $\rho = .122$; $\chi^2(16) = 3{,}876.5$, $P < .001$; Cramér's $V = .07$. The two ratings match exactly for 29.5% of articles and are within one scale point for 74.1%. The most common combinations pair a neutral or somewhat unfavorable headline with a somewhat unfavorable image (16.2% and 15.4% of articles).

Joint extremity. Most articles (78.8%) have no extreme rating in either modality; 15.1% have an extreme headline only, 4.9% an extreme image only, and 1.2% (2,288 articles) both (Table S7). In 0.8% of articles (1,469), the two extremes share the same valence: 0.6% are very unfavorable in both modalities and 0.2% very favorable in both. Nearly all of the remaining 0.4% pair a very unfavorable headline with a very favorable image.

Table S7. Joint extremity of headline and image stance

Category	Articles	% of articles
Neither modality extreme	144,525	78.75
Headline extreme only	27,753	15.12
Image extreme only	8,950	4.88
Both modalities extreme	2,288	1.25
Both very unfavorable (1, 1)	1,028	0.56
Both very favorable (5, 5)	441	0.24
Very unfavorable headline, very favorable image (1, 5)	763	0.42
Very favorable headline, very unfavorable image (5, 1)	56	0.03

Notes. Extreme = rating 1 or 5. Percentages of the analytic sample (N = 183,516 articles). Indented rows break down the articles with both modalities extreme; same-valence extremes (1, 1 and 5, 5) are 0.80% of articles. The full headline × image distribution is in Fig. S8C.

S4.2. Distributions by politician and outlet

By politician. Headline stance is similar for the two parties: the mean on the 1–5 scale is 2.68 for Democrats and 2.67 for Republicans, and 17.0% versus 15.7% of headlines are extreme. Image stance is not similar. Images of Republican politicians were rated somewhat or very unfavorable far more often than images of Democrats (61.8% vs. 37.2%), and their mean image stance is lower (2.62 vs. 3.09) (Fig. S9A–B). Several politicians stand out:

- **Barack Obama**: the most favorable image portrayals; 47.5% of images somewhat favorable and 17.1% very favorable, giving the highest extreme-image share (17.9%).
- **Mitch McConnell and Donald Trump**: the most unfavorable images (64.7% and 60.4% somewhat unfavorable).
- **Joe Biden**: the highest extreme-headline share (18.6%).
- **Mike Pence**: the lowest extreme-headline share (8.8%).

Politician-level shares and payoffs are in Table S17.

By outlet, leaning, and media type. Extreme-headline shares vary far more across outlets (2.1%–59.6%) than extreme-image shares (2.7%–13.0%), and the two are only weakly correlated across the 48 outlets (Spearman ρ = .15; Fig. S9C). By MBFC leaning, the extreme-headline share is:

MBFC leaning	Extreme-headline share
Right	25.7%
Extreme right	17.9%
Left	17.7%
Center-right	11.9%
Center-left	10.0%
Center	6.2%

These are pooled article shares; outlet-level distributions are in Fig. S9D. Extreme-image shares vary little by leaning (5.0%–6.8%).

By media type, extreme headlines are most common in digital-native outlets (22.4%) and magazines (19.0%), followed by cable news (12.4%) and newspapers (11.1%), and rarest on network TV (3.1%).

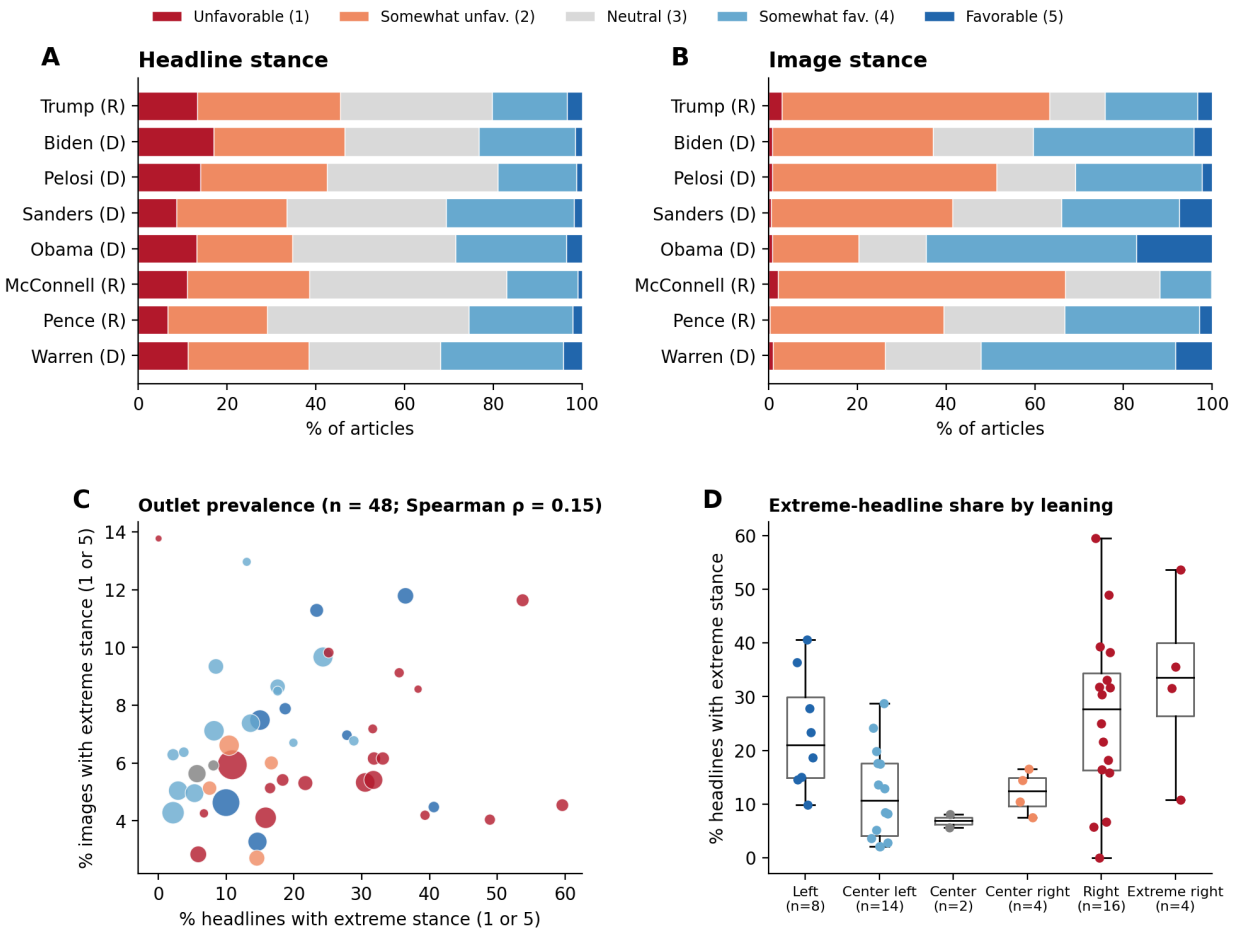


Fig. S9. Stance prevalence across politicians and outlets. (A, B) Distribution of headline and image stance for each politician. (C) Outlet-level percentage of extreme headlines versus extreme images (n = 48 outlets); marker color = MBFC leaning, as in D; marker size proportional to the outlet's articles. (D) Outlet extreme-headline shares by MBFC leaning; boxes show the median and interquartile range across outlets.

S5. Headline and Image Stance Associations with Engagement (RQ1)

RQ1 asks how headline and image stance relate to Facebook engagement. We estimated five OLS models in steps. Models (i)–(iii) describe the shape of the association with categorical stance; model (iv) separates the valence and extremity of each modality; model (v), the main RQ1 model, adds partisan congruency, headline sentiment, and facial expression to model (iv).

Reported in the main text. Fig. 2A shows model (v), and Fig. 2B–D show models (i)–(iii). Model (iv) is not shown in the main text; it is the stance-only comparison for model (v).

- S5.1 specifies all five models.
- S5.2–S5.6 report the models in order, (i) to (v). Each subsection has a coefficient plot and a table with every coefficient (unstandardized b, standardized β, SE, 95% CI, P) and the model fit (sample, R^2, RMSE, log-likelihood, AIC, BIC, fixed effects, standard errors). All models are OLS with outlet-clustered SEs and have no random effects. S5.5 also discusses how headline and image extremity combine; S5.6 discusses partisan congruency, headline sentiment, and facial expression, and how the stance estimates change from model (iv) to model (v).
- S5.7–S5.10 report the robustness checks, grouped by the question each answers (table below). Each check is run on model (v), with model (iv) as a secondary comparison.
- S5.11 compares articles within the same outlet and week, and within the same news story, to ask whether the extremity payoffs reflect which outlets, weeks, politicians, or stories carry extreme content.

Subsection	Question	Checks
S5.7. Collinearity	Are individual coefficients stable when the stance terms are built from the same ratings?	Variance inflation factors; condition numbers
S5.8. Inference with few clusters	Are the standard errors trustworthy with 45 outlet clusters?	t critical values; restricted wild cluster bootstrap
S5.9. Outcome model	Does the payoff depend on the log-OLS specification or on the heavy right tail of engagement?	Top-tail trimming; negative binomial and PPML on raw counts
S5.10. Sample	Is the payoff driven by particular outlets or politicians, or by which articles entered the sample?	Leave one outlet or politician out; single-politician articles; reweighting and restricting by image download success; outlets present in all years
S5.11. Within-outlet and within-story comparisons	Is the payoff explained by differences between outlets, weeks, politicians, or stories?	Outlet×week and politician×week fixed effects; shared-story fixed effects

Across all checks, $|H|$, $|I|$, and $G \times H$ stay positive. $|H|$ and $G \times H$ are significant in every check. $|I|$ is significant in every check except PPML on raw counts (S5.9), which estimates differences in mean engagement and is dominated by the most-engaged articles (β|I| = 0.18, P = .15; β|H| = 0.30, P = .023). $H \times |I|$ and the sentiment terms are the least stable (S5.9–S5.10), and facial intensity $|F|$ is not significant within outlet-weeks (S5.11).

S5.1. Model setup

Outcome. All models use $Y_i = \log(1 + \text{engagement}_i)$, and all standard errors are clustered by outlet (CR1).

Samples and controls. Models (i)–(iv) use the 183,516 analytic articles from 48 outlets with the following controls $\mathbf{X}_i$:

Control	Categories	Reference
MBFC leaning	Left, Center-left, Center, Center-right, Right, Extreme right	Center
Media type	Digital native, newspaper, cable news, network TV, magazine	Cable news
Publication year	2017–2021	2017

Model (v) uses 176,556 articles from 45 outlets. The sample is smaller for two reasons:

- partisan congruency is undefined for the two Center outlets (Politico, The Hill; S3.5);
- OANN (29 articles) has no headline-sentiment scores, because sentiment was scored only for the 47 outlets with at least 100 articles (S6.1).

Because the Center category is empty in this sample, model (v) replaces MBFC leaning with a collapsed Left/Right indicator; media type and year are as above.

- **(i) Categorical headline stance:** $Y_i = \alpha + \sum_{h\in\{1,2,4,5\}} \delta_h \mathbb{1}[h_i = h] + \gamma'\mathbf{X}_i + \varepsilon_i$, with neutral headlines ($h = 3$) as the reference. δ_h is the difference in log engagement between headlines at level h and neutral headlines.
- **(ii) Categorical image stance:** the same model with image categories.
- **(iii) Categorical headline × image:** indicators for all 25 combinations of headline and image stance, with the neutral–neutral cell (3, 3) as the reference. Adjusted predictions for each cell average over the observed controls and are reported as $\exp(\hat{Y}) - 1$ interactions.

Model (iv): valence and extremity. Model (iv) crosses the valence and extremity of the headline with those of the image:

$$Y_i = (H_i + |H_i|) \times (I_i + |I_i|) + \gamma'\mathbf{X}_i + \varepsilon_i,$$

where the product denotes the two main effects of each modality and their four cross-modal products. Written out:

$$\begin{aligned} Y_i = \alpha &+ \beta_H H_i + \beta_{|H|}|H_i| + \beta_I I_i + \beta_{|I|}|I_i| \\ &+ \beta_{HI} H_i I_i + \beta_{H|I|} H_i|I_i| + \beta_{|H|I} |H_i| I_i + \beta_{|H||I|} |H_i||I_i| + \gamma'\mathbf{X}_i + \varepsilon_i. \end{aligned}$$

- $\beta_{|H|}$ (the *headline extremity payoff*) is the average difference in Y between a very favorable or very unfavorable headline and a neutral one, when the image is neutral.
- β_H is half the difference between very favorable and very unfavorable headlines.
- The endpoint contrasts are $\beta_{|H|} + \beta_H$ (very favorable vs. neutral) and $\beta_{|H|} - \beta_H$ (very unfavorable vs. neutral).
- The image terms are interpreted the same way.
- The four cross-modal terms capture valence alignment between modalities ($H \times I$), headline valence combined with image extremity ($H \times |I|$), headline extremity combined with image valence ($|H| \times I$), and joint extremity ($|H| \times |I|$).

When the other modality is not neutral, the payoffs also involve the cross-modal terms. For example, the headline payoff when the image is very unfavorable ($I = -1$, $|I| = 1$) is $\beta_{|H|} - \beta_{|H|I} + \beta_{|H||I|}$. All contrasts are linear combinations of the coefficients with clustered SEs.

$$Y_i = \text{model (iv) terms} + \delta_G G_i + \beta_{H_{cong}} G_i H_i + \beta_{I_{cong}} G_i I_i + \phi_S S_i + \phi_{|S|} |S_i| + \phi_F F_i + \phi_{|F|} |F_i| + \gamma' \mathbf{X}_i + \varepsilon_i,$$

where G is partisan congruency, S headline sentiment, and F facial valence (S3). $\beta_{|H|}$ and $\beta_{|I|}$ are now the extremity payoffs holding congruency, tone, and facial expression constant.

Intermediate congruency model. M1 adds partisan congruency to model (iv): G, $H_{cong} = G \cdot H$, and $I_{cong} = G \cdot I$. It is the second step of the model (iv)-to-(v) comparison in S5.6.

Congruency-specific payoffs. Because $H_{cong} = G \cdot H$, the headline payoff depends on whether the extreme portrayal is congruent. With the image neutral, an extreme headline compared with a neutral one adds:

- $\beta_{|H|} + \beta_{H_{cong}}$ when it is congruent ($G \cdot H = +1$: favorable to a co-partisan or unfavorable to an opponent);
- $\beta_{|H|} - \beta_{H_{cong}}$ when it is incongruent ($G \cdot H = -1$).

Each contrast averages over the favorable and unfavorable cases within the type, which cancels β_H. The difference between the two is $2\beta_{H_{cong}}$. Image contrasts are defined analogously.

S5.2. Model (i): categorical headline stance

Model (i) corresponds to main-text Fig. 2B. Compared with neutral headlines, every non-neutral headline category is associated with higher engagement, in a U-shape (Fig. S10; Table S8; all P < .001). Very unfavorable headlines are associated with 1.46 more log points of engagement than neutral headlines (4.3 times as many interactions) and very favorable headlines with 1.56 more (4.8 times); the somewhat unfavorable and somewhat favorable categories add 0.84 (2.3 times) and 0.62 (1.9 times). The model explains 12.5% of the variance in log engagement.

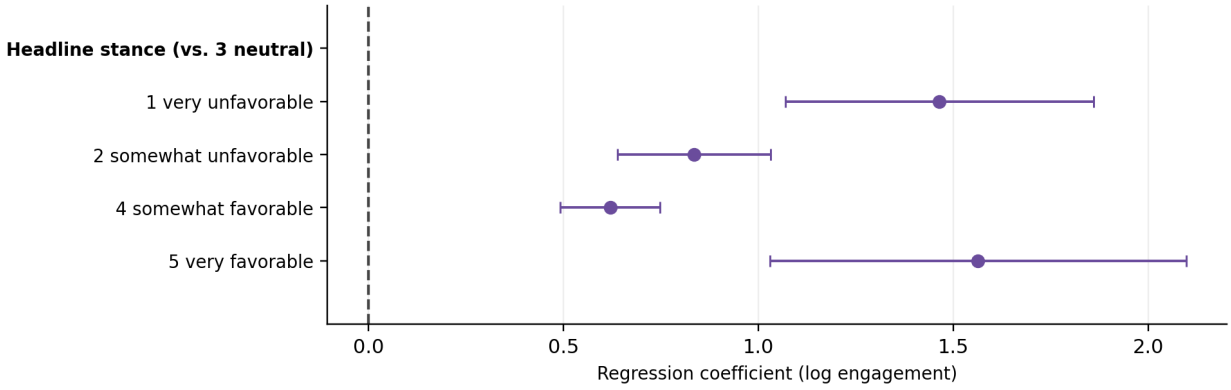


Fig. S10. Model (i): categorical headline stance. Differences in log engagement between each headline stance category and neutral headlines, with 95% CIs (outlet-clustered); 183,516 articles from 48 outlets. Controls are omitted from the plot; all estimates are in Table S8.

Table S8. Model (i): categorical headline stance

A. Coefficients.

Term	b	β	SE	95% CI	P
Headline 1 (very unfavorable) vs. 3	1.46	0.150	0.20	[1.07, 1.86]	<.001
Headline 2 (somewhat unfavorable) vs. 3	0.84	0.113	0.10	[0.64, 1.03]	<.001
Headline 4 (somewhat favorable) vs. 3	0.62	0.073	0.07	[0.49, 0.75]	<.001
Headline 5 (very favorable) vs. 3	1.56	0.072	0.27	[1.03, 2.10]	<.001
Controls					
Leaning: Center-left	−0.04	−0.005	0.43	[−0.88, 0.80]	.925
Leaning: Center-right	−2.20	−0.189	1.05	[−4.26, −0.13]	.037
Leaning: Extreme-right	−2.31	−0.216	0.40	[−3.10, −1.53]	<.001
Leaning: Left	−0.78	−0.095	0.35	[−1.48, −0.09]	.026
Leaning: Right	−0.82	−0.104	0.55	[−1.90, 0.26]	.137
Media type: Digital Native	−0.88	−0.131	0.43	[−1.73, −0.04]	.041
Media type: Magazine	−1.03	−0.078	0.42	[−1.84, −0.21]	.013
Media type: Network TV	−2.80	−0.275	0.81	[−4.38, −1.22]	<.001
Media type: Newspaper	−0.98	−0.105	0.87	[−2.68, 0.73]	.261
Year: 2018	0.15	0.015	0.12	[−0.08, 0.39]	.207
Year: 2019	0.63	0.075	0.17	[0.29, 0.96]	<.001
Year: 2020	1.21	0.160	0.20	[0.81, 1.60]	<.001
Year: 2021	−0.18	−0.023	0.33	[−0.82, 0.46]	.582
Constant					
Intercept	6.58		0.45	[5.71, 7.46]	<.001

B. Model fit.

Statistic	Value
Articles	183,516
Outlets (clusters)	48
R^2	0.125
Adjusted R^2	0.125
Residual SD (RMSE)	3.161
Log-likelihood	–471,613
AIC	943,262
BIC	943,444
Fixed effects (controls)	MBFC leaning, media type, year
Random effects	None
Standard errors	Clustered by outlet (CR1)

Notes. OLS of log(1 + engagement); SEs clustered by outlet (CR1); no random effects. Controls (fixed effects): MBFC leaning (reference Center), media type (reference Cable news), and publication year (reference 2017). b = unstandardized coefficient (difference in log engagement); β = standardized coefficient, b · SD(x) / SD(y) in the estimation sample; 95% CIs and P from outlet-clustered SEs (normal approximation).

S5.3. Model (ii): categorical image stance

Model (ii) corresponds to main-text Fig. 2C. The image pattern is also U-shaped but about half as large as the headline pattern (Fig. S11; Table S9; all P ≤ .001). Very unfavorable and very favorable images are associated with 0.76 (2.1 times) and 0.55 (1.7 times) more log points of engagement than neutral images; the somewhat unfavorable and somewhat favorable categories add 0.35 (1.4 times) and 0.24 (1.3 times). The model explains 10.5% of the variance, less than model (i).

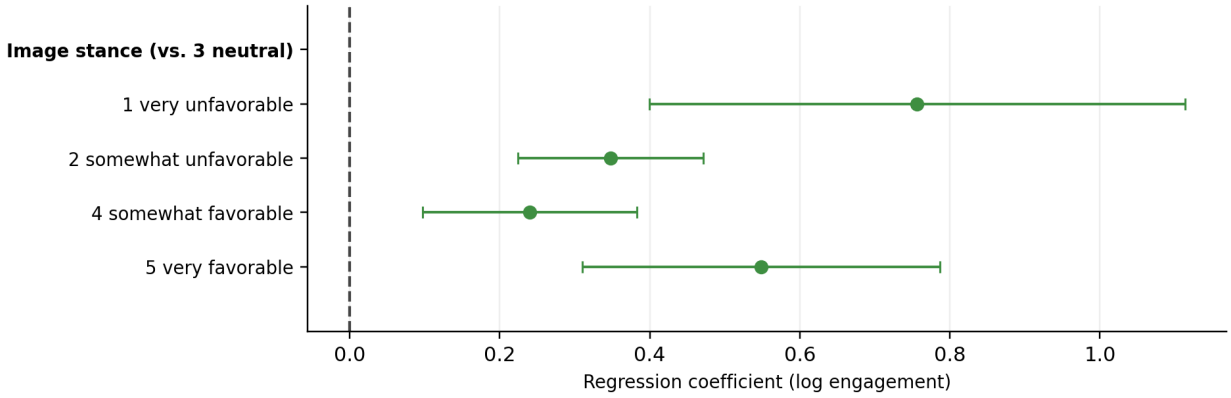


Fig. S11. Model (ii): categorical image stance. Differences in log engagement between each image stance category and neutral images, with 95% CIs (outlet-clustered); 183,516 articles from 48 outlets. Controls are omitted from the plot; all estimates are in Table S9.

Table S9. Model (ii): categorical image stance

A. Coefficients.

Term	b	β	SE	95% CI	P
Image 1 (very unfavorable) vs. 3	0.76	0.030	0.18	[0.40, 1.11]	<.001
Image 2 (somewhat unfavorable) vs. 3	0.35	0.051	0.06	[0.22, 0.47]	<.001
Image 4 (somewhat favorable) vs. 3	0.24	0.032	0.07	[0.10, 0.38]	<.001
Image 5 (very favorable) vs. 3	0.55	0.033	0.12	[0.31, 0.79]	<.001
Controls					
Leaning: Center-left	0.13	0.018	0.43	[−0.72, 0.99]	.759
Leaning: Center-right	−2.06	−0.178	1.09	[−4.19, 0.07]	.058
Leaning: Extreme-right	−2.18	−0.204	0.46	[−3.09, −1.27]	<.001
Leaning: Left	−0.52	−0.063	0.35	[−1.21, 0.16]	.136
Leaning: Right	−0.52	−0.066	0.56	[−1.61, 0.57]	.348
Media type: Digital Native	−0.69	−0.102	0.44	[−1.55, 0.18]	.119
Media type: Magazine	−0.87	−0.066	0.41	[−1.67, −0.07]	.033
Media type: Network TV	−2.91	−0.285	0.82	[−4.51, −1.30]	<.001
Media type: Newspaper	−0.90	−0.097	0.89	[−2.66, 0.85]	.311
Year: 2018	0.16	0.016	0.12	[−0.08, 0.39]	.199
Year: 2019	0.65	0.078	0.18	[0.31, 1.00]	<.001
Year: 2020	1.23	0.163	0.21	[0.82, 1.64]	<.001
Year: 2021	−0.06	−0.008	0.34	[−0.73, 0.61]	.861
Constant					
Intercept	6.57		0.45	[5.70, 7.45]	<.001

B. Model fit.

Statistic	Value
Articles	183,516
Outlets (clusters)	48
R^2	0.105
Adjusted R^2	0.105
Residual SD (RMSE)	3.196
Log-likelihood	–473,625
AIC	947,285
BIC	947,468
Fixed effects (controls)	MBFC leaning, media type, year
Random effects	None
Standard errors	Clustered by outlet (CR1)

Notes. OLS of log(1 + engagement); SEs clustered by outlet (CR1); no random effects. Controls (fixed effects): MBFC leaning (reference Center), media type (reference Cable news), and publication year (reference 2017). b = unstandardized coefficient (difference in log engagement); β = standardized coefficient, b · SD(x) / SD(y) in the estimation sample; 95% CIs and P from outlet-clustered SEs (normal approximation).

S5.4. Model (iii): categorical headline × image stance

Model (iii) corresponds to main-text Fig. 2D. It estimates a separate mean for each of the 25 headline × image combinations; each coefficient is the cell's difference in log engagement from articles neutral in both modalities (Fig. S12; Table S10).

- All 24 cells exceed the neutral–neutral cell (all $P \leq .004$).
- The largest differences are for articles very favorable in both modalities (2.18) and very unfavorable in both (1.99).
- Within every level of image stance, cells with an extreme headline exceed the cell with a neutral headline. Among articles with a neutral headline, extreme images add 0.51 to 0.70, less than half of what extreme headlines add among articles with a neutral image (1.49 to 1.63).

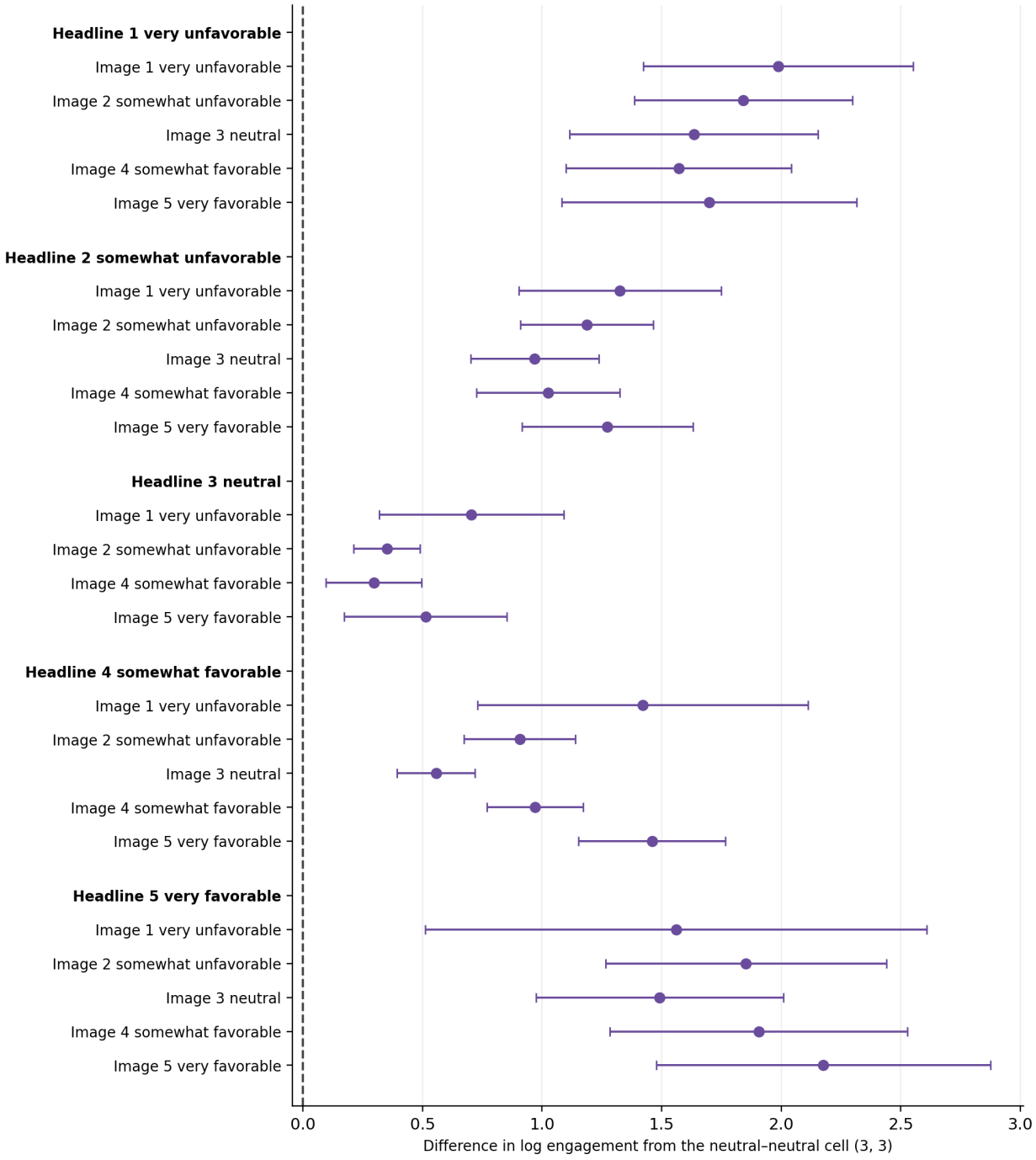


Fig. S12. Model (iii): categorical headline × image stance. Difference in log engagement between each headline × image cell and the neutral–neutral cell (3, 3), with 95% CIs (outlet-clustered); 183,516 articles from 48 outlets. Rows are grouped by headline stance. Controls are omitted from the plot; all estimates are in Table S10.

Table S10. Model (iii): categorical headline × image stance

A. Coefficients.

Term	b	β	SE	95% CI	P
Headline 1, image 1 vs. (3, 3)	1.99	0.044	0.29	[1.42, 2.55]	<.001
Headline 1, image 2 vs. (3, 3)	1.84	0.142	0.23	[1.38, 2.30]	<.001
Headline 1, image 3 vs. (3, 3)	1.63	0.070	0.27	[1.11, 2.15]	<.001
Headline 1, image 4 vs. (3, 3)	1.57	0.085	0.24	[1.10, 2.04]	<.001
Headline 1, image 5 vs. (3, 3)	1.70	0.032	0.31	[1.08, 2.32]	<.001
Headline 2, image 1 vs. (3, 3)	1.33	0.032	0.22	[0.90, 1.75]	<.001
Headline 2, image 2 vs. (3, 3)	1.19	0.127	0.14	[0.91, 1.47]	<.001
Headline 2, image 3 vs. (3, 3)	0.97	0.062	0.14	[0.70, 1.24]	<.001
Headline 2, image 4 vs. (3, 3)	1.03	0.081	0.15	[0.73, 1.32]	<.001
Headline 2, image 5 vs. (3, 3)	1.27	0.037	0.18	[0.91, 1.63]	<.001
Headline 3, image 1 vs. (3, 3)	0.70	0.013	0.20	[0.32, 1.09]	<.001
Headline 3, image 2 vs. (3, 3)	0.35	0.038	0.07	[0.21, 0.49]	<.001
Headline 3, image 4 vs. (3, 3)	0.30	0.026	0.10	[0.10, 0.50]	.004
Headline 3, image 5 vs. (3, 3)	0.51	0.018	0.17	[0.17, 0.85]	.003
Headline 4, image 1 vs. (3, 3)	1.42	0.017	0.35	[0.73, 2.11]	<.001
Headline 4, image 2 vs. (3, 3)	0.91	0.072	0.12	[0.67, 1.14]	<.001
Headline 4, image 3 vs. (3, 3)	0.56	0.032	0.08	[0.39, 0.72]	<.001
Headline 4, image 4 vs. (3, 3)	0.97	0.072	0.10	[0.77, 1.17]	<.001
Headline 4, image 5 vs. (3, 3)	1.46	0.047	0.16	[1.15, 1.77]	<.001
Headline 5, image 1 vs. (3, 3)	1.56	0.008	0.53	[0.51, 2.61]	.004
Headline 5, image 2 vs. (3, 3)	1.85	0.053	0.30	[1.26, 2.44]	<.001
Headline 5, image 3 vs. (3, 3)	1.49	0.027	0.26	[0.97, 2.01]	<.001
Headline 5, image 4 vs. (3, 3)	1.91	0.054	0.32	[1.28, 2.53]	<.001
Headline 5, image 5 vs. (3, 3)	2.18	0.032	0.36	[1.48, 2.87]	<.001
Controls					
Leaning: Center-left	−0.05	−0.006	0.42	[−0.88, 0.79]	.912
Leaning: Center-right	−2.18	−0.188	1.04	[−4.23, −0.14]	.036
Leaning: Extreme-right	−2.32	−0.217	0.40	[−3.10, −1.54]	<.001
Leaning: Left	−0.79	−0.096	0.35	[−1.48, −0.10]	.025
Leaning: Right	−0.81	−0.103	0.55	[−1.89, 0.26]	.139
Media type: Digital Native	−0.89	−0.132	0.43	[−1.73, −0.05]	.037
Media type: Magazine	−1.03	−0.078	0.41	[−1.83, −0.22]	.013
Media type: Network TV	−2.78	−0.273	0.80	[−4.35, −1.22]	<.001
Media type: Newspaper	−0.98	−0.105	0.86	[−2.67, 0.71]	.255
Year: 2018	0.15	0.015	0.12	[−0.09, 0.38]	.215
Year: 2019	0.63	0.075	0.17	[0.30, 0.97]	<.001
Year: 2020	1.21	0.160	0.20	[0.81, 1.61]	<.001
Year: 2021	−0.15	−0.019	0.33	[−0.79, 0.49]	.650
Constant					
Intercept	6.30		0.45	[5.42, 7.18]	<.001

B. Model fit.

Statistic	Value
Articles	183,516
Outlets (clusters)	48

Statistic	Value
R^2	0.127
Adjusted R^2	0.127
Residual SD (RMSE)	3.158
Log-likelihood	–471,406
AIC	942,889
BIC	943,273
Fixed effects (controls)	MBFC leaning, media type, year
Random effects	None
Standard errors	Clustered by outlet (CR1)

Notes. OLS of log(1 + engagement); SEs clustered by outlet (CR1); no random effects. Controls (fixed effects): MBFC leaning (reference Center), media type (reference Cable news), and publication year (reference 2017). The model has one indicator per headline × image cell, with the neutral–neutral cell (3, 3) as the reference; it is identical in fit to the full factorial of the headline and image categories. b = unstandardized coefficient (difference in log engagement); β = standardized coefficient, b · SD(x) / SD(y) in the estimation sample; 95% CIs and P from outlet-clustered SEs (normal approximation).

S5.5. Model (iv): valence and extremity

Model (iv) is the stance-only version of the main model (Fig. S13; Table S11).

- **Extremity.** Both extremity payoffs are large and positive: $\beta_{|H|} = 1.44$ (SE 0.19, P < .001; standardized β = .15) and $\beta_{|I|} = 0.62$ (SE 0.12, P < .001; β = .04). The headline payoff exceeds the image payoff by 0.82 (95% CI 0.50 to 1.14; P < .001).
- **Valence.** Headline valence is small and negative ($\beta_H = -0.26$, P = .018), so unfavorable headlines draw somewhat more engagement than favorable ones. Image valence is not significant ($\beta_I = -0.06$, P = .48).
- **Cross-modal terms.** Two are positive: valence alignment $H \times I = 0.19$ (P < .001) and $H \times |I| = 0.39$ (P < .001). The other two are not significant: $|H| \times I = 0.04$ (P = .69) and $|H| \times |I| = -0.11$ (P = .51). The non-significant joint-extremity term means that headline and image extremity combine roughly additively on the log scale.

Endpoint contrasts (Table S11C). With the other modality neutral, very unfavorable headlines exceed neutral ones by 1.70 log points (5.5 times as many interactions) and very favorable headlines by 1.19 (3.3 times); the difference between the two valences is significant ($2\beta_H = -0.52$, P = .018). For images, the contrasts are 0.68 (2.0 times) and 0.56 (1.7 times), and the valences do not differ ($2\beta_I = -0.13$, P = .48). Relative to articles neutral in both modalities, articles very unfavorable in both have Δ = 2.04 (7.7 times) and articles very favorable in both Δ = 2.25 (9.5 times).

How headline and image extremity combine. The joint-extremity term is small and not significant ($|H| \times |I| = -0.11$, P = .51), so articles extreme in both modalities receive about the sum of the two separate extremity payoffs on the log scale, with no additional payoff for joint extremity. Because the outcome is logged, additive payoffs in log engagement multiply on the scale of raw interactions. The endpoint contrasts illustrate this. For articles very unfavorable in both modalities, the separate headline and image contrasts sum to 2.38 (1.70 + 0.68) and the joint contrast is 2.04; for articles very favorable in both, they sum to 1.75 (1.19 + 0.56) and the joint contrast is 2.25. The two joint contrasts depart from the sums in opposite directions because of the valence terms ($H \times I$ and $H \times |I|$), not because of joint extremity. Model (iii), which imposes no structure, gives the same picture: the very-unfavorable cell (1, 1) is 1.99 above neutral–neutral against a sum of 2.34 for the single-modality cells (1, 3) and (3, 1), and the very-favorable cell (5, 5) is 2.18 against a sum of 2.00 (S5.4). The headline contributes most of the combined payoff: βIHI exceeds βIII by 0.82 (P < .001), and its standardized coefficient is more than three times as large (.15 vs. .04).

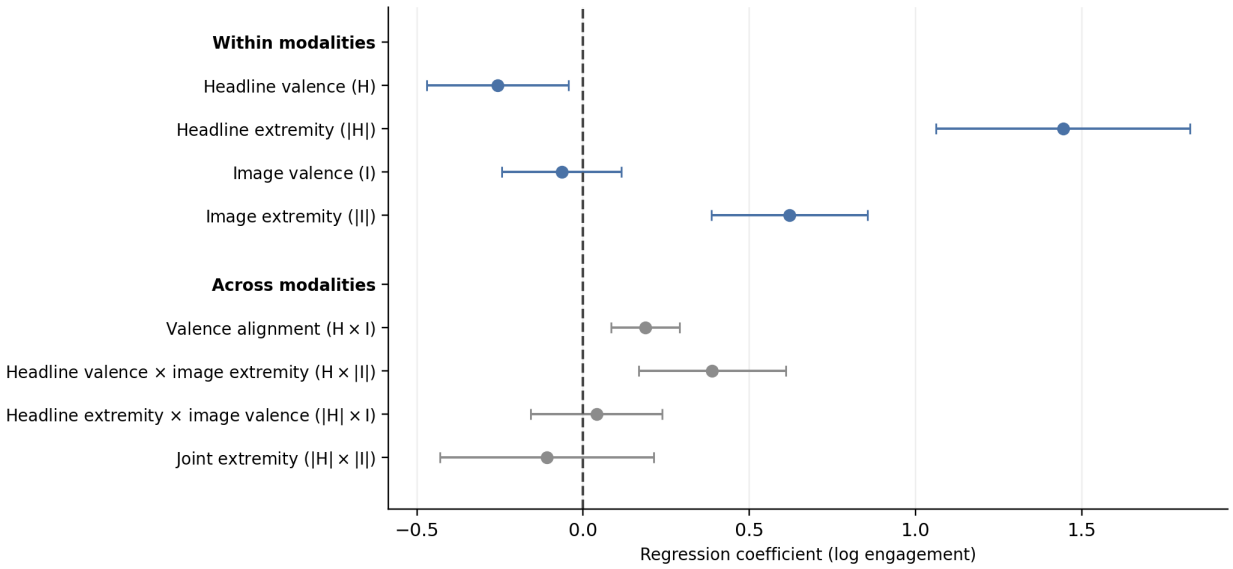


Fig. S13. Model (iv): valence and extremity. Coefficients with 95% CIs (outlet-clustered); 183,516 articles from 48 outlets. Valence (H, I) runs from –1 to 1 and extremity ($|H|$, $|I|$) from 0 to 1. Controls are omitted from the plot; all estimates are in Table S11.

Table S11. Model (iv): valence and extremity, $Y \sim (H + |H|) \times (I + |I|)$

A. Coefficients.

Term	b	β	SE	95% CI	P
H	–0.26	–0.039	0.11	[–0.47, –0.04]	.018
$\|H\|$	1.44	0.147	0.19	[1.06, 1.83]	<.001
I	–0.06	–0.010	0.09	[–0.24, 0.11]	.479
$\|I\|$	0.62	0.044	0.12	[0.39, 0.85]	<.001
$H \times I$	0.19	0.015	0.05	[0.08, 0.29]	<.001
$H \times \|I\|$	0.39	0.030	0.11	[0.17, 0.61]	<.001
$\|H\| \times I$	0.04	0.003	0.10	[–0.16, 0.24]	.691
$\|H\| \times \|I\|$	–0.11	–0.007	0.16	[–0.43, 0.21]	.505
Controls					
Leaning: Center-left	–0.05	–0.007	0.43	[–0.88, 0.78]	.908
Leaning: Center-right	–2.19	–0.189	1.04	[–4.24, –0.14]	.036
Leaning: Extreme-right	–2.32	–0.217	0.40	[–3.10, –1.54]	<.001
Leaning: Left	–0.79	–0.096	0.35	[–1.48, –0.10]	.025
Leaning: Right	–0.81	–0.103	0.55	[–1.89, 0.26]	.138
Media type: Digital Native	–0.90	–0.133	0.43	[–1.74, –0.05]	.037
Media type: Magazine	–1.03	–0.078	0.41	[–1.84, –0.22]	.013
Media type: Network TV	–2.79	–0.274	0.80	[–4.36, –1.22]	<.001
Media type: Newspaper	–0.98	–0.106	0.86	[–2.67, 0.71]	.255
Year: 2018	0.15	0.015	0.12	[–0.09, 0.38]	.217
Year: 2019	0.63	0.075	0.17	[0.30, 0.97]	<.001
Year: 2020	1.20	0.160	0.20	[0.80, 1.61]	<.001
Year: 2021	–0.16	–0.020	0.33	[–0.80, 0.48]	.631
Constant					
Intercept	6.33		0.46	[5.43, 7.23]	<.001

B. Model fit.

Statistic	Value
Articles	183,516
Outlets (clusters)	48
R^2	0.126
Adjusted R^2	0.126
Residual SD (RMSE)	3.158
Log-likelihood	–471,446
AIC	942,935
BIC	943,158
Fixed effects (controls)	MBFC leaning, media type, year
Random effects	None
Standard errors	Clustered by outlet (CR1)

C. Endpoint and cross-modal contrasts.

Contrast	Estimate	SE	95% CI	P	exp(estimate)
Headline: Very favorable vs neutral (image neutral)	1.19	0.19	[0.82, 1.56]	<.001	3.28
Headline: Very unfavorable vs neutral (image neutral)	1.70	0.25	[1.21, 2.20]	<.001	5.48
Headline: Average extreme vs neutral = βIHI	1.44	0.19	[1.06, 1.83]	<.001	4.24
Headline: Favorable minus unfavorable extreme = 2βH	–0.52	0.22	[–0.94, –0.09]	.018	0.60
Image: Very favorable vs neutral (headline neutral)	0.56	0.17	[0.23, 0.88]	<.001	1.74
Image: Very unfavorable vs neutral (headline neutral)	0.68	0.13	[0.43, 0.94]	<.001	1.98
Image: Average extreme vs neutral = βIII	0.62	0.12	[0.39, 0.85]	<.001	1.86
Image: Favorable minus unfavorable extreme = 2βI	–0.13	0.18	[–0.49, 0.23]	.479	0.88
Comparison: βIHI – βIII	0.82	0.16	[0.50, 1.14]	<.001	2.28
Joint: Both very unfavorable vs both neutral	2.04	0.26	[1.52, 2.55]	<.001	7.66
Joint: Both very favorable vs both neutral	2.25	0.27	[1.72, 2.77]	<.001	9.47

Notes. OLS of log(1 + engagement); SEs clustered by outlet (CR1); no random effects. Controls (fixed effects): MBFC leaning (reference Center), media type (reference Cable news), and publication year (reference 2017). Panel C: with the other modality neutral, favorable vs. neutral = βIHI + βH and unfavorable vs. neutral = βIHI – βH. b = unstandardized coefficient (difference in log engagement); β = standardized coefficient, b · SD(x) / SD(y) in the estimation sample; 95% CIs and P from outlet-clustered SEs (normal approximation).

S5.6. Model (v): main RQ1 model

Model (v) is the model in main-text Fig. 2A (Fig. S14; Table S12).

Extremity. Both extremity payoffs remain large and significant after congruency, sentiment, and facial expression are held constant:

- headline extremity: βIHI = 0.80 (SE 0.21, P < .001; standardized β = .08);
- image extremity: βIII = 0.60 (SE 0.12, P < .001; β = .04).

Cross-modal terms. Only $H \times |I|$ is significant (0.40, SE 0.13, P = .002): the advantage of favorable over unfavorable headlines grows as the image becomes more extreme. The other three cross-modal terms are not significant ($H \times I$ = 0.02; $|H| \times I$ = –0.03; $|H| \times |I|$ = –0.21, P = .22).

Partisan congruency, headline sentiment, and facial expression. The three sets of covariates relate to engagement differently.

- *Partisan congruency.* Congruent headline stance is strongly associated with engagement ($G \times H$ = 0.66, SE 0.13, P < .001; β = .09): headlines favorable to co-partisan politicians, or unfavorable to opposing-party politicians, draw more engagement. Neither congruent image stance ($G \times I$ = 0.13, P = .12) nor a co-partisan target (G = –0.10, P = .11) is significant. As a result, the headline extremity payoff is concentrated in congruent portrayals. With the image neutral, a congruent extreme headline exceeds a neutral one by 1.46 (P < .001; $\beta_{|H|} + \beta_{H_{cong}}$, S5.1), whereas an incongruent extreme headline does not differ from a neutral one (0.14, P = .50); the difference is 1.32 (P < .001). The image payoff, by contrast, is present for both congruent (0.73, P < .001) and incongruent portrayals (0.47, P = .003), and the two do not differ significantly (0.26, P = .12).
- *Headline sentiment.* Negative and intense headline tone are each associated with engagement in their own right: S = –0.22 (P = .001) and $|S|$ = 0.81 (P < .001; β = .08, as large as that of headline extremity). Sentiment is also the covariate that most reduces the headline extremity payoff (see below), because extreme headlines tend to use more intense language (S3.3).
- *Facial expression.* Facial valence is not associated with engagement (F = 0.06, P = .27), and more emotionally intense faces are associated with slightly *lower* engagement ($|F|$ = –0.17, P = .002). Facial expression leaves the image extremity payoff essentially unchanged, so image stance is not a proxy for the politician's expression (S3.4).

Valence. Neither headline valence (β_H = –0.05) nor image valence (β_I = 0.02) is significant.

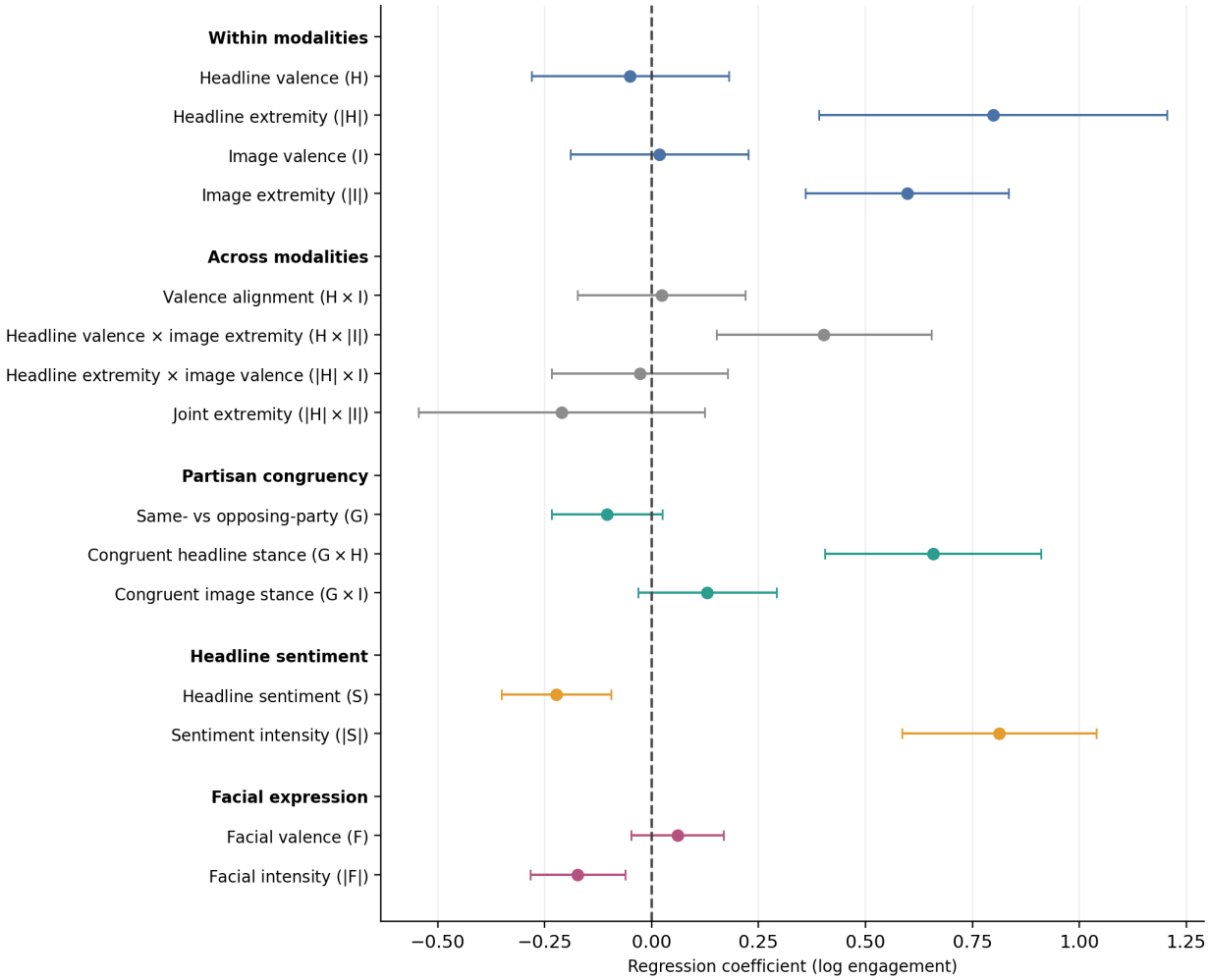


Fig. S14. Model (v): main RQ1 model. Coefficients with 95% CIs (outlet-clustered) from model (iv) with partisan congruency (G, $G \times H$, $G \times I$), headline sentiment (S, $|S|$), and facial expression (F, $|F|$); 176,556 articles from 45 Left/Right outlets. Controls are omitted from the plot; all estimates are in Table S12.

Table S12. Model (v): main RQ1 model

A. Coefficients.

Term	b	β	SE	95% CI	P
H	−0.05	−0.008	0.12	[−0.28, 0.18]	.668
$\|H\|$	0.80	0.081	0.21	[0.39, 1.20]	<.001
I	0.02	0.003	0.11	[−0.19, 0.23]	.864
$\|I\|$	0.60	0.042	0.12	[0.36, 0.84]	<.001
$H \times I$	0.02	0.002	0.10	[−0.17, 0.22]	.819
$H \times \|I\|$	0.40	0.031	0.13	[0.15, 0.65]	.002
$\|H\| \times I$	−0.03	−0.002	0.10	[−0.23, 0.18]	.795
$\|H\| \times \|I\|$	−0.21	−0.013	0.17	[−0.55, 0.12]	.218
G (same- vs opposing-party)	−0.10	−0.030	0.07	[−0.23, 0.03]	.114
$H_{cong} = G \times H$	0.66	0.094	0.13	[0.40, 0.91]	<.001
$I_{cong} = G \times I$	0.13	0.019	0.08	[−0.03, 0.29]	.116
S	−0.22	−0.034	0.07	[−0.35, −0.09]	<.001
$\|S\|$	0.81	0.082	0.12	[0.59, 1.04]	<.001

Term	b	β	SE	95% CI	P
F	0.06	0.012	0.06	[−0.05, 0.17]	.274
$\|F\|$	−0.17	−0.015	0.06	[−0.28, −0.06]	.002
Controls					
Leaning: Right	−1.02	−0.150	0.42	[−1.85, −0.19]	.016
Media type: Digital Native	−0.88	−0.129	0.55	[−1.97, 0.21]	.113
Media type: Magazine	−0.95	−0.073	0.41	[−1.74, −0.15]	.019
Media type: Network TV	−2.31	−0.229	0.67	[−3.62, −1.00]	<.001
Media type: Newspaper	−1.01	−0.110	0.50	[−1.99, −0.04]	.042
Year: 2018	0.05	0.005	0.10	[−0.14, 0.25]	.593
Year: 2019	0.62	0.073	0.17	[0.28, 0.95]	<.001
Year: 2020	1.14	0.150	0.24	[0.67, 1.62]	<.001
Year: 2021	−0.13	−0.016	0.36	[−0.83, 0.57]	.724
Constant					
Intercept	5.80		0.39	[5.03, 6.57]	<.001

B. Model fit.

Statistic	Value
Articles	176,556
Outlets (clusters)	45
R^2	0.118
Adjusted R^2	0.118
Residual SD (RMSE)	3.191
Log-likelihood	−455,391
AIC	910,832
BIC	911,084
Fixed effects (controls)	Left/Right leaning, media type, year
Random effects	None
Standard errors	Clustered by outlet (CR1)

Notes. OLS of log(1 + engagement); SEs clustered by outlet (CR1); no random effects. Controls (fixed effects): collapsed Left/Right leaning (reference Left), media type (reference Cable news), and publication year (reference 2017). Center outlets are excluded because congruency is undefined, and OANN because it has no sentiment scores. b = unstandardized coefficient (difference in log engagement); β = standardized coefficient, b · SD(x) / SD(y) in the estimation sample; 95% CIs and P from outlet-clustered SEs (normal approximation).

From model (iv) to model (v). The table below adds the model (v) covariates in blocks, on the same 176,556 articles (SEs clustered by outlet in parentheses; Table S12 gives all terms of model (v)):

Specification	$\|H\|$	$\|I\|$	H	I	$\|H\| \times \|I\|$	R^2
Model (iv), stance terms only	1.57 (0.27)	0.63 (0.12)	−0.33 (0.13)	−0.03 (0.09)	−0.16 (0.17)	.100
+ partisan congruency (M1)	1.11 (0.23)	0.61 (0.12)	−0.18 (0.12)	−0.01 (0.09)	−0.18 (0.17)	.109
+ congruency + facial expression	1.12 (0.23)	0.59 (0.12)	−0.18 (0.12)	−0.03 (0.11)	−0.18 (0.17)	.110
+ congruency + headline sentiment	0.80 (0.21)	0.62 (0.12)	−0.05 (0.12)	0.04 (0.09)	−0.21 (0.17)	.117
Model (v)	0.80 (0.21)	0.60 (0.12)	−0.05 (0.12)	0.02 (0.11)	−0.21 (0.17)	.118
Model (v), six emotion probabilities instead of F, $\|F\|$	0.80 (0.21)	0.60 (0.12)	−0.05 (0.12)	0.03 (0.10)	−0.21 (0.17)	.118

- **Headline extremity** shrinks by about half, from 1.57 to 0.80. Partisan congruency and headline sentiment each account for part of the drop; facial expression accounts for none of it.

- **Image extremity** barely changes (0.63 to 0.60).

The headline shrinkage reflects substantive overlap that model (v) is designed to separate: extreme headlines are more often congruent ($r[|H|, G \times H]$ = .52) and more intense in tone ($r[|H|, |S|]$ = .45; S3.3). It is not estimation instability: the standard error of $|H|$ falls from 0.27 to 0.21 as covariates are added (see also the collinearity diagnostics in S5.7).

The same pattern holds on the 47-outlet sample:

- adding S and $|S|$ to model (iv) reduces β|H| from 1.44 to 1.09 (–24%);
- adding F and $|F|$ changes β|I| by less than 7%.

S5.7. Robustness: collinearity

Purpose. Model (v) has 15 focal terms: valence and extremity for each modality, their four cross-modal products, partisan congruency, headline sentiment, and facial expression. Several are built from the same ratings, so they could be collinear enough to make individual coefficients unstable. We computed variance inflation factors (VIFs) for the 176,556 articles in model (v), using generalized VIFs (Fox & Monette, 1992) for the categorical controls, together with the condition number of the design matrix.

Terms	VIF in model (v)	VIF without cross-modal products
Headline valence H, extremity $\lvert H\rvert$	6.07, 6.01	1.86, 1.81
Image valence I, extremity $\lvert I\rvert$	2.78, 2.54	1.21, 1.06
Cross-modal products $H \times I$, $H \times \lvert I\rvert$, $\lvert H\rvert \times I$, $\lvert H\rvert \times \lvert I\rvert$	1.62, 5.60, 3.09, 7.30	—
Partisan congruency G, $G \times H$, $G \times I$	1.28, 1.45, 1.34	1.26, 1.44, 1.05
Headline sentiment S, $\lvert S\rvert$	1.83, 1.65	1.81, 1.64
Facial expression F, $\lvert F\rvert$	1.34, 1.13	1.33, 1.13
Controls: outlet leaning, media type, year (GVIF$^{1/(2df)}$)	1.19, 1.05, 1.01	—

Note. VIF = $1/(1 - R_j^2)$ from regressing term j on all other predictors. The model (v) column includes the controls; the last column drops the four cross-modal products and is computed among the focal terms (including the controls changes no VIF by more than 0.07). In the additive stance model alone (H, $\lvert H\rvert$, I, $\lvert I\rvert$), every VIF is at most 1.22.

No VIF reaches the conventional threshold of 10. The largest, 7.30, belongs to $\lvert H\rvert \times \lvert I\rvert$. The condition number is 20.2 for unit-length columns with the intercept (Belsley et al., 1980; values above 30 signal harmful collinearity) and 8.1 for standardized predictors. The measures themselves are not collinear:

- valence and extremity of the same modality correlate at *r* = –.41 for headlines and .07 for images;
- headline stance and headline sentiment correlate at *r* = .52 (S3.3);
- image stance and facial valence correlate at *r* = .35.

The moderately elevated VIFs come entirely from the cross-modal products. Because 82% of images are non-neutral ($\lvert I\rvert = 0.5$ or 1), $H \times \lvert I\rvert$ closely tracks H (*r* = .88), and $\lvert H\rvert \times \lvert I\rvert$ tracks $\lvert H\rvert$ (*r* = .78). This is the expected, nonessential collinearity between a product term and its components. It enlarges the standard errors of the individual components, by at most $\sqrt{7.3} \approx 2.7$ times, but it does not bias the estimates, and it does not affect the payoff contrasts, which are linear combinations of coefficients (S5.1). Adding the model (v) covariates to model (iv) also lowers, rather than inflates, the standard error of $\lvert H\rvert$ (S5.6).

S5.8. Robustness: inference with few clusters

Purpose. Standard errors are clustered by outlet, and model (v) has only 45 outlet clusters. With few clusters, clustered SEs can understate uncertainty and normal-approximation P values can be too small.

We re-tested every model (v) term with t_{44} critical values and a restricted wild cluster bootstrap-t (Table S13):

- the null hypothesis is imposed for each coefficient;
- residuals are perturbed with outlet-level Rademacher weights;
- B = 1,999 replications.

Every term that is significant under the normal approximation remains so under the bootstrap: |H| (P = .003), |I| (P < .001), $H \times |I|$ (P = .012), $G \times H$ (P < .001), S (P = .002), $|S|$ (P < .001), and $|F|$ (P = .006). In model (iv), both extremity terms (bootstrap P < .001), $H \times I$ (P = .001), and $H \times |I|$ (P = .008) remain significant, and headline valence is weaker but still significant (P = .028, vs. .018 under the normal approximation).

Table S13. Few-cluster inference for model (v)

Term	β	t (CR1)	P normal	P t(44)	P wild bootstrap	Model (iv): P wild bootstrap
H	–0.05	–0.43	.668	.670	.668	.028
$\|H\|$	0.80	3.85	<.001	<.001	.003	<.001
I	0.02	0.17	.864	.865	.865	
$\|I\|$	0.60	4.93	<.001	<.001	<.001	<.001
$H \times I$	0.02	0.23	.819	.820	.874	.001
$H \times \|I\|$	0.40	3.14	.002	.003	.012	.008
$\|H\| \times I$	–0.03	–0.26	.795	.796	.783	
$\|H\| \times \|I\|$	–0.21	–1.23	.218	.224	.257	
G	–0.10	–1.58	.114	.121	.200	
$G \times H$	0.66	5.10	<.001	<.001	<.001	
$G \times I$	0.13	1.57	.116	.123	.108	
S	–0.22	–3.42	<.001	.001	.002	
$\|S\|$	0.81	7.00	<.001	<.001	<.001	
F	0.06	1.09	.274	.280	.344	
$\|F\|$	–0.17	–3.06	.002	.004	.006	

Notes. Model (v) (176,556 articles, 45 outlets; Table S12) with outlet-clustered CR1 SEs. P t(44) uses t critical values with G – 1 = 44 degrees of freedom. Restricted wild cluster bootstrap-t: null imposed for each coefficient, Rademacher weights at the outlet level, B = 1,999. The last column gives the same bootstrap for model (iv) (183,516 articles, 48 outlets) where the term exists.

S5.9. Robustness: outcome model

Purpose. Engagement is heavily right-skewed, and the main models use OLS on log(1 + engagement). A few viral articles, or the log transformation itself, could produce the extremity payoff. We therefore re-estimated model (v) after trimming the top of the engagement distribution and with two count models on raw engagement, negative binomial and Poisson pseudo-maximum likelihood (PPML) with outlet and year fixed effects (Table S14; Fig. S15). The count models estimate proportional differences in *mean* engagement, so they give most weight to the most-engaged articles; log-OLS estimates differences in log engagement and is closer to a comparison of typical articles.

- *Top-tail trimming.* Dropping the top 1%–10% of engagement lowers β|H| from 0.80 to 0.73 and β|I| from 0.60 to 0.48. $H \times |I|$ (0.35–0.38, P ≤ .013), $G \times H$ (0.53–0.64), and the sentiment and facial terms all remain significant.
- *Negative binomial.* On raw counts (log E[count] scale), the payoffs are smaller but remain significant: β|H| = 0.33 (P = .004), β|I| = 0.32 (P < .001), $H \times |I|$ = 0.30 (P = .005), and $G \times H$ = 0.37 (P < .001). $|S|$ = 0.62 and $|F|$ = –0.08 keep their signs, but sentiment valence S drops to zero (–0.01, P = .81). Because the count model weights the most-engaged articles most heavily, the negativity term in the log-OLS model is driven by articles in the lower and middle part of the distribution.
- *PPML.* PPML is consistent for the conditional mean whatever the distribution of engagement, provided the mean is correctly specified (Santos Silva & Tenreyro, 2006). We estimated it with outlet and year fixed effects in place of the outlet-level controls. It gives the weakest extremity payoffs of any check. The headline payoff is β|H| = 0.30 (P = .023), or 35% higher mean engagement for extreme than for neutral headlines, and the image payoff is β|I| = 0.18 (P = .15). $G \times H$ (0.32, P < .001), $|S|$ (0.59, P < .001), and $|F|$ (–0.06, P = .032) remain significant; $H \times |I|$ is positive but not significant (0.25, P = .11), and S is again zero (0.03, P = .50).
- *Reading.* All outcome models agree on the direction of every extremity term. They differ in magnitude and precision. Extreme headlines and images are reliably associated with more engagement for the typical article (log-OLS, trimming, negative binomial). In mean engagement, which is dominated by a small number of viral articles, the headline payoff is smaller but still significant, and the image payoff cannot be distinguished from zero. Within outlet-weeks and politician-weeks, a PPML version of the additive stance model gives β|H| = 0.72 (P < .001) and β|I| = 0.18 (P = .002) (S5.11).
- *Categorical stance.* Negative binomial versions of models (i) and (ii) keep the U-shape: extreme headline categories exceed neutral by 0.79 (very unfavorable) and 0.76 (very favorable), and extreme image categories by 0.27 and 0.37 (Table S14B).

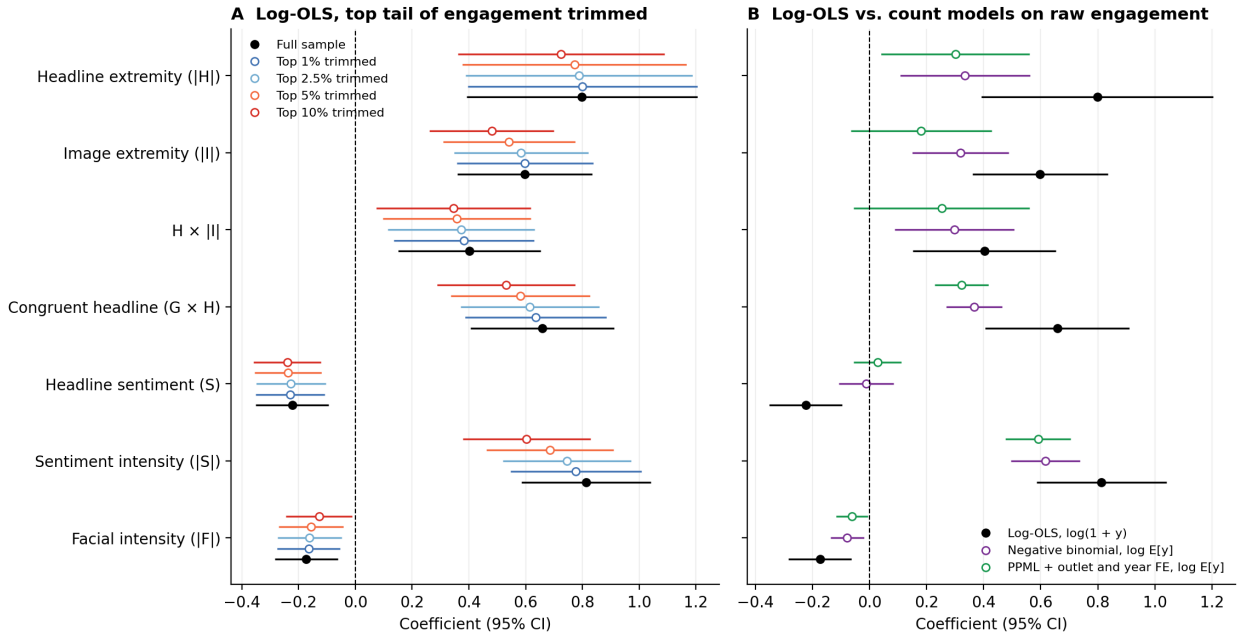


Fig. S15. Model (v) under alternative outcome models. Key model (v) terms with 95% CIs (outlet-clustered). (A) OLS on log(1 + engagement) in the full sample (filled) and after dropping the top 1%, 2.5%, 5%, or 10% of engagement (open). (B) Log-OLS baseline (filled) and two count models of raw engagement (open; log E[count] scale, so magnitudes are not directly comparable with log-OLS): negative binomial, and PPML with outlet and year fixed effects. Estimates are in Table S14A.

Table S14. Model (v) under alternative outcome models

A. Model (v) under alternative outcome models, β (P).

Term	Log-OLS (baseline)	Top 1% trimmed	Top 2.5% trimmed	Top 5% trimmed	Top 10% trimmed	Negative binomial (raw counts)	PPML, outlet and year FE (raw counts)
$\|H\|$	0.80 (<.001)	0.80 (<.001)	0.79 (<.001)	0.77 (<.001)	0.73 (<.001)	0.33 (.004)	0.30 (.023)
$\|I\|$	0.60 (<.001)	0.60 (<.001)	0.58 (<.001)	0.54 (<.001)	0.48 (<.001)	0.32 (<.001)	0.18 (.149)
$H \times \|I\|$	0.40 (.002)	0.38 (.002)	0.37 (.005)	0.36 (.007)	0.35 (.013)	0.30 (.005)	0.25 (.107)
$G \times H$	0.66 (<.001)	0.64 (<.001)	0.61 (<.001)	0.58 (<.001)	0.53 (<.001)	0.37 (<.001)	0.32 (<.001)
S	−0.22 (<.001)	−0.23 (<.001)	−0.23 (<.001)	−0.24 (<.001)	−0.24 (<.001)	−0.01 (.808)	0.03 (.504)
$\|S\|$	0.81 (<.001)	0.78 (<.001)	0.75 (<.001)	0.69 (<.001)	0.60 (<.001)	0.62 (<.001)	0.59 (<.001)
$\|F\|$	−0.17 (.002)	−0.16 (.004)	−0.16 (.005)	−0.16 (.008)	−0.13 (.031)	−0.08 (.009)	−0.06 (.032)
Articles	176,556	174,790	172,142	167,728	158,902	176,556	176,556

B. Negative binomial versions of models (i) and (ii), raw engagement counts.

Model	Contrast	β (log E[count])	SE	P
(i) Headline categorical	1 vs 3 (unfav.)	0.79	0.09	<.001
(i) Headline categorical	2 vs 3	0.39	0.05	<.001
(i) Headline categorical	4 vs 3	0.34	0.04	<.001
(i) Headline categorical	5 vs 3 (fav.)	0.76	0.10	<.001
(ii) Image categorical	1 vs 3 (unfav.)	0.27	0.12	.021
(ii) Image categorical	2 vs 3	0.16	0.04	<.001
(ii) Image categorical	4 vs 3	0.11	0.04	.011
(ii) Image categorical	5 vs 3 (fav.)	0.37	0.06	<.001

Notes. Panel A: model (v) terms and controls with outlet-clustered SEs (45 outlets). Trimmed columns: OLS on log(1 + engagement) after dropping articles above the 99th, 97.5th, 95th, or 90th percentile of engagement. Negative binomial and PPML (Poisson pseudo-maximum likelihood): raw counts, log E[count] scale, so coefficients are not on the same scale as the log-OLS columns. PPML replaces the outlet leaning, outlet type, and year controls with outlet and year fixed effects. Panel B: categorical stance with RQ1 controls and outlet-clustered SEs on 183,516 articles; contrasts are relative to the neutral category.

S5.10. Robustness: sample

Purpose. The analytic sample is dominated by a few outlets and politicians, and which articles entered it depends on deduplication and on whether the image could be downloaded (S1.7–S1.8). We therefore checked whether the results depend on any single outlet or politician, and whether they change when the sample is restricted or reweighted to address how articles entered it.

Influential outlets and politicians (Table S15A; Fig. S16). Model (v) was re-fitted 45 times, each time omitting one outlet, and 8 times, each time omitting one politician.

- *Outlets.* β|H| ranges from 0.64 to 0.88 and β|I| from 0.52 to 0.64. No key term changes sign or loses significance.
- *Breitbart.* Breitbart is the largest outlet (16,292 articles, 8.9% of the sample) and has the largest headline payoff (3.00; S6.3). Dropping it produces the largest single-outlet change in β|H|, from 0.80 to 0.64, still P < .001. In model (iv), the same drop moves β|H| from 1.44 to 1.26.
- *Politicians.* β|H| ranges from 0.78 to 0.92 and β|I| from 0.57 to 0.69. Dropping Biden (31% of articles) moves β|H| to 0.92 and $H \times |I|$ to 0.25 (P = .032), the largest change in any key term. Dropping Trump (42%) moves β|I| to 0.69. No key term loses significance.
- *Model (iv).* β|H| ranges from 1.26 to 1.49 across outlet fits and from 1.31 to 1.58 across politician fits.

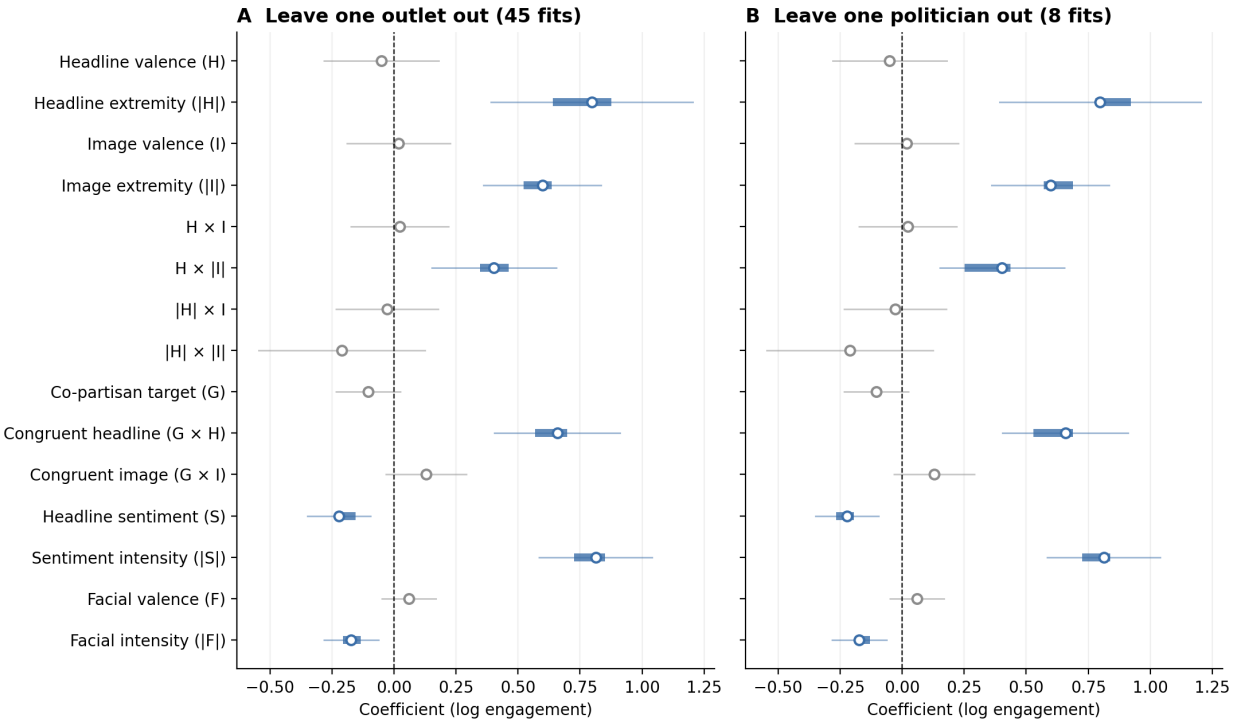


Fig. S16. Influence of individual outlets and politicians on model (v). (A) Leave one outlet out (45 fits); (B) leave one politician out (8 fits). Thick bars, range of the estimates across fits; open circles and thin lines, full-sample estimate and 95% CI. Blue, P < .05 in the full sample. Estimates are in Table S15A.

Sample composition (Table S15B).

Multi-politician images. Some images show more than one target politician. Deduplication assigns each such article to one of them by row order (S1.7). Restricting to the 168,841 model (v) articles with a single

matched politician leaves model (v) essentially unchanged: β|H| = 0.82, β|I| = 0.60, $H \times |I|$ = 0.39, and $G \times H$ = 0.66. Model (iv) is likewise unchanged (β|H| = 1.44, β|I| = 0.63; 175,584 articles).

Image availability and outlet inclusion. Image download success differs across outlets and years and is lower for right-leaning outlets (S1.8). If articles whose images survived differ from those whose images were lost, the outlet-years with the lowest download success are the most exposed. We re-estimated model (v) in four ways:

1. weighting each article by the inverse of its outlet-year download success, so that poorly covered outlet-years count as much as they would with complete images;
2. the same weights, capped at 10 so that a few outlet-years with under 10% success do not dominate;
3. restricting to outlet-years with at least 50% download success (141,663 articles, 37 outlets);
4. restricting to outlets with analytic articles in all five years (140,219 articles, 30 outlets).

The extremity effects are stable across all four checks: β|H| ranges from 0.78 to 0.94 (baseline 0.80) and β|I| from 0.53 to 0.62 (baseline 0.60), all P < .001. $G \times H$ (0.54–0.70), $|S|$, and $|F|$ remain significant in every check (all P ≤ .012). Two terms are less stable under uncapped inverse weights, where a handful of outlet-years with under 10% download success receive weights of up to about 200:

- $H \times |I|$ falls to 0.24 (P = .11) and holds in the other three checks (0.32–0.53, P ≤ .014);
- headline sentiment S falls to –0.14 (P = .09).

For model (iv), β|H| ranges from 1.39 to 1.61 and β|I| from 0.61 to 0.68 across the four checks (all P ≤ .002). The extremity results are therefore not an artifact of which images could be downloaded.

Table S15. Influential units and sample composition for model (v)

A. Leave-one-out influence.

Procedure	Term	Full	Range	Largest shift (dropped)	Sign flips	Lose P < .05	Model (iv): range
Leave one outlet out (45 fits)	$\|H\|$	0.80	0.64 to 0.88	–0.16 (Breitbart)	0	0	1.26 to 1.49
Leave one outlet out (45 fits)	$\|I\|$	0.60	0.52 to 0.64	–0.08 (CBS)	0	0	0.53 to 0.65
Leave one outlet out (45 fits)	$H \times \|I\|$	0.40	0.35 to 0.46	0.06 (Townhall)	0	0	0.31 to 0.43
Leave one outlet out (45 fits)	$G \times H$	0.66	0.57 to 0.70	–0.09 (Breitbart)	0	0	
Leave one outlet out (45 fits)	S	–0.22	–0.24 to –0.16	0.07 (Breitbart)	0	0	
Leave one outlet out (45 fits)	$\|S\|$	0.81	0.73 to 0.85	–0.09 (Breitbart)	0	0	
Leave one outlet out (45 fits)	$\|F\|$	–0.17	–0.21 to –0.13	0.04 (Breitbart)	0	0	
Leave one politician out (8 fits)	$\|H\|$	0.80	0.78 to 0.92	0.12 (Joe Biden)	0	0	1.31 to 1.58
Leave one politician out (8 fits)	$\|I\|$	0.60	0.57 to 0.69	0.09 (Donald Trump)	0	0	0.57 to 0.75
Leave one politician out (8 fits)	$H \times \|I\|$	0.40	0.25 to 0.44	–0.15 (Joe Biden)	0	0	0.26 to 0.43

Procedure	Term	Full	Range	Largest shift (dropped)	Sign flips	Lose P < .05	Model (iv): range
Leave one politician out (8 fits)	$G \times H$	0.66	0.53 to 0.69	−0.13 (Donald Trump)	0	0	
Leave one politician out (8 fits)	S	−0.22	−0.27 to −0.20	−0.04 (Joe Biden)	0	0	
Leave one politician out (8 fits)	$\|S\|$	0.81	0.73 to 0.84	−0.09 (Donald Trump)	0	0	
Leave one politician out (8 fits)	$\|F\|$	−0.17	−0.18 to −0.13	0.04 (Donald Trump)	0	0	

B. Sample composition and image coverage, β (P).

Term	Baseline	One matched politician	Weighted (1/success)	Weighted, capped at 10	Outlet-years ≥50% success	Outlets in all 5 years
$\|H\|$	0.80 (<.001)	0.82 (<.001)	0.78 (<.001)	0.78 (<.001)	0.87 (<.001)	0.94 (<.001)
$\|I\|$	0.60 (<.001)	0.60 (<.001)	0.53 (<.001)	0.59 (<.001)	0.60 (<.001)	0.62 (<.001)
$H \times \|I\|$	0.40 (.002)	0.39 (.003)	0.24 (.112)	0.32 (.014)	0.43 (.004)	0.53 (.001)
$G \times H$	0.66 (<.001)	0.66 (<.001)	0.55 (.002)	0.69 (<.001)	0.54 (<.001)	0.70 (<.001)
S	−0.22 (<.001)	−0.22 (.001)	−0.14 (.090)	−0.21 (<.001)	−0.25 (<.001)	−0.23 (.002)
$\|S\|$	0.81 (<.001)	0.80 (<.001)	1.12 (<.001)	0.84 (<.001)	0.75 (<.001)	0.81 (<.001)
$\|F\|$	−0.17 (.002)	−0.15 (.010)	−0.44 (.007)	−0.20 (.002)	−0.18 (<.001)	−0.16 (.012)
Articles / outlets	176,556 / 45	168,841 / 45	176,551 / 45	176,551 / 45	141,663 / 37	140,219 / 30
Model (iv): $\|H\|$	1.44 (<.001)	1.44 (<.001)	1.57 (<.001)	1.48 (<.001)	1.39 (<.001)	1.61 (<.001)
Model (iv): $\|I\|$	0.62 (<.001)	0.63 (<.001)	0.65 (.002)	0.68 (<.001)	0.61 (<.001)	0.65 (<.001)

Notes. All rows re-estimate model (v) (176,556 articles, 45 outlets; Table S12) with its controls and outlet-clustered SEs; model (iv) columns and rows give the corresponding stance-only estimates (183,516 articles, 48 outlets). Panel A: model (v) re-fitted omitting one outlet or one politician at a time; shift = estimate without the named unit minus the full-sample estimate. Panel B: one matched politician excludes articles whose image matched more than one of the eight politicians; download success = images with a detected face / NewsWhip articles for each outlet-year (Table S1); weights 1/success normalized to mean 1, the capped version sets success to at least 0.10. Model (iv) rows: baseline 183,516 articles; one matched politician 175,584 articles.

S5.11. Within-outlet and within-story comparisons

Purpose. Models (i)–(v) compare articles across outlets and years with outlet-level controls. The extremity payoffs could therefore reflect which outlets use extreme stance (for example, outlets with large or highly engaged audiences), which weeks carry extreme coverage (for example, weeks of high news salience), or which politicians and stories attract it. We re-estimated the payoffs with fixed effects that remove each of these sources of variation in turn (Table S16; Fig. S17):

- *outlet×week* fixed effects remove anything common to an outlet in a given week (audience size, platform changes, traffic shocks, the stories the outlet chose to cover);
- *politician×week* fixed effects remove anything common to a politician in a given week (news salience, scandals);
- *shared-story* fixed effects compare articles about the same story, defined as articles about the same politician within ±1 day whose headlines have TF-IDF cosine similarity of at least 0.5 and that were covered by at least two outlets (10,506 events covering 30,473 articles).

These models use the 181,471 articles that remain after removing the second copy of 997 NewsWhip articles that appear twice in the data and 1,048 articles whose publication year differs from their file year. The comparison is between articles from the same outlet in the same week, net of how much attention the politician received that week.

Results.

- *Model (v).* Within outlet-weeks and politician-weeks, β|H| is essentially unchanged (0.80 to 0.78) and β|I| falls from 0.60 to 0.47; both remain P < .001. $H \times |I|$ (0.28, P = .009), $G \times H$ (0.52, P < .001), S, and $|S|$ remain significant. Facial intensity $|F|$ shrinks to −0.03 (P = .34): its association with engagement in model (v) reflects differences between outlets or weeks rather than between articles within them.
- *Model (iv) and the additive stance model.* The headline payoff barely moves as fixed effects are added (additive model: 1.42 with controls, 1.31 within outlet-weeks and politician-weeks, 1.32 within outlet×politician×weeks). The image payoff falls more, from 0.55 to 0.36, but remains significant in every specification.
- *Shared stories.* Comparing outlets' coverage of the same story lowers the headline payoff to about 1.00 in the additive model (1.27 in model iv) and leaves the image payoff at about 0.41 (0.59 in model iv). The estimates are similar at cosine thresholds of 0.4 and 0.6.
- *Count model.* PPML on raw counts with outlet×week and politician×week fixed effects (additive model) gives β|H| = 0.72 (P < .001) and β|I| = 0.18 (P = .002).

The extremity payoffs are therefore not explained by which outlets, weeks, politicians, or stories carry extreme content. These are still conditional associations: the fixed effects do not remove article-level features that co-occur with extreme stance, such as the topic angle within a story, the time of day, or how sensational the story itself is.

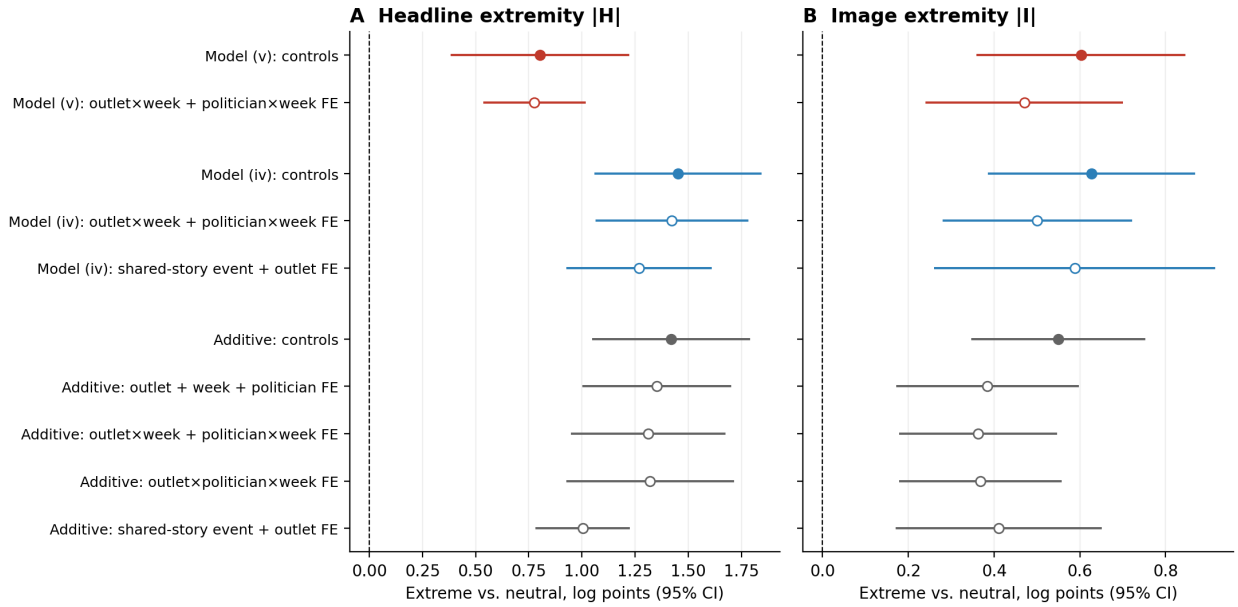


Fig. S17. Headline and image extremity payoffs as fixed effects are added. Differences in log(1 + engagement) between extreme and neutral stance, with 95% CIs (outlet-clustered). Filled circles, controls only (MBFC leaning, outlet type, year); open circles, fixed-effect models. Red, model (v) terms; blue, model (iv) terms; gray, additive model (H, $|H|$, I, $|I|$). Shared-story rows use cosine ≥ 0.5. Estimates are in Table S16.

Table S16. RQ1 extremity payoffs within outlet-weeks and within shared stories

A. Headline and image extremity payoffs as fixed effects are added.

Terms	Fixed effects	IHI (SE)	P IHI	III (SE)	P III	Articles	Outlets
Model (v)	Controls	0.80 (0.21)	<.001	0.60 (0.12)	<.001	174,584	45
Model (v)	Outlet×week + politician×week FE	0.78 (0.12)	<.001	0.47 (0.11)	<.001	174,170	45
Model (iv)	Controls	1.45 (0.19)	<.001	0.63 (0.12)	<.001	181,471	48
Model (iv)	Outlet×week + politician×week FE	1.42 (0.18)	<.001	0.50 (0.11)	<.001	181,029	48
Model (iv)	Outlet×month + politician×month FE	1.43 (0.17)	<.001	0.48 (0.11)	<.001	181,420	48
Model (iv)	Shared-story event + outlet FE	1.27 (0.17)	<.001	0.59 (0.16)	<.001	30,471	47
Additive	Controls	1.42 (0.18)	<.001	0.55 (0.10)	<.001	181,471	48
Additive	Outlet + week + politician FE	1.35 (0.17)	<.001	0.38 (0.10)	<.001	181,471	48
Additive	Outlet×week + politician×week FE	1.31 (0.18)	<.001	0.36 (0.09)	<.001	181,029	48
Additive	Outlet×politician×week + politician×week FE	1.32 (0.19)	<.001	0.37 (0.09)	<.001	169,018	48
Additive	Shared-story event + outlet FE (cosine ≥ 0.4)	1.10 (0.11)	<.001	0.41 (0.10)	<.001	43,387	48

Terms	Fixed effects	IHI (SE)	P IHI	III (SE)	P III	Articles	Outlets
Additive	Shared-story event + outlet FE (cosine ≥ 0.5)	1.00 (0.11)	<.001	0.41 (0.12)	.001	30,471	47
Additive	Shared-story event + outlet FE (cosine ≥ 0.6)	1.00 (0.12)	<.001	0.44 (0.13)	.002	20,583	47

B. Model (v) key terms, β (P).

Term	Controls only	Outlet×week + politician×week FE
$\|H\|$	0.80 (<.001)	0.78 (<.001)
$\|I\|$	0.60 (<.001)	0.47 (<.001)
$H \times \|I\|$	0.40 (.003)	0.28 (.009)
$G \times H$	0.66 (<.001)	0.52 (<.001)
S	−0.22 (.002)	−0.13 (.002)
$\|S\|$	0.82 (<.001)	0.82 (<.001)
$\|F\|$	−0.17 (.004)	−0.03 (.335)

Notes. OLS on log(1 + engagement) with outlet-clustered SEs, estimated on the longitudinal sample (181,471 articles; the second copy of 997 NewsWhip articles that appear twice and 1,048 articles whose publication year differs from their file year removed from the 183,516 articles). Controls: MBFC leaning, outlet type, year. Additive: H, IHI, I, III. Model (iv) and (v) terms as in S5.1; controls absorbed by the fixed effects are dropped. Singleton cells are dropped, so articles differ slightly across rows. Shared-story events: articles about the same politician within ±1 day whose headlines have TF-IDF cosine similarity at or above the threshold, covered by at least two outlets. Model (iv) event row uses cosine ≥ 0.5.

S6. Heterogeneity Across Politicians and News Outlets (RQ2)

RQ2 asks how the prevalence and engagement payoff of extreme stances vary across politicians and news outlets. This section estimates the model (iv) extremity payoffs separately for each of the eight politicians and each of the 47 outlets with at least 100 articles. S7 then asks whether outlet payoffs are associated with how often the outlet publishes extreme stances (prevalence) and with other outlet characteristics.

- **Politicians:** the headline payoff is positive for all eight politicians, and the image payoff for five (S6.2; Fig. S18; Table S17).
- **Outlets:** the headline payoff is positive for 46 of 47 outlets and the image payoff for 31 (S6.3; Fig. S19; Table S18).

Additional analyses. Separate within-politician models check the pooled politician estimates (Table S17).

S6.1. Model setup

Both analyses estimate the model (iv) payoffs separately for each politician or outlet. Each uses a single pooled model in which every stance term is interacted with politician or outlet indicators. Pooling keeps the controls common across groups and allows the group-specific slopes to be compared within one model.

Politician-specific model. With p indexing politicians and Donald Trump as the reference:

$$Y_i = \sum_p \mathbb{1}[p_i = p]\Big(\alpha_p + \beta_{H,p}H_i + \beta_{|H|,p}|H_i| + \beta_{I,p}I_i + \beta_{|I|,p}|I_i| + \beta_{HI,p}H_iI_i + \beta_{H|I|,p}H_i|I_i| + \beta_{|H|I,p}|H_i|I_i + \beta_{|H||I|,p}|H_i||I_i|\Big) + \gamma'\mathbf{X}_i + \varepsilon_i.$$

- Controls $\mathbf{X}_i$: outlet partisan leaning (MBFC, five categories), media type, and publication year, as in model (iv). Politicians are not nested in outlets, so the leaning control keeps politician payoffs from reflecting which outlets cover each politician.
- Standard errors are clustered by outlet.
- Sample: 183,516 articles, 48 outlets.
- Each politician's payoff $\beta_{|H|,p}$ (and $\beta_{|I|,p}$) is the reference slope plus that politician's interaction, with SE computed from the clustered covariance matrix.

For comparison, Table S17 also reports estimates from a separate model (iv) fitted within each politician's articles.

Outlet-specific model. The same structure, with outlet k in place of politician p:

- Controls: politician and year fixed effects. Outlet leaning and media type are absorbed by the outlet intercepts α_k.
- Standard errors are heteroskedasticity-robust (HC1), not clustered, because outlet is the unit whose heterogeneity is being estimated.
- Eligibility: outlets need at least 100 analytic articles in total; the threshold counts all articles, not extreme-headline articles. OANN (29 articles) is the only outlet excluded.
- Sample: 47 outlets, 183,487 articles.

Estimands. For politician p (or outlet k), $\beta_{|H|,p}$ is the difference in log engagement between an extreme and a neutral headline when the image is neutral, and $\beta_{|I|,p}$ is the analogous image payoff. The outlet payoffs $\widehat{\beta}_{|H|,k}$ also serve as outcomes in S7.

S6.2. Politicians: prevalence and payoffs

Prevalence. Extreme-headline shares range from 8.8% (Pence) to 18.6% (Biden), and extreme-image shares from 2.5% (McConnell) to 17.9% (Obama). Trump accounts for 42% of articles and Biden for 31%; each of the other six accounts for 3.6%–6.6%.

Headline payoff. The headline payoff is positive and significant for all eight politicians, ranging from 1.20 (Biden) to 1.81 (Obama) (Table S17). Separate within-politician models give similar estimates (1.18–1.78).

Image payoff. The image payoff is positive for seven of eight politicians and significant for five: Trump, Biden, Pelosi, Obama, and McConnell (Table S17). It is not significant for Pence, Warren, or Sanders (the only negative estimate, –0.03). Obama's pooled-model image payoff (1.14) is about twice his separate-model estimate (0.61), so it is sensitive to specification.

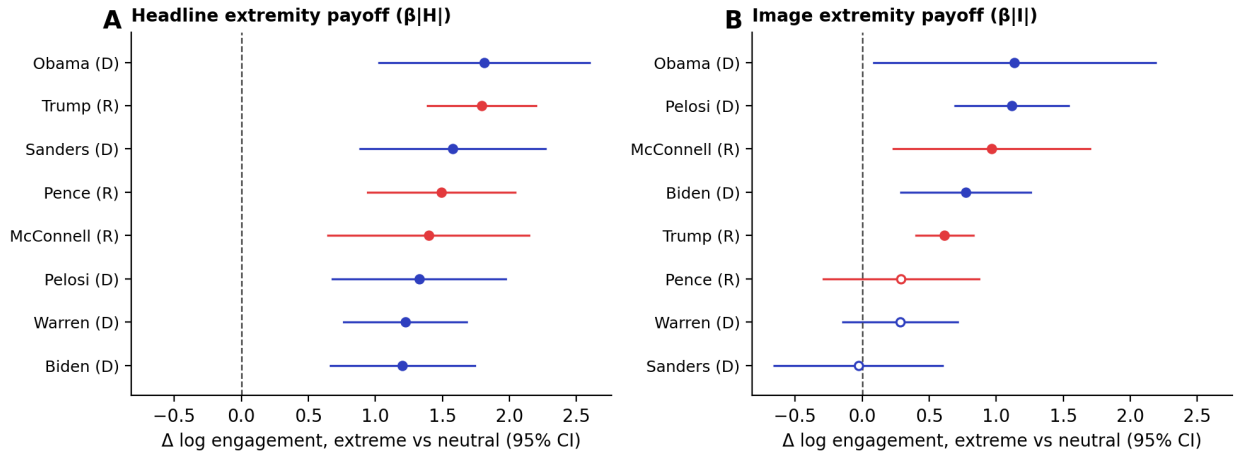


Fig. S18. Politician-specific engagement payoffs. (A) Headline extremity payoff βIHI and (B) image extremity payoff βIII for each politician, with 95% CIs, from the pooled politician-interacted model (S6.1). Color = party.

Table S17. Politician-specific extreme-stance prevalence and engagement payoffs

Politician	Party	Articles	Extreme H (%)	Extreme I (%)	βIHI [95% CI]	P (IHI)	βIII [95% CI]	P (III)	Separate-model βIHI	Separate-model βIII
Donald Trump	R	76,984	16.7	6.3	1.79 [1.39, 2.20]	<.001	0.62 [0.40, 0.83]	<.001	1.78	0.60
Joe Biden	D	56,998	18.6	5.1	1.20 [0.67, 1.74]	<.001	0.77 [0.29, 1.26]	.002	1.22	0.77
Nancy Pelosi	D	12,145	15.3	3.2	1.33 [0.68, 1.97]	<.001	1.12 [0.69, 1.54]	<.001	1.26	1.04
Bernie Sanders	D	8,080	10.4	8.0	1.58 [0.89, 2.27]	<.001	–0.03 [–0.66, 0.60]	.930	1.49	–0.07
Barack Obama	D	7,883	16.8	17.9	1.81 [1.03, 2.60]	<.001	1.14 [0.09, 2.19]	.034	1.41	0.61
Mitch McConnell	R	7,623	12.1	2.5	1.40 [0.65, 2.15]	<.001	0.97 [0.23, 1.70]	.010	1.53	1.03
Mike Pence	R	7,275	8.8	3.2	1.49 [0.94, 2.04]	<.001	0.29 [–0.29, 0.87]	.327	1.42	0.34
Elizabeth Warren	D	6,528	15.3	9.3	1.22 [0.76, 1.68]	<.001	0.28 [–0.14, 0.71]	.194	1.18	0.27

Notes. βIHI (βIII) = politician-specific difference in log engagement between an extreme and a neutral

headline (image), with the other modality neutral, from one pooled model interacting all terms of model (iv) with politician indicators; controls: outlet partisan leaning (MBFC), outlet type, and year; SEs clustered by outlet (Trump = reference, its slopes are the main effects). Headline payoffs are positive and significant for all eight politicians. Image payoffs are positive for 7 of 8 and significant ($P < .05$) for 5 of 8. Separate-model columns re-estimate model (iv) within each politician's articles (same controls) as a check.

S6.3. Outlets: prevalence and payoffs

Prevalence. Among the 47 outlets, extreme-headline shares range from 2.1% (CBS, PBS) to 59.6% (RedState), and extreme-image shares from 2.7% to 13.0%. Extreme-headline articles per outlet range from 41 to 1,976; four outlets have fewer than 100.

Headline payoff. The headline payoff is positive for 46 of 47 outlets and significant for 33 (Table S18). The only negative estimate is Daily Caller's (−0.84). The largest are Breitbart (3.00), ABC (2.81), and USA Today (2.28).

Image payoff. The image payoff is positive for 31 of 47 outlets but significant for only 10, and less precisely estimated. Estimates range from −2.19 (The Federalist) to 3.19 (The Hill, with wide uncertainty).

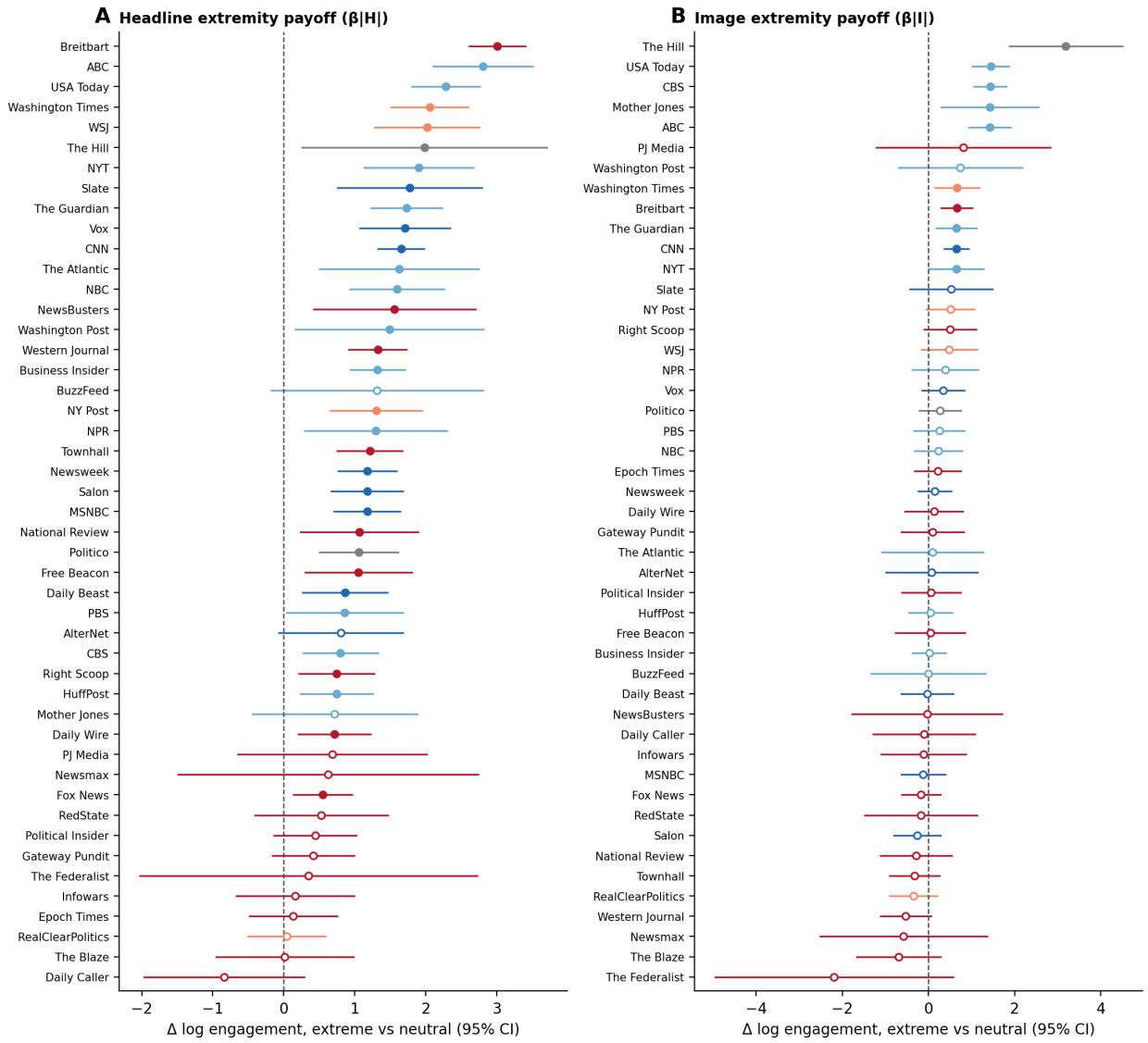


Fig. S19. Outlet-specific engagement payoffs. (A) Headline extremity payoff β|H| and (B) image extremity payoff β|I| for the 47 outlets with at least 100 articles, with 95% CIs, from the pooled outlet-interacted model (S6.1). Filled markers, $P < .05$; color, MBFC leaning. OANN (29 articles) is excluded.

Table S18. Outlet-specific extreme-stance prevalence and engagement payoffs

Outlet	Leaning	Articles	Extreme H: n (%)	Extreme I: n (%)	βIHI [95% CI]	βIII [95% CI]
Breitbart	Extreme right	16,292	1,763 (10.8)	971 (6.0)	3.00 [2.61, 3.40]	0.66 [0.30, 1.02]
CNN	Left	13,958	1,383 (9.9)	648 (4.6)	1.65 [1.33, 1.98]	0.65 [0.37, 0.93]
CBS	Center-left	8,819	188 (2.1)	379 (4.3)	0.80 [0.27, 1.32]	1.43 [1.05, 1.81]
Fox News	Right	8,020	1,268 (15.8)	331 (4.1)	0.55 [0.14, 0.96]	−0.17 [−0.62, 0.27]
Newsweek	Left	7,189	1,077 (15.0)	539 (7.5)	1.18 [0.76, 1.59]	0.15 [−0.23, 0.53]
Washington Times	Center-right	7,164	746 (10.4)	474 (6.6)	2.05 [1.51, 2.60]	0.67 [0.16, 1.17]
USA Today	Center-left	7,154	586 (8.2)	510 (7.1)	2.28 [1.80, 2.76]	1.44 [1.02, 1.87]
HuffPost	Center-left	6,881	1,665 (24.2)	666 (9.7)	0.74 [0.24, 1.25]	0.05 [−0.46, 0.55]
Townhall	Right	6,492	1,976 (30.4)	346 (5.3)	1.21 [0.75, 1.67]	−0.32 [−0.90, 0.25]
ABC	Center-left	6,399	182 (2.8)	323 (5.0)	2.80 [2.10, 3.51]	1.42 [0.93, 1.91]
MSNBC	Left	6,196	903 (14.6)	204 (3.3)	1.17 [0.71, 1.64]	−0.13 [−0.64, 0.39]
Western Journal	Right	6,038	1,913 (31.7)	328 (5.4)	1.32 [0.91, 1.73]	−0.53 [−1.11, 0.05]
NBC	Center-left	5,909	309 (5.2)	293 (5.0)	1.59 [0.93, 2.26]	0.23 [−0.33, 0.79]
Business Insider	Center-left	5,703	775 (13.6)	422 (7.4)	1.32 [0.94, 1.70]	0.02 [−0.37, 0.41]
Politico	Center	5,462	310 (5.7)	308 (5.6)	1.06 [0.51, 1.61]	0.27 [−0.21, 0.75]
Epoch Times	Right	4,530	262 (5.8)	129 (2.8)	0.13 [−0.48, 0.75]	0.21 [−0.32, 0.75]
Salon	Left	4,328	1,576 (36.4)	511 (11.8)	1.17 [0.67, 1.68]	−0.27 [−0.81, 0.28]
RealClearPolitics	Center-right	4,046	584 (14.4)	110 (2.7)	0.04 [−0.51, 0.59]	−0.35 [−0.90, 0.21]
NYT	Center-left	3,762	317 (8.4)	352 (9.4)	1.90 [1.13, 2.67]	0.64 [0.01, 1.28]
The Guardian	Center-left	3,660	643 (17.6)	317 (8.7)	1.73 [1.23, 2.23]	0.65 [0.18, 1.12]
National Review	Right	3,423	740 (21.6)	182 (5.3)	1.06 [0.23, 1.89]	−0.29 [−1.12, 0.54]
WSJ	Center-right	3,069	229 (7.5)	158 (5.1)	2.02 [1.28, 2.75]	0.48 [−0.16, 1.13]
NY Post	Center-right	2,871	476 (16.6)	173 (6.0)	1.30 [0.65, 1.95]	0.51 [−0.06, 1.07]
Daily Beast	Left	2,779	649 (23.4)	314 (11.3)	0.86 [0.26, 1.46]	−0.03 [−0.63, 0.57]
Right Scoop	Right	2,517	800 (31.8)	155 (6.2)	0.74 [0.21, 1.27]	0.50 [−0.11, 1.10]
Daily Wire	Right	2,477	819 (33.1)	153 (6.2)	0.71 [0.20, 1.22]	0.13 [−0.54, 0.80]
Gateway Pundit	Extreme right	2,325	1,248 (53.7)	271 (11.7)	0.42 [−0.16, 0.99]	0.09 [−0.64, 0.82]
RedState	Right	2,213	1,318 (59.6)	101 (4.6)	0.53 [−0.41, 1.47]	−0.18 [−1.48, 1.13]
Free Beacon	Right	2,137	390 (18.2)	116 (5.4)	1.05 [0.30, 1.80]	0.04 [−0.76, 0.85]
Vox	Left	1,990	372 (18.7)	157 (7.9)	1.71 [1.07, 2.34]	0.34 [−0.15, 0.84]
PBS	Center-left	1,986	41 (2.1)	125 (6.3)	0.86 [0.04, 1.67]	0.25 [−0.33, 0.84]
AlterNet	Left	1,626	660 (40.6)	73 (4.5)	0.80 [−0.07, 1.68]	0.07 [−0.99, 1.14]
Daily Caller	Right	1,557	256 (16.4)	80 (5.1)	−0.84 [−1.97, 0.29]	−0.10 [−1.29, 1.08]
PJ Media	Right	1,478	723 (48.9)	60 (4.1)	0.68 [−0.65, 2.01]	0.81 [−1.21, 2.83]
The Hill	Center	1,469	119 (8.1)	87 (5.9)	1.98 [0.26, 3.70]	3.19 [1.88, 4.50]
The Blaze	Right	1,302	326 (25.0)	128 (9.8)	0.01 [−0.95, 0.98]	−0.69 [−1.67, 0.28]
Slate	Left	1,248	347 (27.8)	87 (7.0)	1.77 [0.76, 2.79]	0.53 [−0.43, 1.49]
NPR	Center-left	1,222	45 (3.7)	78 (6.4)	1.30 [0.30, 2.29]	0.39 [−0.38, 1.16]
Mother Jones	Center-left	1,149	331 (28.8)	78 (6.8)	0.72 [−0.44, 1.87]	1.42 [0.29, 2.56]
NewsBusters	Right	1,116	439 (39.3)	47 (4.2)	1.56 [0.42, 2.69]	−0.03 [−1.77, 1.71]
Political Insider	Extreme right	1,095	389 (35.5)	100 (9.1)	0.44 [−0.14, 1.02]	0.06 [−0.63, 0.74]
Washington Post	Center-left	953	167 (17.5)	81 (8.5)	1.49 [0.16, 2.81]	0.74 [−0.69, 2.17]
Infowars	Extreme right	932	294 (31.5)	67 (7.2)	0.16 [−0.67, 0.99]	−0.12 [−1.10, 0.87]
Newsmax	Right	749	50 (6.7)	32 (4.3)	0.62 [−1.49, 2.73]	−0.58 [−2.51, 1.36]
The Atlantic	Center-left	731	145 (19.8)	49 (6.7)	1.62 [0.50, 2.74]	0.09 [−1.09, 1.27]
BuzzFeed	Center-left	663	86 (13.0)	86 (13.0)	1.31 [−0.18, 2.80]	−0.00 [−1.33, 1.33]
The Federalist	Right	408	156 (38.2)	35 (8.6)	0.35 [−2.03, 2.72]	−2.19 [−4.96, 0.57]
oann	Right	29	0 (0.0)	4 (13.8)	excluded	excluded

Notes. Sorted by number of articles. Payoffs from one pooled model interacting all terms of model (iv) with outlet indicators (CNN = reference), with politician and year fixed effects and HC1 robust SEs. Headline payoff is positive for 46 of 47 outlets (significant for 33); image payoff is positive for 31 of 47 (significant for 10). OANN (29 articles) was excluded from outlet models (threshold: ≥100 articles). Extreme-stance counts are among analytic articles.

S7. Extreme-Stance Prevalence, Outlet Characteristics, and Engagement Payoffs

This section asks whether outlet characteristics are associated with the size of an outlet's extremity payoffs, using the outlet-specific payoffs from S6 (RQ2). The main characteristic is the outlet's extreme-stance prevalence (S7.2). Credibility, audience ideology, and audience size come from external sources (S3.2); with 47 outlets and strongly overlapping characteristics, those analyses are exploratory (S7.4–S7.6). Each is reported for the headline payoff, shown in main-text Fig. 3, and for the image payoff.

Reported in the main text (Fig. 3).

- **Prevalence (Fig. 3A).** Outlets that publish extreme headlines more often have smaller headline payoffs; there is no comparable pattern for images (S7.2; Table S20).
- **Credibility (Fig. 3B).** Headline payoffs are larger for outlets with higher NewsGuard scores (S7.4).
- **Audience ideology (Fig. 3C).** Headline payoffs follow an inverted U across audience ideology (S7.5).
- **Audience size (Fig. 3D).** Headline payoffs are larger for outlets with more Facebook followers (S7.6).
- full coefficient tables for the prevalence–payoff models (S7.2; Table S20);
- the prevalence–payoff comparison for politicians and for outlet images (S7.2; Figs. S20 and S21);
- correlations among the outlet characteristics (S7.1; Table S19);
- for each characteristic, the image-payoff counterpart of Fig. 3B–D and article-level mixed models for both payoffs (S7.4–S7.6).

Robustness. S7.3 tests whether the prevalence–payoff association depends on imprecise outlet payoffs or on individual outlets, including Breitbart (Table S21; Fig. S22).

Summary and within-outlet comparison. S7.7 summarizes whether credibility, audience ideology, or audience size accounts for the prevalence–payoff association. S7.8 asks whether the association holds within outlets over time, holding stable outlet characteristics fixed (Table S25; Fig. S26).

S7.1. Model setup

Prevalence. The main characteristic is the outlet's extreme-stance share: the percentage of its articles with an extreme headline (share$_k^H$) or an extreme image (share$_k^I$). Shares are expressed per +10 percentage points (pp). The sample is the 47 outlets of the outlet-specific model (S6.1; 183,487 articles). Three analyses relate share to the outlet payoffs.

Outlet-level correlation. Spearman correlation between $\hat{\beta}_{|H|,k}$ and share$_k^H$ (and between $\hat{\beta}_{|I|,k}$ and share$_k^I$).

Outlet-level regression. OLS across the 47 outlets,

$$\hat{\beta}_{|H|,k} = a + b\,\text{share}_k^H + e_k,$$

with heteroskedasticity-robust (HC3) SEs and t_{45} inference. The image analogue regresses $\widehat{\beta}_{|I|,k}$ on share_k^I. Because the outcome is itself an estimate whose precision varies across outlets, the HC3 SEs allow the residual variance to differ across outlets.

Article-level mixed model. This model uses the articles directly instead of the estimated payoffs. For article i in outlet k,

$$Y_{ik} = \text{model (iv) terms} + \theta_H\,\text{share}_k^H + \theta_I\,\text{share}_k^I + \lambda_H|H_{ik}|\,\text{share}_k^H + \lambda_I|I_{ik}|\,\text{share}_k^I + \gamma'\mathbf{X}_{ik} + u_{0k} + u_{1k}|H_{ik}| + u_{2k}|I_{ik}| + \varepsilon_{ik}.$$

- $\mathbf{X}_{ik}$ contains the RQ1 controls (S5.1) and politician fixed effects.
- (u_{0k}, u_{1k}, u_{2k}) are an outlet random intercept and random slopes on $|H|$ and $|I|$, with unstructured covariance.
- Shares are centered at their unweighted 47-outlet means (20.6% for headlines, 6.6% for images).
- The model is estimated by REML.
- λ_H is the change in the headline payoff per +10 pp of extreme-headline share, and λ_I the image analogue.

Other outlet characteristics (measures defined in S3.2).

Characteristic	Outlet-level scaling	Mixed-model scaling	Outlets
NewsGuard score	per 10 points	centered, per 10 points	47
Bakshy audience alignment	raw	centered	45
Facebook followers	log10 (per tenfold increase)	log, standardized	47

The Bakshy alignment sample is smaller because data are missing for two outlets. The same three analyses are applied to each characteristic Z_k, for the headline payoff $\widehat{\beta}_{|H|,k}$ and the image payoff $\widehat{\beta}_{|I|,k}$:

- **Outlet-level correlation.** Spearman correlation of Z_k with each payoff.
- **Outlet-level regression.** OLS of each payoff on Z_k, with HC3 SEs and t inference. For alignment, a quadratic $\widehat{\beta}_k = a + b_1 Z_k + b_2 Z_k^2$ is also fitted, and the curvature is tested by the HC3 t test on b_2.
- **Article-level mixed model.** The mixed model above, with Z_k added as a main effect and interacted with both $|H|$ and $|I|$ (alongside the share interactions), on the outlets for which Z is available.

Overlap among characteristics. Outlets with high extreme-headline shares tend to have lower NewsGuard scores (ρ = −.42), more one-sided audiences (|alignment|, ρ = .72), and smaller Facebook audiences (ρ = −.55). NewsGuard scores are also lower for more conservative audiences (ρ = −.52 with alignment) and for more one-sided ones (ρ = −.62 with |alignment|) (Table S19). Because of this overlap, no association in S7.4–S7.6 can be attributed to a single characteristic.

Table S19. Spearman correlations among outlet coverage, prevalence, payoff, and characteristics

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1) Total articles	—								
(2) Extreme-headline articles	0.59***	—							

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(3) Extreme-headline share	−0.36*	0.48***	—						
(4) Extreme-image share	−0.24	−0.03	0.22	—					
(5) Headline payoff βIHI	0.30*	−0.00	−0.38**	0.11	—				
(6) Image payoff βIII	0.19	−0.20	−0.39**	0.03	0.55***	—			
(7) log Facebook followers	0.27	−0.19	−0.55***	0.06	0.38**	0.27	—		
(8) NewsGuard score	0.10	−0.25	−0.42**	0.25	0.59***	0.30*	0.42**	—	
(9) Audience alignment (Bakshy)	0.00	0.25	0.30*	−0.26	−0.30*	−0.28	−0.33*	−0.52***	—
(10) |Audience alignment|	−0.20	0.44**	0.72***	−0.12	−0.43**	−0.45**	−0.56***	−0.62***	0.52***

Notes. Outlets in the outlet-level sample (n = 47; pairwise n = 45 with Bakshy alignment, including followers × alignment). * P < .05, ** P < .01, *** P < .001. Pairwise n and exact P values are in the source file.

S7.2. Extreme-stance prevalence and payoff

Table S20 reports the full outlet-level OLS estimates for headlines and images (panel A) and all coefficients of the article-level mixed model (panel B).

Headlines. Outlets that publish extreme headlines more often have smaller headline payoffs:

- outlet-level correlation: ρ = –.38 (P = .008);
- outlet-level OLS: –0.20 per +10 pp of share (95% CI [–0.32, –0.08], P = .002); an outlet at the average share (20.6%) has an estimated headline payoff of 1.13;
- article-level mixed model: –0.18 per +10 pp (95% CI [–0.32, –0.05], P = .009).

The mixed model also shows substantial between-outlet variation in the headline payoff: the SD of the random |H| slope is 0.66.

Images. There is no comparable relationship for images (ρ = .03, P = .83; Fig. S21). The slopes are negative but imprecise: –0.37 per +10 pp (95% CI [–1.07, 0.32], P = .29) in outlet-level OLS and –0.35 (95% CI [–1.00, 0.30], P = .29) in the mixed model.

Politicians. Fig. S20 shows the same comparison across the eight politicians, for headlines and images. With eight politicians, this comparison is descriptive.

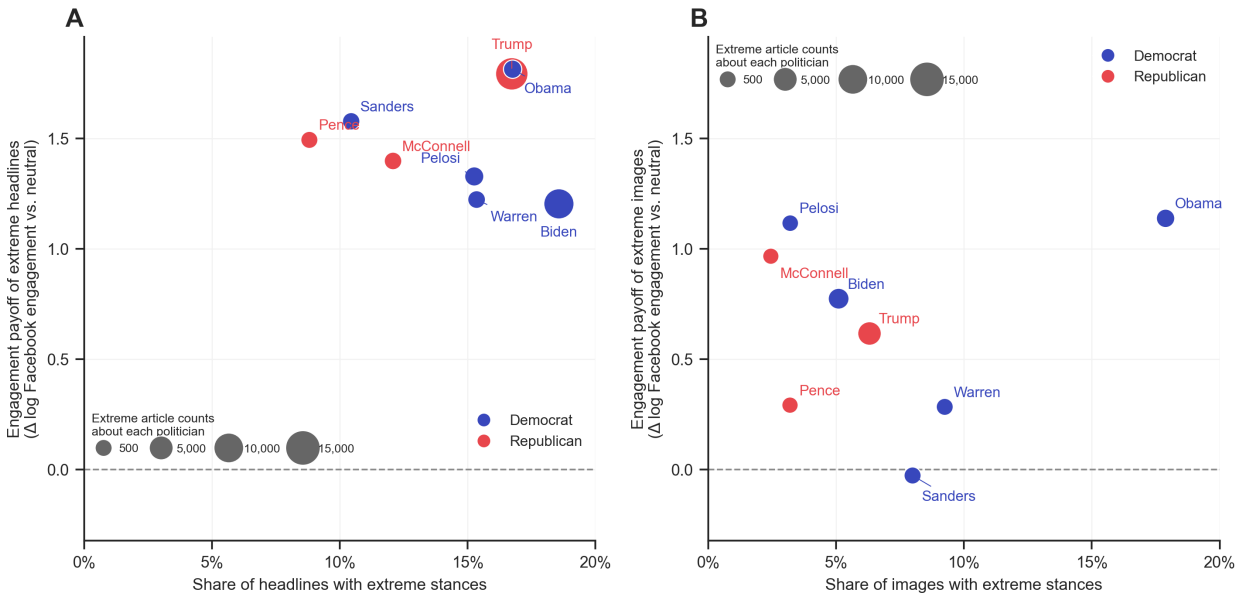


Fig. S20. Extreme-stance prevalence and engagement payoff across politicians. Share of (A) headlines or (B) images with extreme stances about each politician versus the corresponding extremity payoff from the politician-specific model (S6.1; Table S17). Marker size = number of extreme articles about the politician; color = party.

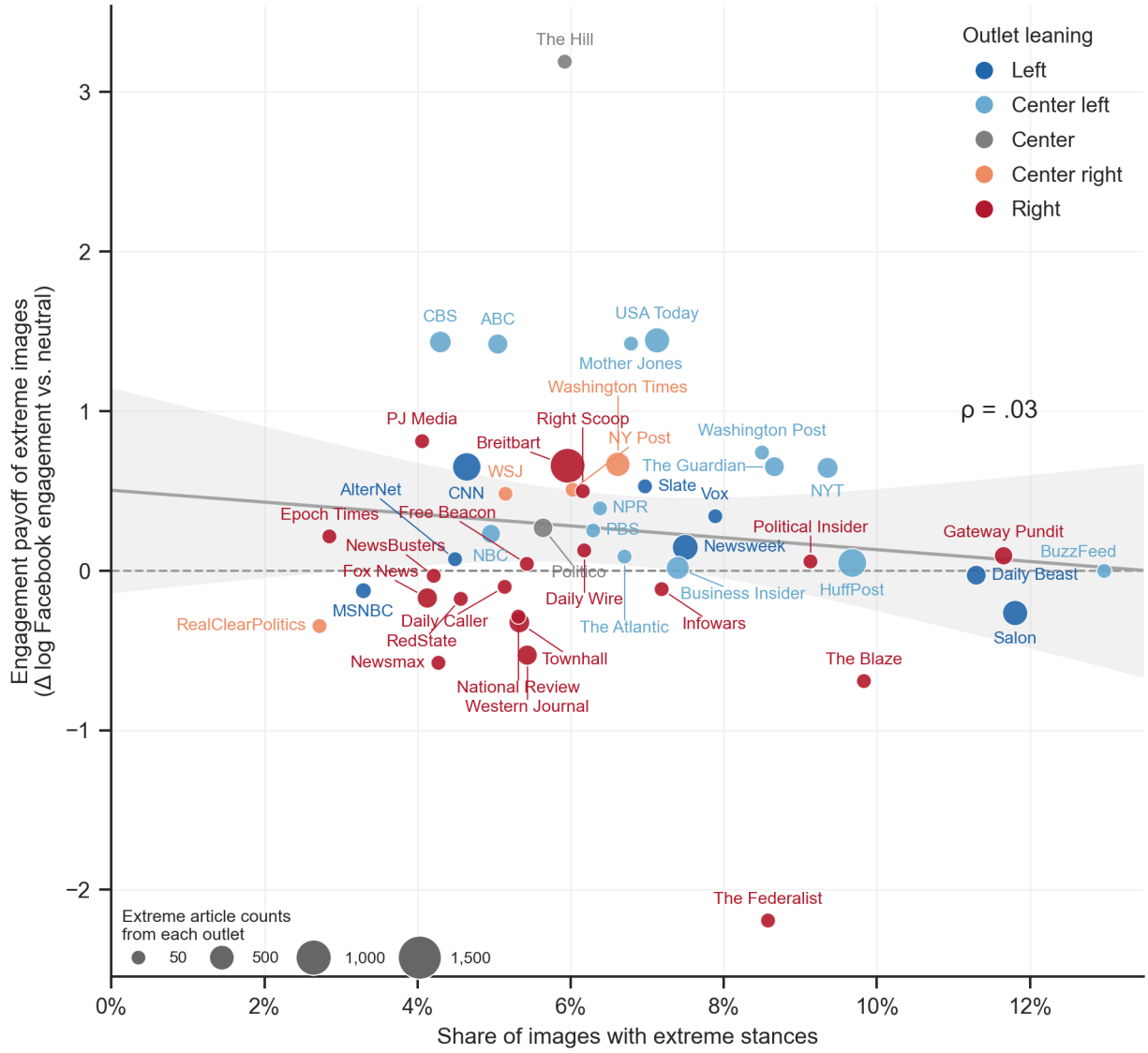


Fig. S21. Extreme-image prevalence and the image engagement payoff across outlets. The image counterpart of main-text Fig. 3A. Share of each outlet's articles with an extreme image versus its image extremity payoff βIII (S6.1; Table S18) for the 47 outlets with at least 100 articles. Marker size = number of extreme-image articles from the outlet; color = MBFC leaning. Line is the OLS fit with 95% band; ρ = Spearman correlation.

Table S20. Extreme-stance prevalence and engagement payoff: full estimates

A. Outlet-level OLS: outlet payoff on extreme-stance share.

Term	Headline: b (SE)	Headline: 95% CI	Headline: P	Image: b (SE)	Image: 95% CI	Image: P
Intercept (payoff at mean share)	1.13 (0.10)	[0.92, 1.33]	<.001	0.26 (0.11)	[0.03, 0.49]	.026
Share (per +10 pp)	−0.20 (0.06)	[−0.32, −0.08]	.002	−0.37 (0.34)	[−1.07, 0.32]	.285
R^2	0.14			0.01		
Outlets	47			47		
Spearman ρ (P)	−0.38 (.008)			0.03 (.825)		

B. Article-level mixed model.

Term	b	SE	95% CI	P
Stance				
H	–0.19	0.03	[–0.25, –0.13]	<.001
$\|H\|$	1.06	0.11	[0.84, 1.27]	<.001
I	–0.12	0.02	[–0.17, –0.08]	<.001
$\|I\|$	0.28	0.09	[0.11, 0.45]	.002
$H \times I$	0.09	0.03	[0.03, 0.15]	.003
$H \times \|I\|$	0.23	0.06	[0.10, 0.35]	<.001
$\|H\| \times I$	0.16	0.04	[0.07, 0.25]	<.001
$\|H\| \times \|I\|$	0.03	0.10	[–0.15, 0.22]	.723
Prevalence				
Extreme-headline share	0.15	0.24	[–0.31, 0.62]	.523
Extreme-image share	2.13	1.35	[–0.52, 4.78]	.115
$\|H\|\times$ extreme-headline share (λ_H)	–0.18	0.07	[–0.32, –0.05]	.009
$\|I\|\times$ extreme-image share (λ_I)	–0.35	0.33	[–1.00, 0.30]	.289
Controls				
Leaning: Center-left	–1.34	1.41	[–4.11, 1.43]	.342
Leaning: Center-right	–3.37	1.80	[–6.90, 0.17]	.062
Leaning: Extreme-right	–3.21	1.63	[–6.40, –0.01]	.049
Leaning: Left	–2.21	1.48	[–5.12, 0.70]	.136
Leaning: Right	–2.22	1.50	[–5.17, 0.73]	.140
Media type: Digital Native	–0.59	0.99	[–2.54, 1.36]	.551
Media type: Magazine	–0.59	1.21	[–2.97, 1.79]	.627
Media type: Network TV	–0.80	1.43	[–3.59, 2.00]	.576
Media type: Newspaper	–0.31	1.48	[–3.21, 2.60]	.836
Year: 2018	0.19	0.03	[0.14, 0.24]	<.001
Year: 2019	0.66	0.02	[0.61, 0.71]	<.001
Year: 2020	1.17	0.02	[1.12, 1.22]	<.001
Year: 2021	–0.12	0.03	[–0.17, –0.07]	<.001
Politician: Bernard Sanders	–0.27	0.05	[–0.37, –0.18]	<.001
Politician: Donald Trump	–0.48	0.04	[–0.55, –0.41]	<.001
Politician: Elizabeth Warren	–0.41	0.05	[–0.51, –0.32]	<.001
Politician: Joe Biden	–0.62	0.04	[–0.69, –0.54]	<.001
Politician: Mike Pence	–0.50	0.05	[–0.59, –0.40]	<.001
Politician: Mitch McConnell	–0.22	0.05	[–0.32, –0.13]	<.001
Politician: Nancy Pelosi	0.09	0.04	[0.00, 0.17]	.047
Constant				
Intercept	7.94	1.67	[4.68, 11.21]	<.001
Outlet random effects and residual				
SD of intercept	1.72			
SD of $\|H\|$ slope	0.66			
SD of $\|I\|$ slope	0.47			
Corr(intercept, $\|H\|$ slope)	–0.20			
Corr(intercept, $\|I\|$ slope)	–0.50			
Corr($\|H\|$ slope, $\|I\|$ slope)	0.33			
Residual SD	2.92			
Articles / outlets	183,487 / 47			
REML log-likelihood	-457,182.8			

Notes. Panel A: OLS of the outlet-specific payoffs (Table S18) on the outlet's extreme-stance share, headlines (βIHI on extreme-headline share) and images (βIII on extreme-image share); share centered at the unweighted 47-outlet mean (20.6% headlines, 6.6% images) and scaled per +10 pp; HC3 SEs with t(45) inference. Panel B: one mixed model with both prevalence interactions (S7.1); outlet random intercept and random slopes on IHI and III with unstructured covariance; REML. Shares centered and scaled as in panel A. Reference categories: Center (leaning), Cable news (media type), 2017 (year), Barack Obama (politician).

S7.3. Robustness of the prevalence–payoff association

This subsection checks whether the negative association between extreme-headline prevalence and the headline payoff (S7.2) depends on imprecise outlet payoffs or on individual outlets, and whether the null image association changes under the same checks. Each sample is analyzed with the three analyses of S7.1 and with a precision-weighted meta-regression (Table S21). Whether the association reflects stable outlet characteristics is addressed in S7.7 and S7.8.

Precision of outlet payoffs. The outlet-level analyses treat each outlet's estimated payoff $\beta_{|H|}$ as known, although payoffs based on few extreme-headline articles are estimated imprecisely. Three checks address this:

- **Meta-regression.** A random-effects meta-regression weights each outlet by $1/(SE^2 + \tau^2)$, where SE is the standard error of its payoff and τ^2 the between-outlet variance in payoffs, so imprecise payoffs count less. The slope is –0.21 per 10 pp (95% CI [–0.36, –0.05], P = .009), nearly identical to the unweighted OLS estimate (–0.20).
- **Mixed model.** The article-level mixed model (S7.1) estimates outlet payoffs and the share interaction jointly and shrinks imprecise outlet payoffs toward the mean, so it does not depend on the two-step estimates. The interaction is –0.18 (P = .009).
- **Thresholds.** Restricting the sample to outlets with at least 50, 100, or 200 extreme-headline articles, or at least 1,000 or 2,000 articles in total (29–45 outlets), gives slopes of –0.19 to –0.23 (OLS), –0.20 to –0.23 (meta-regression), and –0.20 to –0.25 (mixed model), with Spearman correlations of –.40 to –.45; all P ≤ .021.

Influential outlets. The outlet-level slope was re-fitted 47 times, each time omitting one outlet. It ranges from –0.22 to –0.17 per 10 pp and is significant in every fit (all P ≤ .005; Fig. S22A). The most influential outlet is ABC, which combines a high payoff with a very low extreme-headline share. Two outlets warrant separate checks:

- **Breitbart** is the largest outlet (16,292 articles, 8.9% of the sample) and has the largest headline payoff (3.00). Without it, the estimates are –0.18 (OLS, P = .004), –0.18 (meta-regression, P = .011), and –0.16 (mixed model, P = .015).
- **Daily Caller** is the only outlet with a negative headline payoff. Without it, the estimates are –0.21 (OLS, P = .001), –0.21 (meta-regression, P = .005), and –0.19 (mixed model, P = .004).

Images. The image association is not significant in any sample or analysis (Table S21, panel B). Thresholds use extreme-image articles. Estimates are mostly negative (OLS –0.05 to –0.42; meta-regression –0.11 to –0.45; mixed model –0.43 to 0.06; all P ≥ .20), and Spearman correlations are near zero (–.09 to .12). Omitting any one outlet gives slopes from –0.49 to –0.19 (all P ≥ .16; Fig. S22B). The mixed model for the 20 outlets with at least 200 extreme-image articles did not converge.

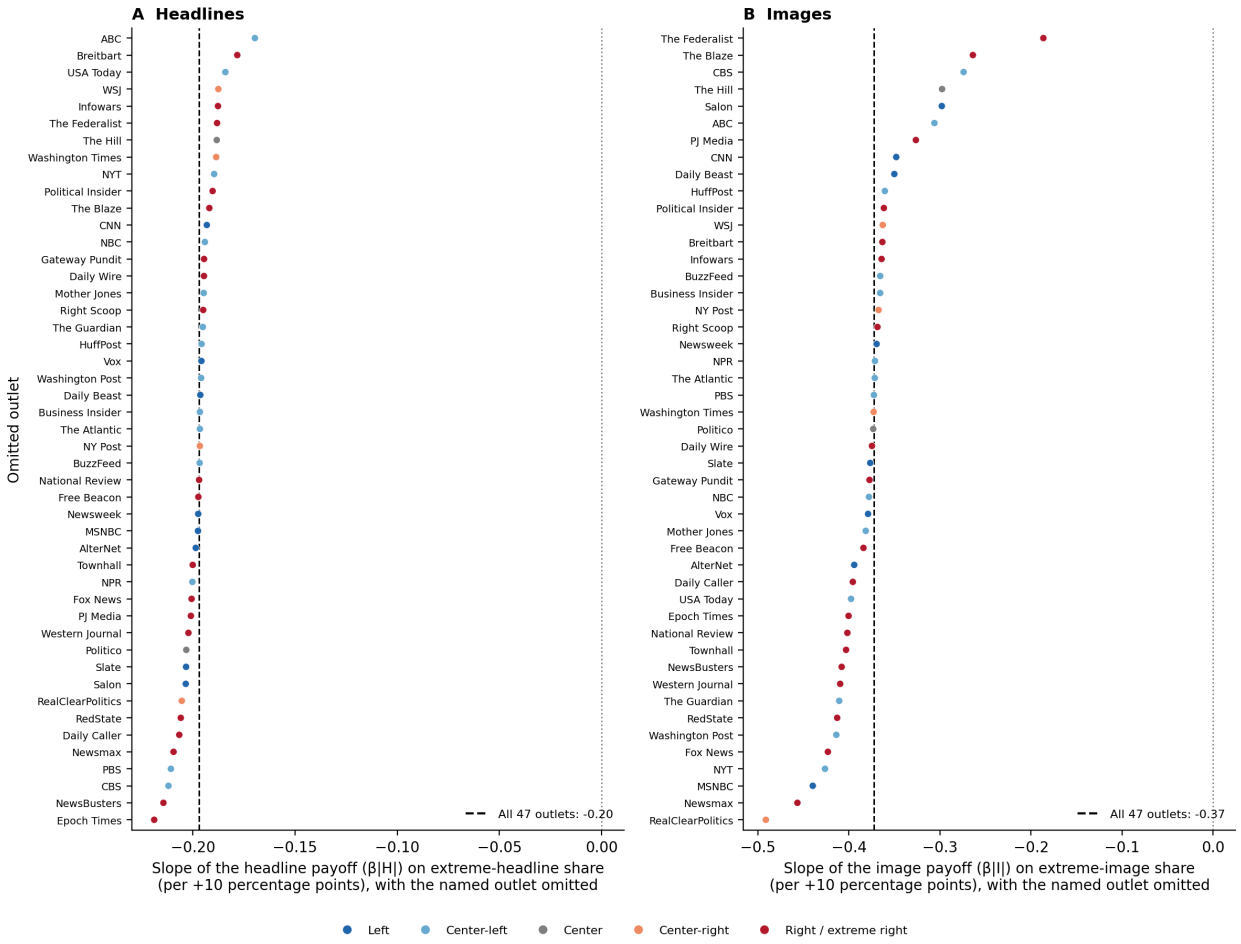


Fig. S22. Leave-one-outlet-out prevalence–payoff slopes. Outlet-level OLS slope of the payoff on extreme-stance share (per +10 pp), re-estimated 47 times, each time omitting the outlet named on the y-axis. (A) Headline payoff βIHI on extreme-headline share. (B) Image payoff βIII on extreme-image share. Colors indicate the omitted outlet's partisan leaning. Dashed line, full-sample slope; dotted line, zero.

Table S21. Prevalence–payoff association: precision of outlet payoffs and influential outlets

A. Headlines (payoff βIHI; extreme-headline share)

Sample	Outlets	Outlet OLS, HC3	Spearman ρ	Meta-regression	Mixed model, payoff × share
All outlets (S7.2)	47	−0.20 [−0.32, −0.08], P = .002	−0.38 (P = .008)	−0.21 [−0.36, −0.05], P = .009	−0.18 [−0.32, −0.05], P = .009
Outlets with ≥50 extreme-headline articles	45	−0.22 [−0.35, −0.08], P = .002	−0.41 (P = .005)	−0.23 [−0.38, −0.07], P = .006	−0.20 [−0.34, −0.05], P = .008
Outlets with ≥100 extreme-headline articles	43	−0.23 [−0.37, −0.09], P = .002	−0.45 (P = .002)	−0.23 [−0.39, −0.07], P = .006	−0.21 [−0.36, −0.07], P = .005
Outlets with ≥200 extreme-headline articles	37	−0.19 [−0.34, −0.05], P = .010	−0.41 (P = .011)	−0.21 [−0.39, −0.04], P = .017	−0.25 [−0.39, −0.10], P = .001
Outlets with ≥1,000 articles in total	41	−0.19 [−0.31, −0.06], P = .004	−0.40 (P = .009)	−0.20 [−0.36, −0.04], P = .016	−0.21 [−0.35, −0.07], P = .004
Outlets with ≥2,000 articles in total	29	−0.23 [−0.38, −0.07], P = .005	−0.44 (P = .018)	−0.23 [−0.41, −0.04], P = .021	−0.23 [−0.36, −0.10], P < .001
Excluding Breitbart	46	−0.18 [−0.30, −0.06], P = .004	−0.37 (P = .011)	−0.18 [−0.32, −0.04], P = .011	−0.16 [−0.29, −0.03], P = .015

Sample	Outlets	Outlet OLS, HC3	Spearman ρ	Meta-regression	Mixed model, payoff × share
Excluding Daily Caller	46	−0.21 [−0.33, −0.09], P = .001	−0.40 (P = .005)	−0.21 [−0.36, −0.07], P = .005	−0.19 [−0.32, −0.06], P = .004
Leave one outlet out (47 fits)	46	range −0.22 to −0.17; all P ≤ .005			

B. Images (payoff βIII; extreme-image share)

Sample	Outlets	Outlet OLS, HC3	Spearman ρ	Meta-regression	Mixed model, payoff × share
All outlets (S7.2)	47	−0.37 [−1.07, 0.32], P = .285	0.03 (P = .825)	−0.26 [−1.05, 0.53], P = .513	−0.35 [−1.00, 0.30], P = .289
Outlets with ≥50 extreme-image articles	43	−0.31 [−0.97, 0.34], P = .341	0.00 (P = .992)	−0.27 [−1.07, 0.54], P = .508	−0.43 [−1.09, 0.23], P = .201
Outlets with ≥100 extreme-image articles	33	−0.23 [−0.97, 0.50], P = .520	0.01 (P = .947)	−0.24 [−1.06, 0.57], P = .546	−0.15 [−1.03, 0.74], P = .743
Outlets with ≥200 extreme-image articles	20	−0.42 [−1.43, 0.58], P = .388	−0.09 (P = .705)	−0.45 [−1.58, 0.67], P = .410	−0.34† [−1.57, 0.89], P = .587
Outlets with ≥1,000 articles in total	41	−0.25 [−1.00, 0.51], P = .512	0.07 (P = .672)	−0.24 [−1.10, 0.63], P = .584	−0.41 [−1.13, 0.32], P = .272
Outlets with ≥2,000 articles in total	29	−0.05 [−0.83, 0.73], P = .893	0.12 (P = .540)	−0.11 [−1.00, 0.78], P = .806	0.06 [−0.66, 0.78], P = .870
Excluding Breitbart	46	−0.36 [−1.06, 0.33], P = .296	0.04 (P = .815)	−0.25 [−1.05, 0.55], P = .536	−0.32 [−0.98, 0.35], P = .354
Excluding Daily Caller	46	−0.40 [−1.10, 0.31], P = .263	0.01 (P = .935)	−0.27 [−1.08, 0.53], P = .493	−0.35 [−1.01, 0.30], P = .292
Leave one outlet out (47 fits)	46	range −0.49 to −0.19; all P ≥ .16			

Notes. Estimates are per +10 pp of the outlet's extreme-stance share, with 95% CIs. Outlet OLS: payoff on share, HC3 SEs (S7.2). Meta-regression: random-effects meta-regression of the payoff on share, weighting each outlet by $1/(SE^2 + \tau^2)$, with REML τ^2 and Knapp–Hartung SEs (all outlets: headlines $\tau = 0.583$, images $\tau = 0.477$). Mixed model: IHI × share or III × share interaction in the article-level mixed model of S7.1, refitted on each sample. † Model did not converge. Thresholds count articles per outlet. Daily Caller is the only outlet with a negative headline payoff.

S7.4. Credibility

Headlines. Higher-credibility outlets have larger headline payoffs (ρ = .59, P < .001; n = 47). In outlet-level OLS, each additional 10 NewsGuard points is associated with a headline payoff 0.12 larger (SE 0.02, P < .001; R^2 = .23) (Table S22A; main-text Fig. 3B).

Images. The image payoff shows a weaker association in the same direction (ρ = .30, P = .040): 0.10 per 10 points (SE 0.05, P = .042; R^2 = .14) (Fig. S23).

Mixed model. With NewsGuard interacted with IHI and III alongside share (Table S22B), the IHI × NewsGuard interaction is 0.08 per 10 points (P = .044) and the share interaction attenuates to −0.13 (P = .095). The III × NewsGuard interaction is 0.04 (P = .17).

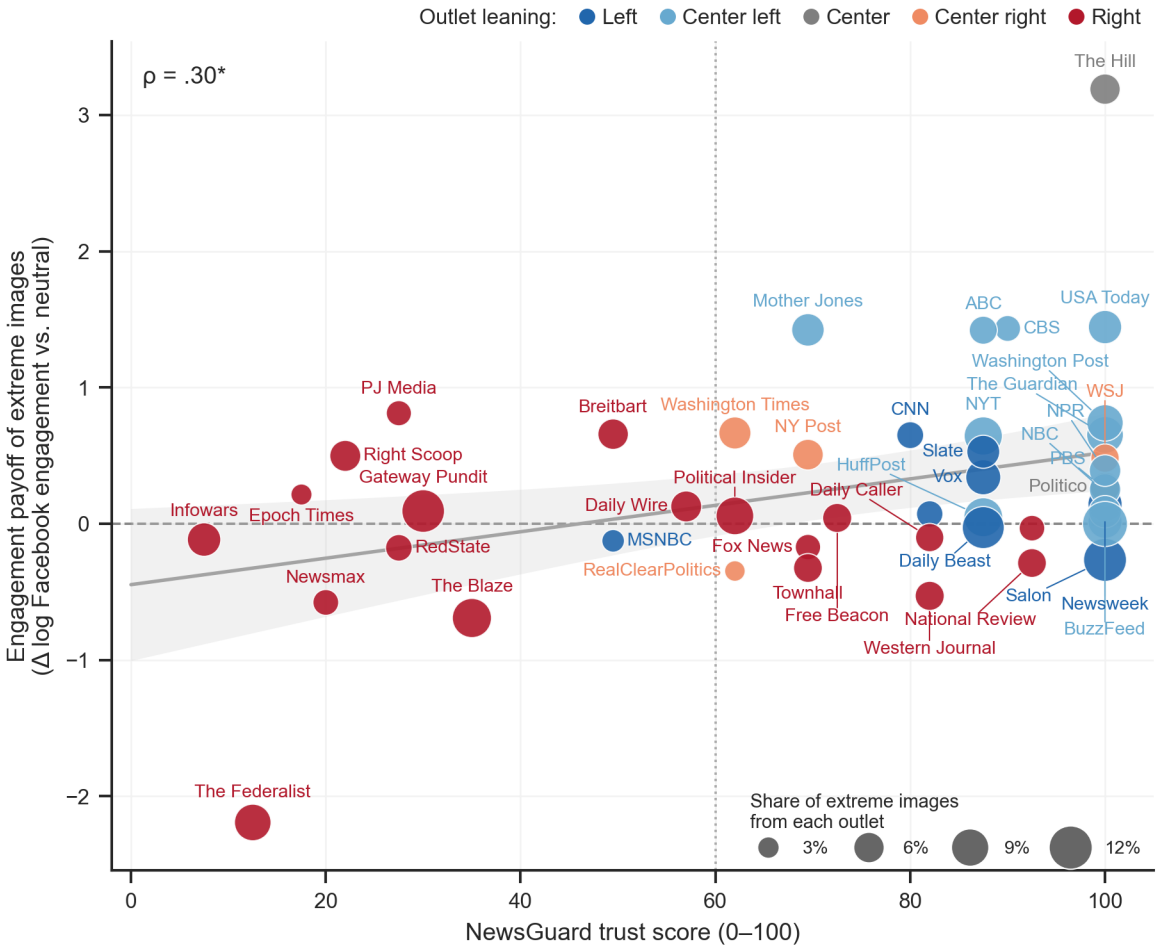


Fig. S23. NewsGuard trust score and the image extremity payoff. Image counterpart of main-text Fig. 3B. NewsGuard score (18 March 2025 snapshot) versus the outlet image payoff βIII for 47 outlets; marker size = share of the outlet's images with extreme stances; color = MBFC leaning; dotted line = NewsGuard's 60-point credibility threshold. Line is the OLS fit with 95% band; ρ = Spearman correlation. * P < .05.

Table S22. NewsGuard credibility and the extremity payoffs

A. Outlet level: association with the outlet payoffs.

Statistic	Headline payoff $\beta_{|H|}$	Image payoff $\beta_{|I|}$
Spearman ρ (P)	0.59 (<.001)	0.30 (.040)
OLS: NewsGuard score (per 10 points), b (SE) [95% CI]	0.12 (0.02) [0.08, 0.17], P <.001	0.10 (0.05) [0.00, 0.19], P .042
R^2 (NewsGuard score)	0.23	0.14
Outlets	47	47

B. Article level: mixed model with |H| and |I| interacted with the characteristic and with extreme-stance share (183,487 articles, 47 outlets).

Term	b	SE	95% CI	P
$\|H\|\times$ extreme-headline share (per 10 pp)	–0.13	0.08	[–0.28, 0.02]	.095
$\|I\|\times$ extreme-image share (per 10 pp)	–0.37	0.33	[–1.03, 0.28]	.261
NewsGuard (per 10 points)	0.01	0.15	[–0.28, 0.29]	.969
$\|H\|\times$ NewsGuard (per 10 points)	0.08	0.04	[0.00, 0.15]	.044
$\|I\|\times$ NewsGuard (per 10 points)	0.04	0.03	[–0.02, 0.10]	.167

Notes. Panel A: outlet-specific payoffs (Table S18); OLS with HC3 SEs and t inference. Panel B: the prevalence–payoff mixed model (S7.1; Table S20) with the characteristic added as a main effect and interacted with |H| and |I|; stance terms, controls, and politician fixed effects are not shown. NewsGuard score from the 18 March 2025 snapshot (S3.2). Mixed model: NewsGuard centered at the 47-outlet mean (72.7) and scaled per 10 points.

S7.5. Audience ideology

Headlines. The headline payoff declines linearly with Bakshy alignment (ρ = –.30, P = .048; OLS slope –0.33, P = .043; n = 45), but a quadratic fits better: squared term $b_2 = -0.79$ (SE 0.32, P = .017; R^2 = .18), with the vertex near the center of the scale (alignment ≈ –0.09). Payoffs are therefore highest for outlets with ideologically mixed audiences and lower at both ends (main-text Fig. 3C). Consistent with this, |alignment| correlates with the headline payoff at ρ = –.43 (P = .004) (Table S23A).

Images. The image payoff shows the same inverted U: squared term $b_2 = -1.02$ (SE 0.50, P = .049; R^2 = .22), vertex ≈ –0.04, and |alignment| ρ = –.45 (P = .002) (Fig. S24).

Mixed model. The linear interactions with alignment are not significant for either payoff (|H|: –0.21, P = .18; |I|: –0.22, P = .087), while the headline share interaction remains significant (–0.19, P = .010) (Table S23B).

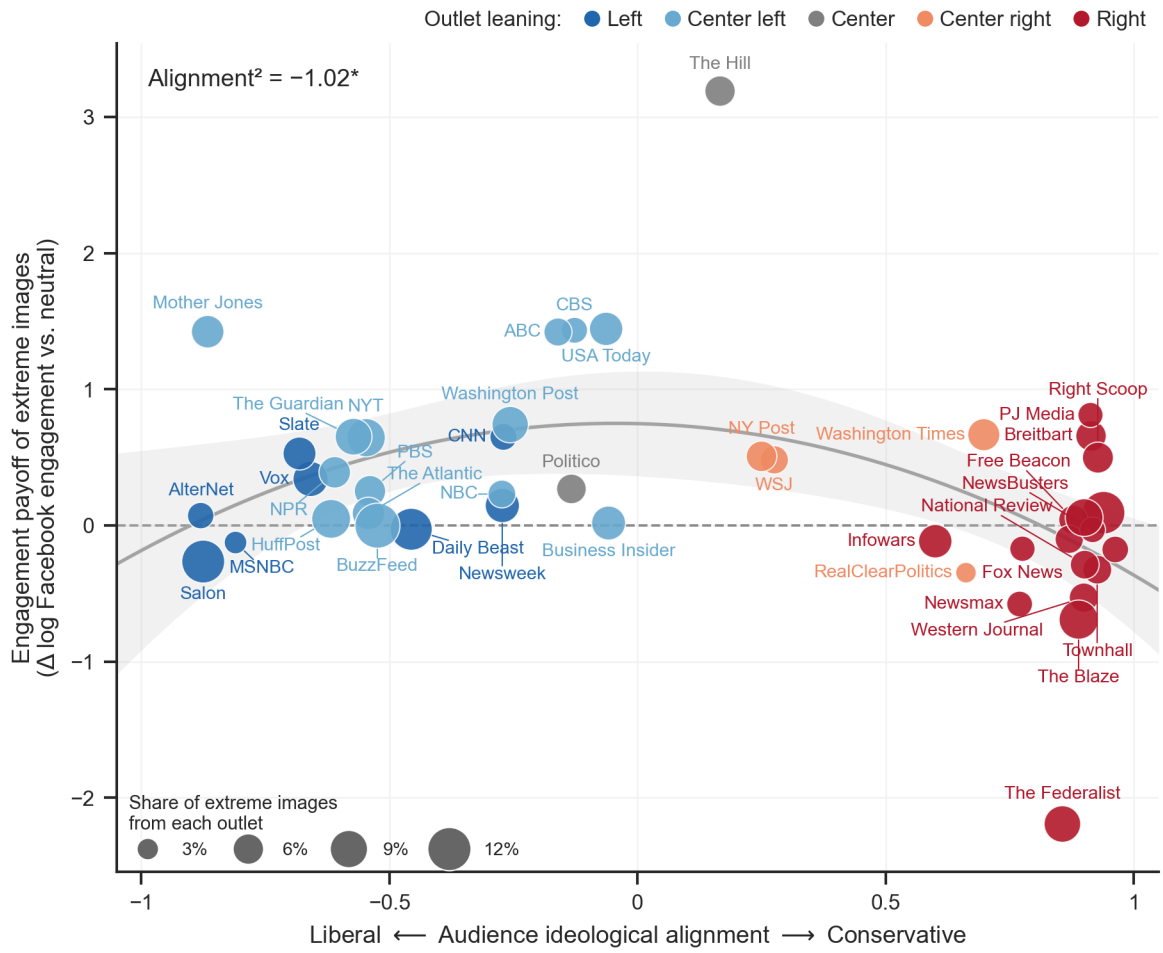


Fig. S24. Audience ideological alignment and the image extremity payoff. Image counterpart of main-text Fig. 3C. Bakshy et al. audience alignment (–1 liberal, +1 conservative) versus the outlet image payoff β|I| for 45 outlets; marker size = share of the outlet's images with extreme stances; color = MBFC leaning. Curve is the quadratic OLS fit with 95% band; the label gives the squared term (HC3). * P < .05.

Table S23. Audience ideology and the extremity payoffs

A. Outlet level: association with the outlet payoffs.

Statistic	Headline payoff β|H|	Image payoff β|I|
Spearman ρ (P)	−0.30 (.048)	−0.28 (.067)
Linear OLS: alignment, b (SE) [95% CI]	−0.33 (0.16) [−0.65, −0.01], P .043	−0.34 (0.14) [−0.63, −0.05], P .022
Quadratic OLS: alignment, b (SE) [95% CI]	−0.14 (0.14) [−0.43, 0.15], P .332	−0.09 (0.20) [−0.50, 0.32], P .655
Quadratic OLS: alignment², b (SE) [95% CI]	−0.79 (0.32) [−1.43, −0.15], P .017	−1.02 (0.50) [−2.04, −0.01], P .049
R^2 (Quadratic OLS)	0.18	0.22
Spearman ρ with |alignment| (P)	−0.43 (.004)	−0.45 (.002)
Outlets	45	45

B. Article level: mixed model with |H| and |I| interacted with the characteristic and with extreme-stance share (176,480 articles, 45 outlets).

Term	b	SE	95% CI	P
$\|H\|\times$ extreme-headline share (per 10 pp)	−0.19	0.07	[−0.33, −0.05]	.010
$\|I\|\times$ extreme-image share (per 10 pp)	−0.48	0.35	[−1.16, 0.20]	.167
Alignment	0.42	1.30	[−2.12, 2.96]	.745
$\|H\|\times$ alignment	−0.21	0.16	[−0.52, 0.10]	.183
$\|I\|\times$ alignment	−0.22	0.13	[−0.46, 0.03]	.087

Notes. Panel A: outlet-specific payoffs (Table S18); OLS with HC3 SEs and t inference. Panel B: the prevalence–payoff mixed model (S7.1; Table S20) with the characteristic added as a main effect and interacted with |H| and |I|; stance terms, controls, and politician fixed effects are not shown. Bakshy et al. audience alignment, −1 (liberal) to +1 (conservative), for 45 outlets (S3.2). The quadratic headline fit is the main-text Fig. 3C curve. Mixed model: alignment centered at the 45-outlet mean.

S7.6. Audience size

Headlines. Outlets with more Facebook followers have larger headline payoffs (ρ = .38, P = .009; n = 47). In outlet-level OLS, each tenfold increase in followers is associated with a headline payoff 0.46 larger (SE 0.15, P = .004; R^2 = .13) (Table S24A; main-text Fig. 3D).

Images. The image payoff points in the same direction but is not significantly associated with followers (ρ = .27, P = .066; OLS 0.39 per tenfold, SE 0.26, P = .14) (Fig. S25).

Mixed model. Neither followers interaction is significant (IHI: 0.11 per SD, P = .34; III: 0.08, P = .29), and the share interaction attenuates to –0.15 (P = .073) (Table S24B).

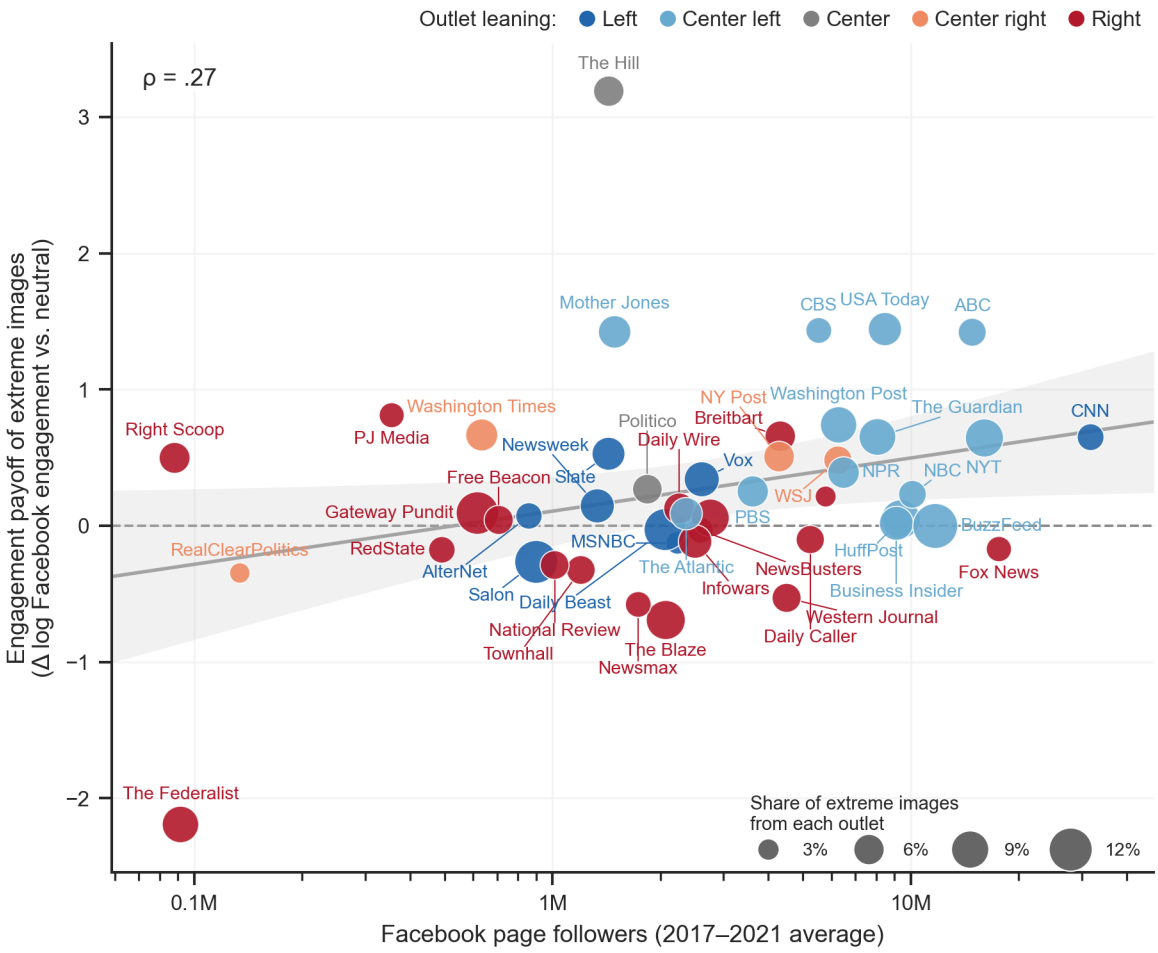


Fig. S25. Facebook audience size and the image extremity payoff. Image counterpart of main-text Fig. 3D. Facebook page followers (2017–2021 average, log scale) versus the outlet image payoff βIII for 47 outlets; marker size = share of the outlet's images with extreme stances; color = MBFC leaning. Line is the OLS fit on log10 followers with 95% band; ρ = Spearman correlation.

Table S24. Facebook audience size and the extremity payoffs

A. Outlet level: association with the outlet payoffs.

Statistic	Headline payoff βIHI	Image payoff βIII
Spearman ρ (P)	0.38 (.009)	0.27 (.066)
OLS: log10 Facebook followers, b (SE) [95% CI]	0.46 (0.15) [0.16, 0.76], P .004	0.39 (0.26) [–0.13, 0.91], P .138
R^2 (log10 Facebook followers)	0.13	0.09
Outlets	47	47

B. Article level: mixed model with IHI and III interacted with the characteristic and with extreme-stance share (183,487 articles, 47 outlets).

Term	b	SE	95% CI	P
$\|H\|\times$ extreme-headline share (per 10 pp)	–0.15	0.08	[–0.31, 0.01]	.073
$\|I\|\times$ extreme-image share (per 10 pp)	–0.34	0.33	[–0.98, 0.31]	.306
Log followers (per SD)	0.70	0.36	[0.00, 1.39]	.049
$\|H\|\times$ log followers (per SD)	0.11	0.12	[–0.12, 0.35]	.339
$\|I\|\times$ log followers (per SD)	0.08	0.08	[–0.07, 0.24]	.293

Notes. Panel A: outlet-specific payoffs (Table S18); OLS with HC3 SEs and t inference. Panel B: the prevalence–payoff mixed model (S7.1; Table S20) with the characteristic added as a main effect and interacted with IHI and III; stance terms, controls, and politician fixed effects are not shown. Facebook page followers, 2017–2021 average (S3.2); outlet-level OLS per tenfold increase. Mixed model: log followers standardized across the 47 outlets (per SD).

S7.7. Outlet characteristics and the prevalence–payoff association

This subsection asks whether credibility, audience ideology, or audience size accounts for the negative association between extreme-headline prevalence and the headline payoff (S7.2).

Overlap. Outlets with high extreme-headline shares have lower NewsGuard scores (ρ = –.42), more one-sided audiences (|alignment|, ρ = .72), and smaller Facebook audiences (ρ = –.55) (Table S19). The characteristics that predict larger payoffs therefore tend to belong to outlets that publish fewer extreme headlines.

Adjustment in the mixed model. Adding each characteristic, interacted with |H| and |I|, to the prevalence–payoff mixed model changes the share interactions as follows (Tables S20 and S22–S24):

Characteristic added	$\|H\|\times$ extreme-headline share	$\|I\|\times$ extreme-image share	Outlets
None (S7.2)	–0.18 (P = .009)	–0.35 (P = .29)	47
NewsGuard credibility	–0.13 (P = .095)	–0.37 (P = .26)	47
Audience ideology (alignment)	–0.19 (P = .010)	–0.48 (P = .17)	45
Audience size (log followers)	–0.15 (P = .073)	–0.34 (P = .31)	47

- The headline share interaction stays negative in every model. Credibility and audience size reduce it by about 15–30% and push it just past P = .05; audience ideology leaves it unchanged.
- The characteristics' own |H| interactions are small and mostly non-significant (credibility 0.08 per 10 points, P = .044; ideology –0.21, P = .18; followers 0.11 per SD, P = .34).
- The image share interaction is negative and non-significant in every model.

Reading. Extreme-headline prevalence, credibility, and audience size describe largely the same outlets. Each adjusted model adds three outlet-level parameters to a comparison across only 45–47 outlets, so the modest attenuation of the share interaction is consistent with a loss of precision and does not show that credibility or audience size explains the prevalence–payoff association. Share remains the more consistent predictor of the headline payoff, but these cross-sectional comparisons cannot separate the characteristics or establish causal effects. S7.8 addresses stable outlet differences directly by comparing each outlet with itself over time.

S7.8. Within-outlet association over time

The analyses above compare outlets with one another. Outlets with many extreme headlines also differ in credibility, audience ideology, audience size, and much else, and with 47 outlets these differences cannot be fully separated (S7.7). This subsection instead compares each outlet *with itself* over time: in months after an outlet published more extreme headlines than usual, was its headline payoff smaller than usual? Because an outlet's credibility, audience ideology, audience size, leaning, and media type barely change over 2017–2021, they cannot explain such a within-outlet pattern. The analysis is a robustness check on the association in S7.2, not a causal test.

Sample. 181,471 articles from 47 outlets. Relative to the main sample, 997 duplicated articles and 1,048 articles with an uncertain publication month are removed. Months are calendar months (New York time).

Exposure: the outlet's recent share. $P_{k,t}$ is the percentage of outlet k's articles with an extreme headline in the three calendar months before month t (month t itself excluded), per +10 pp. Each window must contain at least 50 articles, so each outlet's first months are dropped (168,532 articles in the primary model). Most variation in the share is between outlets (SD 14 pp); an outlet's recent share varies less around its own average (SD 5 pp).

Primary model. For article i about politician p in outlet k and month t,

$$Y_{ikt} = \text{model (iv) stance terms} + \theta P_{k,t} + \delta\,|H_i|\,P_{k,t} + \alpha_k + \tau_t + \pi_p + \gamma_k|H_i| + \mu_t|H_i| + \varepsilon_{ikt}.$$

- Y_{ikt}: log engagement; $|H_i|$: headline extremity, from 0 (neutral) to 1 (very favorable or very unfavorable), as in model (iv).
- α_k, τ_t, π_p: outlet, month, and politician fixed effects. The outlet fixed effects absorb everything stable about an outlet (leaning, media type, credibility, audience ideology, audience size); the month fixed effects absorb the rise and fall of engagement over 2017–2021.
- $\gamma_k|H_i|$: each outlet's own average headline payoff. Stable differences in payoffs between outlets, the basis of S7.2, therefore do not enter δ.
- $\mu_t|H_i|$: each month's platform-wide headline payoff. Changes in the payoff that affect all outlets at once therefore do not enter δ.
- δ: the estimate of interest (below).
- SEs are clustered by outlet (47 clusters) with t_{46} intervals; a restricted wild cluster bootstrap-t (Webb weights, 9,999 draws) is reported as a check.

What δ estimates. δ asks whether an outlet's headline payoff is below its own average in months when its recent share is above its own average, beyond any change shared by all outlets that month. A negative δ means that the prevalence–payoff association holds within outlets and is not explained by stable differences between them.

What it cannot establish. δ would be a causal effect only if nothing else that changes the outlet's payoff moved together with its recent share. This fails if, for example, an outlet's editorial strategy and its audience's response drift together over time. Two checks test this:

- **Lead test.** Replace (or supplement) the recent share with the share in the three months *after* month

t. If a higher share lowered the later payoff, the future share should not predict the current payoff.

- **Outlet-specific trends.** Allow each outlet its own gradual trend in engagement and in the headline payoff (or compare only months within the same quarter). If the association reflects slow drift rather than month-to-month changes, it should shrink.

Results. Building up the model step by step (Table S25; Fig. S26):

1. Between outlets, using each outlet's share over the whole period: –0.30 per 10 pp with controls (P = .058) and –0.27 with outlet fixed effects (P = .007), matching S7.2.
2. Within outlets, using the recent share with outlet, month, and politician fixed effects: –0.33 (P < .001).
3. Primary model, adding outlet- and month-specific headline payoffs: **–0.56 per 10 pp** (95% CI [–0.94, –0.19], P = .004; wild bootstrap P = .002).

When an outlet's recent share of extreme headlines is 10 percentage points above its usual level, its headline payoff is about 0.56 smaller. The estimate is stable across exposure windows (1, 6, and 12 months and the previous calendar quarter: –0.36 to –0.79, all P ≤ .023), minimum window sizes (1 to 200 articles: –0.49 to –0.67), excluding the 2018 and 2020 election periods or spring 2020, adding the model (v) covariates (–0.64), and leaving out any one outlet (–0.61 to –0.38). It is driven by unfavorable-extreme headlines (–0.53, P = .006; favorable-extreme: 0.16, P = .60). The image analogue is not significant (0.18, P = .28), matching the cross-sectional image result.

Limits. The two checks show that the analysis cannot establish that a higher share leads to a smaller payoff:

- **The future predicts as well as the past.** The following three months' share predicts the current payoff almost as strongly as the recent share (–0.49, P = .005). Entered together, the recent share remains significant (–0.54, P = .004) and the following share is weaker (–0.24, P = .063), but the two do not differ significantly (P = .17).
- **Slow trends explain much of it.** Outlet-specific trends in engagement halve the estimate (–0.30, P = .12); trends in both engagement and the headline payoff remove it (–0.00, P = .98). Comparing only months within the same quarter or month gives about –0.22 (P = .15 and .13).
- Estimated by Poisson pseudo-maximum likelihood on raw engagement counts instead of log engagement, the estimate is near zero (–0.08, P = .37).

Reading. The negative prevalence–payoff association is not an artifact of comparing different kinds of outlets: it also appears when each outlet is compared with itself over time. But the within-outlet association reflects gradual drift. Outlets whose extreme-headline share rose over the period tended to see their headline payoff fall, and vice versa. The data cannot tell whether a rising share erodes the payoff, a falling payoff prompts more extreme headlines, or both follow a third outlet-level change. The relationship is sensitive to outlet-specific time trends and does not establish a causal effect of prevalence on engagement.

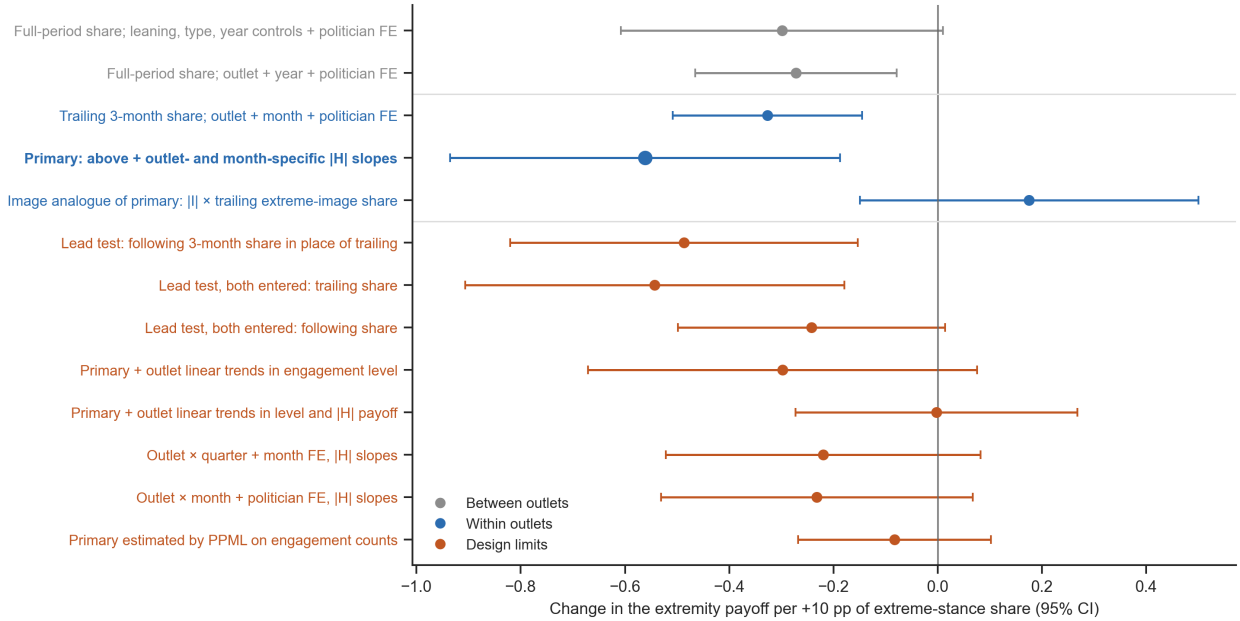


Fig. S26. Prevalence–payoff association between and within outlets over time. Change in the headline payoff per +10 pp of the outlet's extreme-headline share, with 95% CIs (outlet-clustered SEs). Grey: comparisons between outlets, using each outlet's share over the whole period. Blue: comparisons within outlets, using the share in the previous three months; the primary model is in bold, and the image row uses the image payoff and extreme-image share. Orange: checks of what the within-outlet comparison can establish (lead test, outlet-specific trends, outlet × period fixed effects, Poisson model). Estimates are in Table S25.

Table S25. Prevalence–payoff association between and within outlets over time

Group	Specification	δ per 10 pp [95% CI]	SE	P	Wild bootstrap P	Articles
Between outlets	Full-period share; leaning, type, year controls + politician FE	−0.30 [−0.61, 0.01]	0.15	.058		181,442
	Full-period share; outlet + year + politician FE	−0.27 [−0.47, −0.08]	0.10	.007		181,442
Within outlets	Trailing 3-month share; outlet + month + politician FE	−0.33 [−0.51, −0.15]	0.09	<.001	.003	168,532
	Primary: above + outlet- and month-specific |H| slopes	−0.56 [−0.94, −0.19]	0.19	.004	.002	168,532
	Image analogue of primary: |I| × trailing extreme-image share	0.18 [−0.15, 0.50]	0.16	.283		168,532
Design limits	Lead test: following 3-month share in place of trailing	−0.49 [−0.82, −0.15]	0.17	.005		167,923
	Lead test, both entered: trailing share	−0.54 [−0.91, −0.18]	0.18	.004		157,377
	Lead test, both entered: following share	−0.24 [−0.50, 0.01]	0.13	.063		157,377

Group	Specification	δ per 10 pp [95% CI]	SE	P	Wild bootstrap P	Articles
	Primary + outlet linear trends in engagement level	−0.30 [−0.67, 0.08]	0.19	.115		168,532
	Primary + outlet linear trends in level and IHI payoff	−0.00 [−0.27, 0.27]	0.13	.984		168,532
	Outlet × quarter + month FE, IHI slopes	−0.22 [−0.52, 0.08]	0.15	.149		168,532
	Outlet × month + politician FE, IHI slopes	−0.23 [−0.53, 0.07]	0.15	.125	.156	168,530
	Primary estimated by PPML on engagement counts	−0.08 [−0.27, 0.10]	0.09	.370		168,532

Notes. δ = change in the headline extremity payoff (Δ log engagement, extreme vs neutral headline) per +10 pp of the outlet's extreme-headline share; the image row uses the image payoff and extreme-image share. Between outlets: full-period share. Within outlets and design limits: share of the outlet's articles with extreme headlines in the three complete calendar months before the article's month (windows ≥ 50 articles). Primary model as in S7.8. Lead test: the share in the three months after the article's month. SEs clustered by outlet (47 clusters), t(46) intervals; wild bootstrap: restricted wild cluster bootstrap-t, Webb weights, 9,999 draws. The longitudinal sample (181,471 articles) removes 997 duplicate articles and 1,048 articles whose month is uncertain; trailing windows are unavailable for the first months of each outlet's series.

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